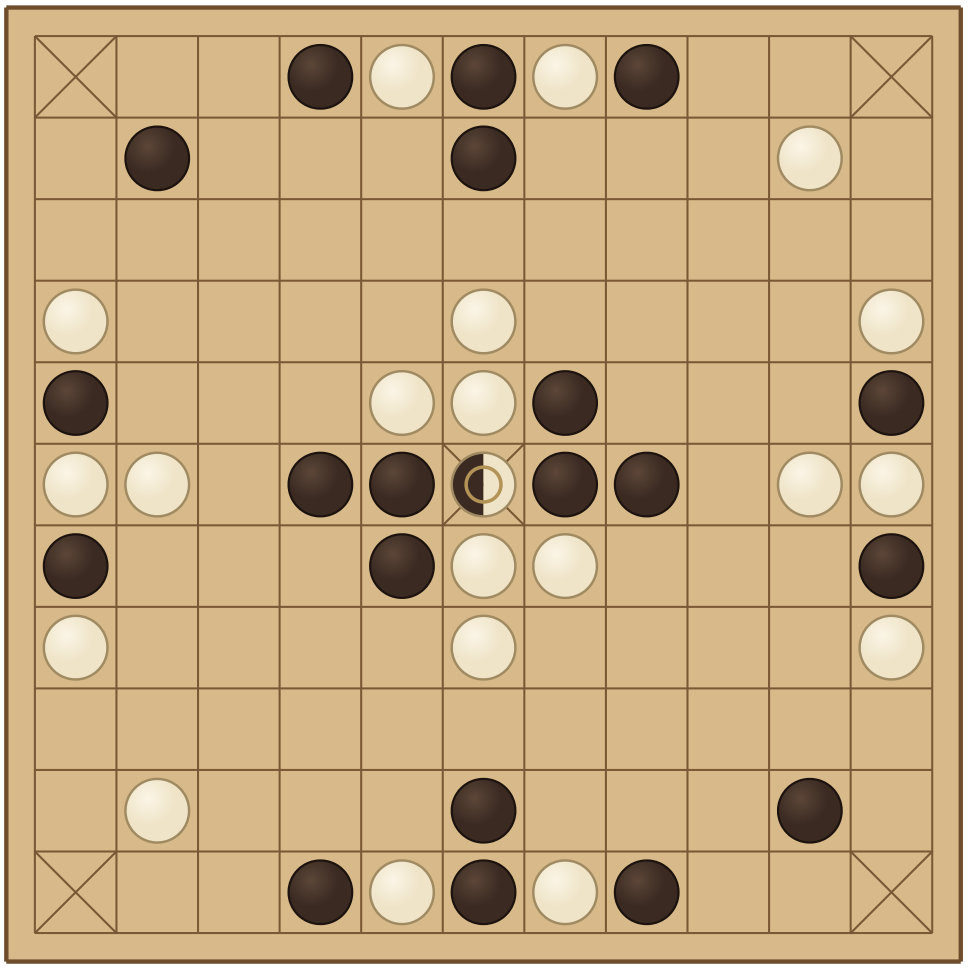
NORMAN TAFL

The player's manual

From beginner to strong player



A game by Guillaume Enguehard

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How to use this manual

The manual follows a four-level progression. Each chapter presents an idea, shows it on the board, then ends with exercises. The solutions are gathered at the end of the book.

Solve the exercises on a real board without moving the pieces: this is how you learn to calculate. When you find a solution, check every reply of your opponent before reading the answer.

The exercise positions come from high-level games, and every solution has been carefully checked.

- **Level 1 · Beginner** (part I) — the rules, capturing, the king, the threat rule.
- **Level 2 · Intermediate** (part II) — winning material, avoiding traps, defending, winning in one or two moves.
- **Level 3 · Advanced** (parts III and IV) — opening theory, control and occupation of squares, the value of pawns, the functions of pawn structures, sacrifices, moving the pair out, plans.
- **Level 4 · Strong player** (parts V to VII) — long combinations, calculation, endgame theory, annotated games.

Notation

A move is written **start-arrival**, for example **d1-d3**. An exclamation mark points out a good move, a question mark a mistake. Under each diagram, “Black to play” or “White to play” tells you whose turn it is.

LEVEL 1 · BEGINNER

The basics

This part introduces the rules step by step, with short exercises after each idea. By the end, you will know how to move your pieces, capture, protect your king and recognise a win.

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1 The board and the pieces

2 Capturing

3 The king and the pair

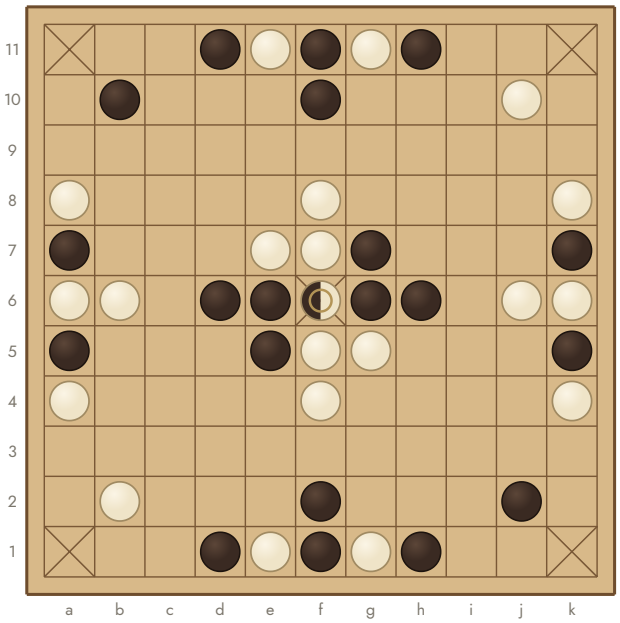
4 Threats and winning

1 The board and the pieces

LEVEL 1 · BEGINNER

NORMAN TAFL is played by two players on a board of eleven by eleven squares. Files are named **a** to **k** and ranks **1** to **11**, so every square has a name, such as **f6**, the central square.

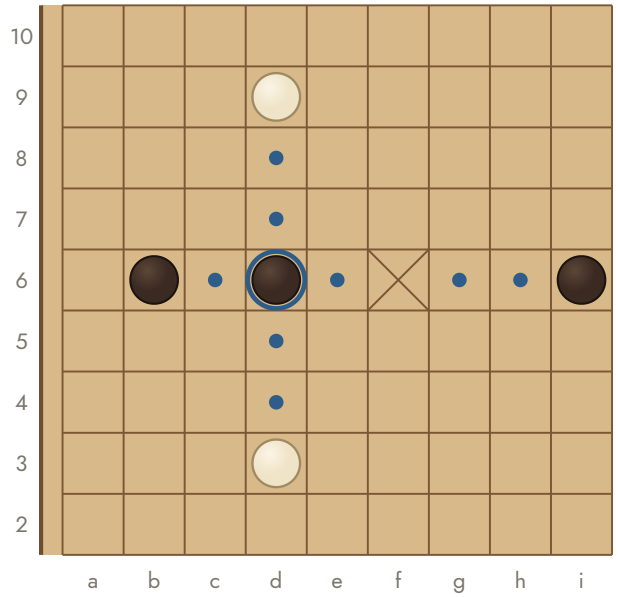
Five squares are **special** and marked with a cross: the four corners and the **throne**, f6. Each side, Black (ebony) and White (ivory), has twenty pawns and a king. Both kings start stacked on the throne: this is the **pair**.



The starting position. Both sides are set up the same way: a quarter-turn of the board turns one into the other.

Moving

- Every piece moves in a straight line, sideways or up and down, over as many empty squares as you like, like a rook in chess.
- Pieces never jump over another piece.
- No piece may stop on the throne f6; pieces may cross it when it is empty.
- Only kings may enter a corner.
- **Black plays first**, then players take turns.



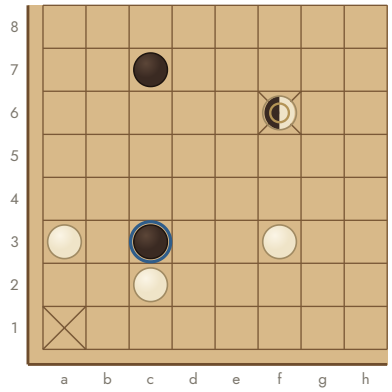
The squares the pawn on d6 can reach (dots). It is stopped by the pieces on b6, i6, d9 and d3; it crosses the empty throne f6 but cannot stop on it.

Notation

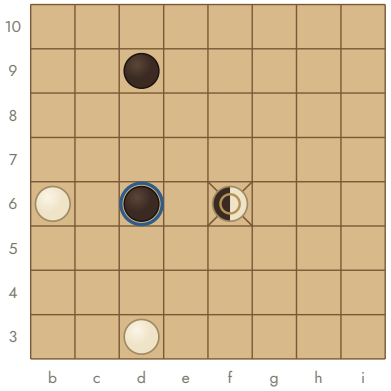
A move is written with its starting and arrival squares: **d1-d3** means the piece on d1 moves to d3. Under each diagram, "Black to play" or "White to play" tells you whose turn it is.

> **REMEMBER**
>
> Everything happens along straight lines. Before each move, look at the ranks and files: they are the roads of your pieces and of your opponent's.

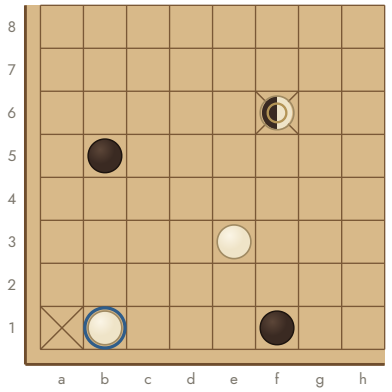
Exercises



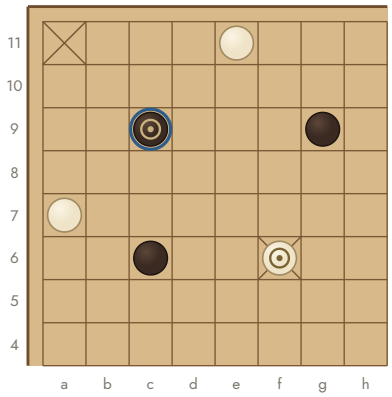
1.1 How many squares can the pawn on c3 reach?



1.2 How many squares can the pawn on d6 reach?



1.3 How many squares can the pawn on b1 reach?



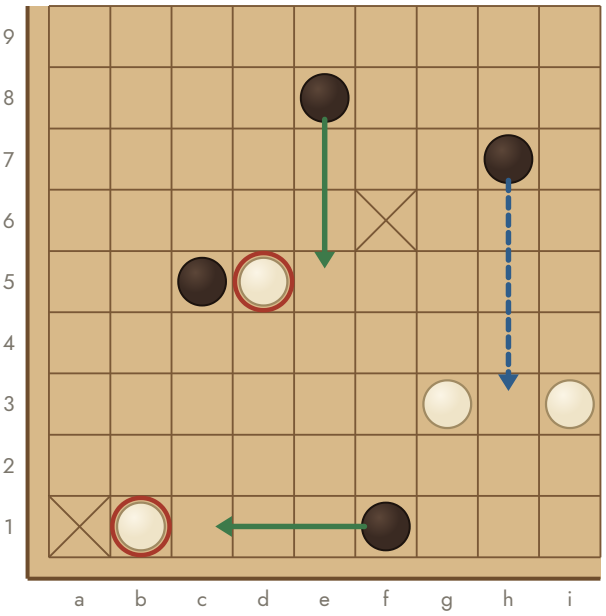
1.4 How many squares can the king on c9 reach?

2 Capturing

LEVEL 1 · BEGINNER

A pawn captures an enemy piece by **taking it in a pincer**: it moves next to it, and on the other side, on the same line, there is **one of your pawns** or **an empty special square** (a corner, or the throne once the kings have left it). The captured piece is removed from the board.

- Only pawns capture. A king never captures and never helps to capture.
- Only the arriving pawn captures. A piece that moves between two enemies on its own is not captured.
- One move can capture several pieces, in several directions.
- The supporting piece is called the **anvil**; the arriving one, the **hammer**.

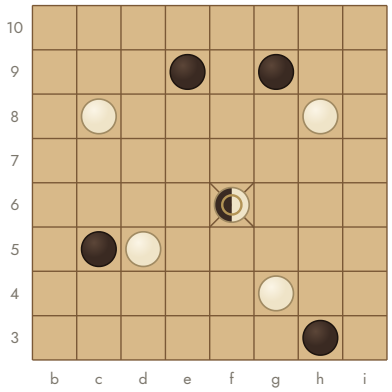


Black can capture d5 with e8-e5 (anvil c5), or b1 with f1-c1 (anvil: the empty corner a1). On h3, the black pawn moves between two white pawns and is not captured.

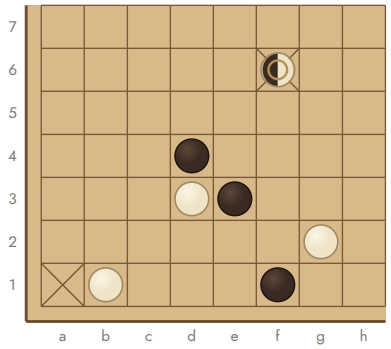
REMEMBER

A piece is in danger as soon as an enemy pawn can reach the empty square on its other side, with an anvil opposite. Look for these squares, for your pieces and for your opponent's.

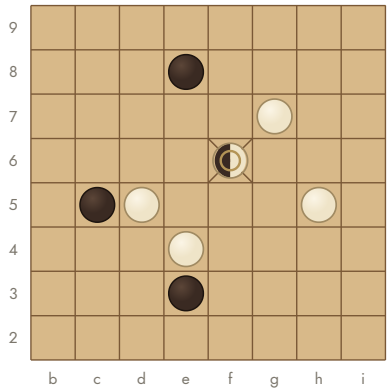
Exercises



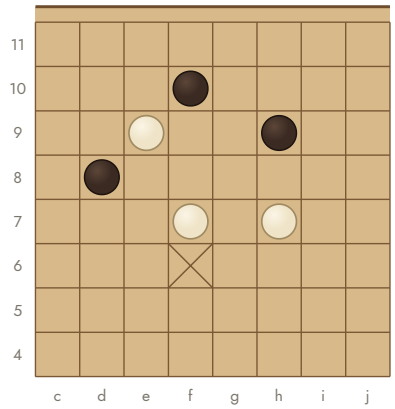
2.1 Black to play and capture.



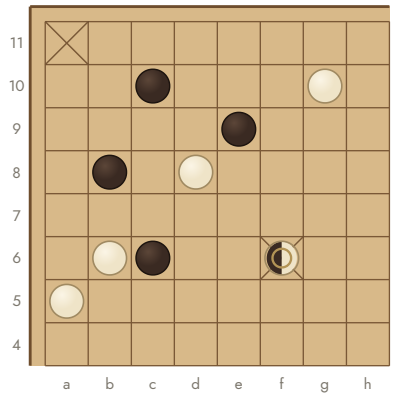
2.2 Black to play and capture.



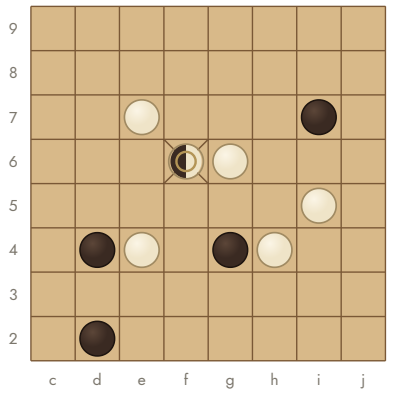
2.3 Black to play and capture.



2.4 Black to play and capture.



2.5 White to play and capture.

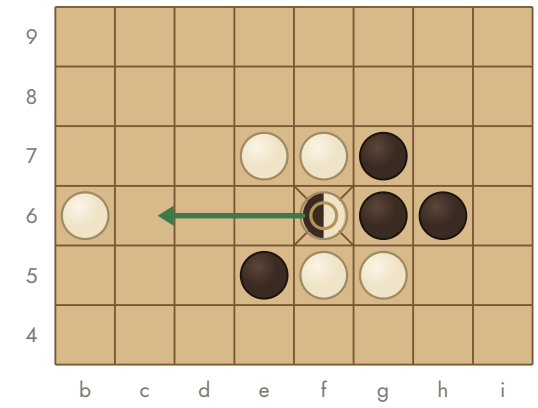


2.6 Black to play and capture.

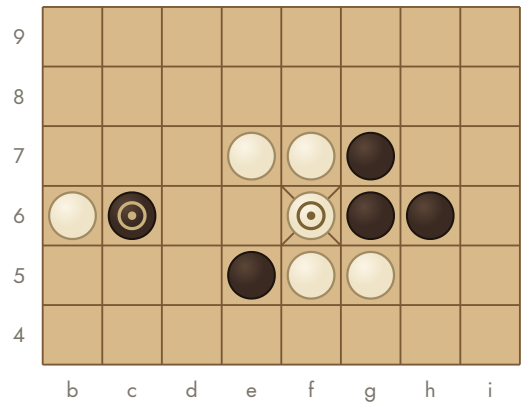
3 The king and the pair

LEVEL 1 · BEGINNER

As long as both kings are stacked on f6, neither king moves on its own: either player may move the **pair**. The first to do so **takes his own king** to the arrival square and **leaves the enemy king on the throne**. The pair itself cannot be captured.



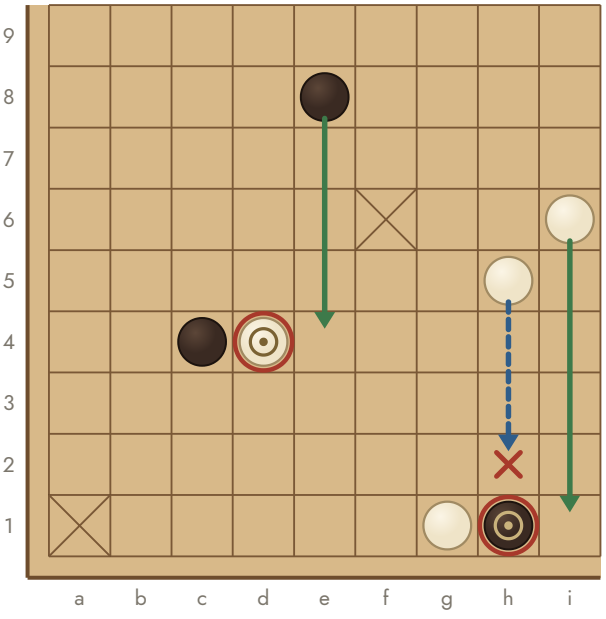
Black moves the pair from f6 to c6.



The black king is on c6; the white king stays on f6.

Capturing the king

- The king is captured like a pawn: between two enemy pawns, or between an enemy pawn and an empty special square.
- On the throne f6, the king **cannot be captured**.
- On an edge, it can only be captured **along the edge**: the edge itself is not an anvil.
- Right next to a corner, **a single pawn** is enough, since the empty corner is the anvil.

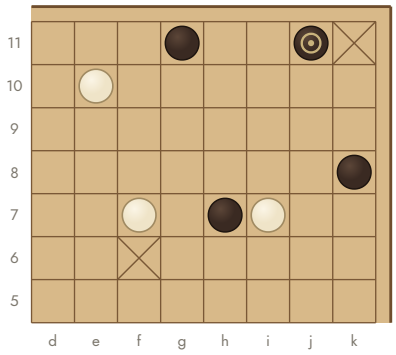


The white king on d4 and the black king on h1 can be captured. A white pawn arriving on h2 would not capture the black king: the edge is not an anvil.

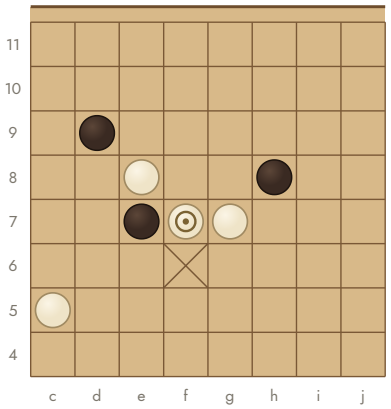
REMEMBER

Your king is safe on the throne, and fairly safe on an edge. It is most exposed in the middle of the board, where it can be captured in two directions, and right next to a corner.

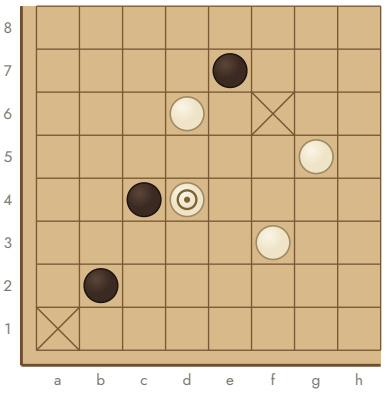
Exercises



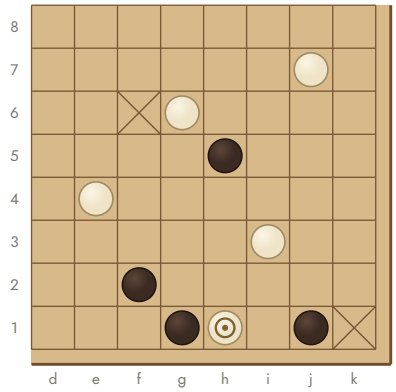
3.1 White to play and capture the king.



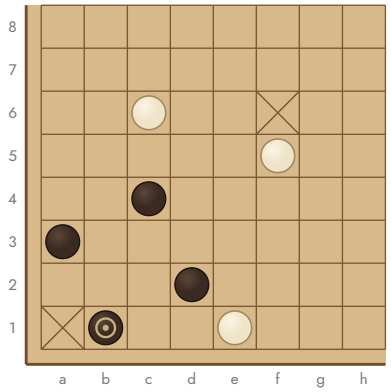
3.2 Black to play and capture the king.



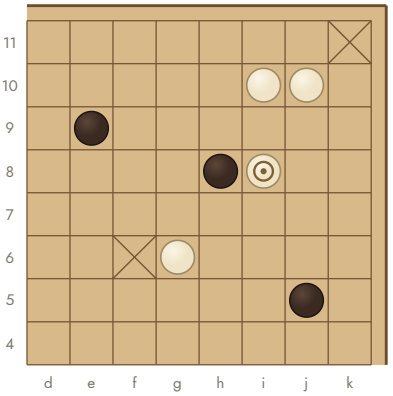
3.3 Black to play and capture the king.



3.4 Black to play and capture the king.



3.5 White to play and capture the king.



3.6 Black to play and capture the king.

4 Threats and winning

LEVEL 1 · BEGINNER

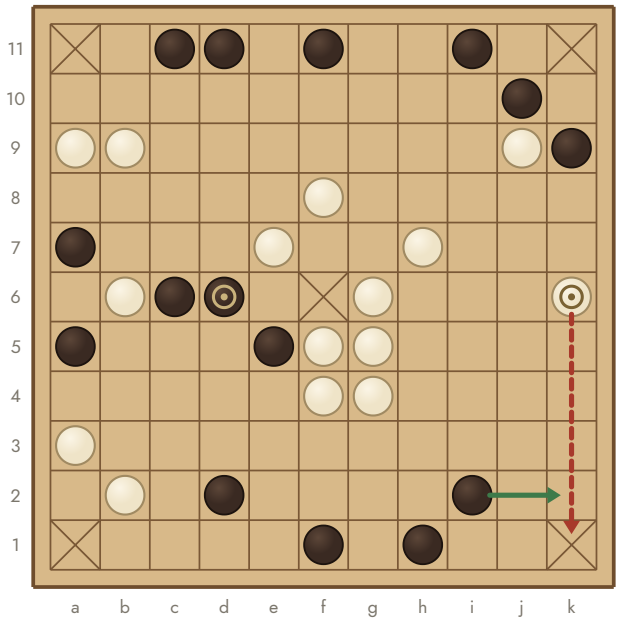
You win when your king reaches a **corner**, when you **capture the enemy king**, or when your opponent **has no allowed move**. A position repeated for the third time, with the same player to move, is a draw.

The threat rule

If your opponent could win on his next move, you **must stop the threat**. A move that leaves him an immediate win is not allowed. If no defence exists, you have lost: this is an **unstoppable win**. So games almost never end with a king entering a corner or being captured: they stop as soon as a threat can no longer be stopped.

The four defences

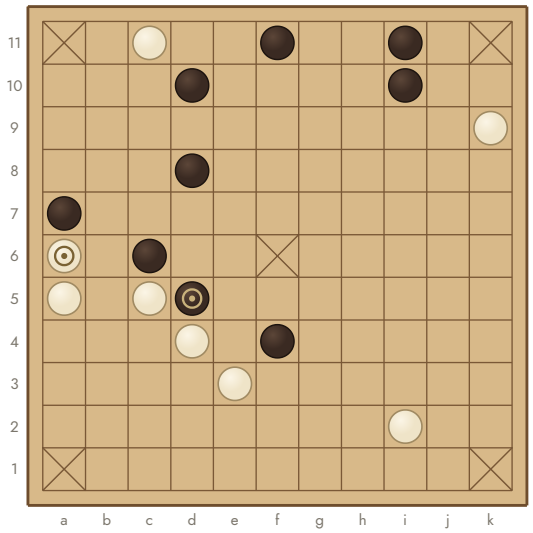
- **Move the threatened king** out of the pincer.
- **Close the line** to the corner by putting a piece on it.
- **Occupy the capture square**, so that the enemy pawn can no longer land there.
- **Capture** the threatening piece, or the one serving as anvil.



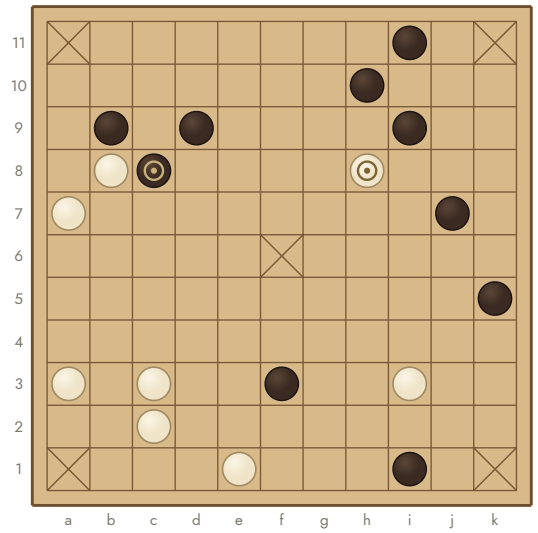
The white king threatens to reach k1. Black has a single allowed move: i2-k2, which closes the k-file.

> **REMEMBER**
>
> Winning means making a threat your opponent cannot stop. Every threat cuts down his allowed moves: it is the main tool of the game.

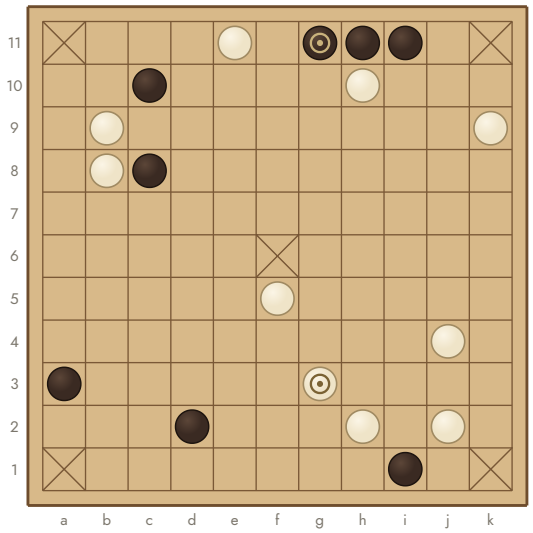
Exercises



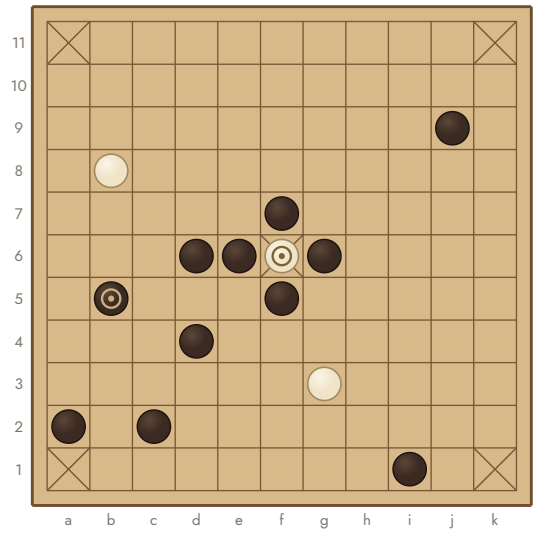
4.1 White is preparing an unstoppable threat. Black to play: find the defence.



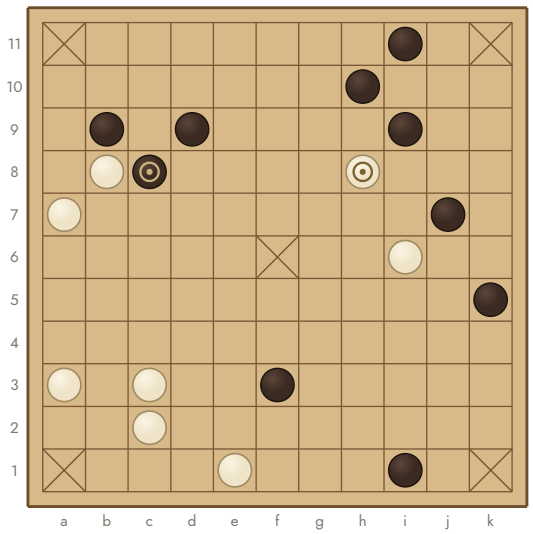
4.2 Black is preparing an unstoppable threat. White to play: find the defence.



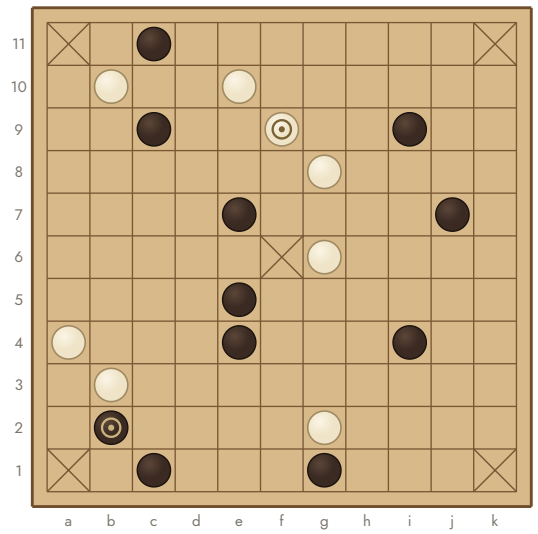
4.3 White is preparing an unstoppable threat. Black to play: find the defence.



4.4 Black to play and win.



4.5 Black to play and win.



4.6 Black to play and win.

II

LEVEL 2 · INTERMEDIATE

Tactics

Tactics are the moves that win something at once: a pawn, time, or the game. The positions in this part come from real games.

---

5 Winning material

6 Poisoned captures

7 Combinations

8 Defending and preventing

9 The unstoppable threat

10 Winning in two moves

5 Winning material

LEVEL 2 · INTERMEDIATE

Material is the most reliable asset in the game: every captured pawn brings you closer to victory, and the bigger the lead, the easier it is to cash in. To find a capture, look for enemy pieces that have **an empty square on one side and a possible anvil on the other**.

Easy targets

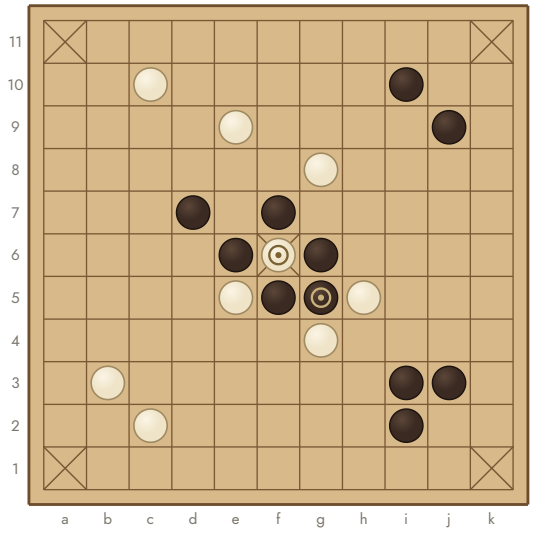
- A piece standing **next to an empty corner**: the corner is the anvil, a single pawn is enough.
- A piece standing **next to the empty throne**, once the kings have left.
- Two pieces lined up with **a gap between them** and anvils on both sides: a single pawn can capture both.
- A piece whose only friendly neighbour has just left, leaving an empty square opposite an enemy anvil.

The double attack

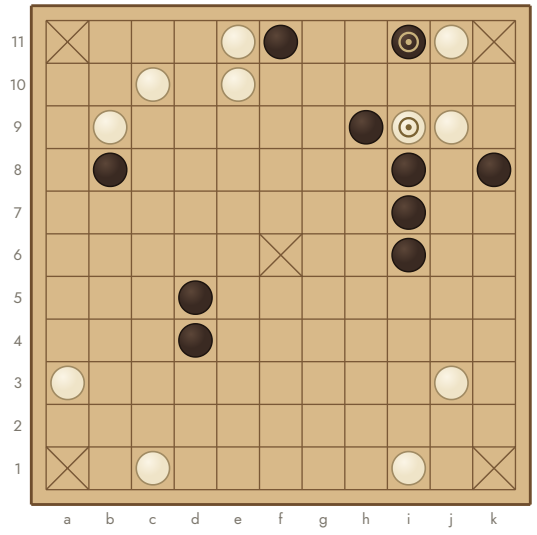
A move that threatens **two captures at once** often wins a pawn: your opponent can only stop one. The best-placed pawn is the one that can reach, next move, two different capture squares.

> **REMEMBER**
>
> Before each move, list the enemy pieces you can capture, and your own pieces that can be captured. A safe capture is almost always better than a waiting move.

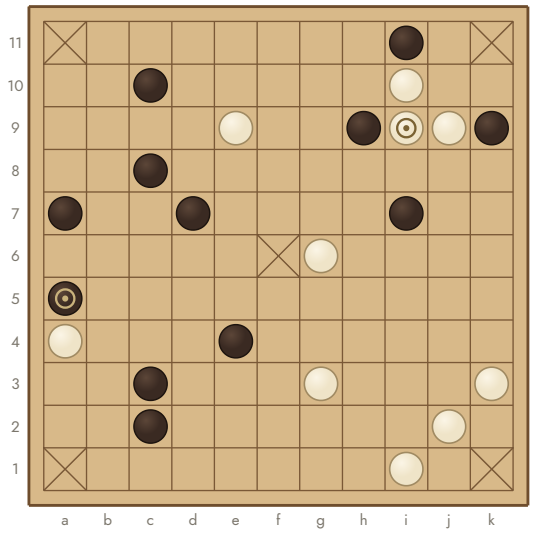
Exercises



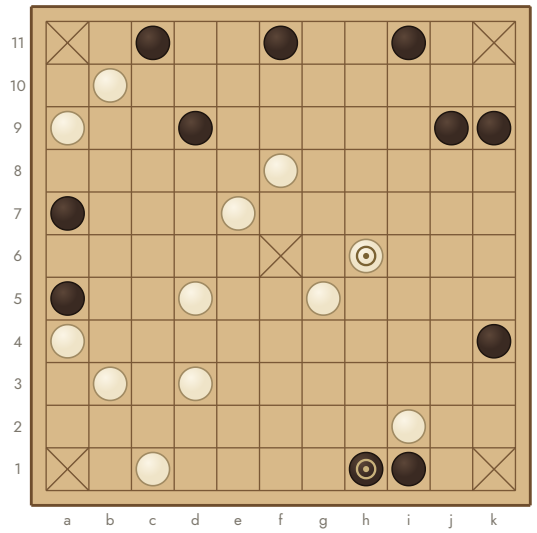
5.1 Black to play and capture a pawn.



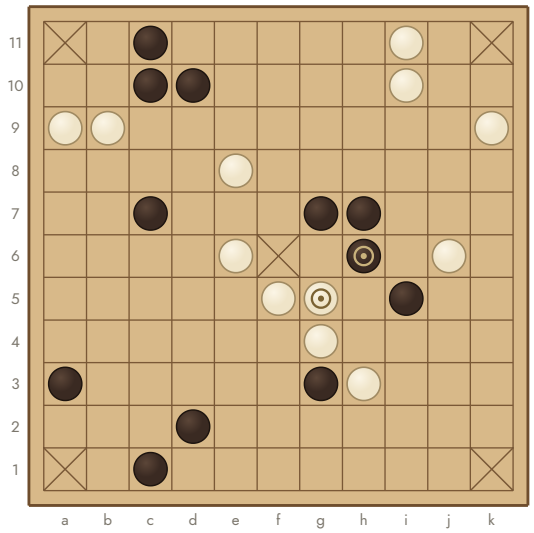
5.2 Black to play and capture a pawn.



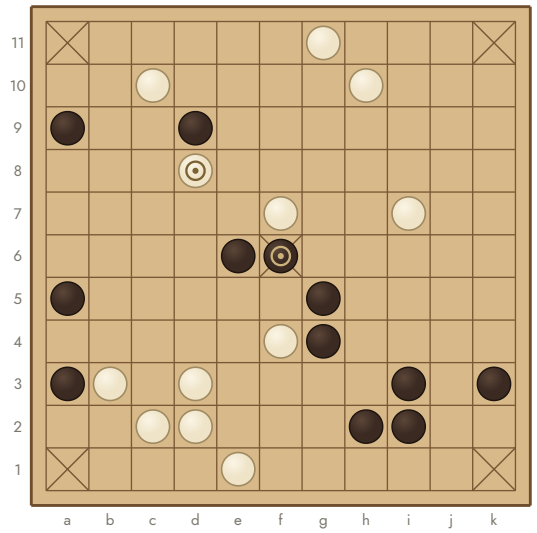
5.3 Black to play and capture a pawn.



5.4 Black to play and capture a pawn.



5.5 Black to play and capture a pawn.



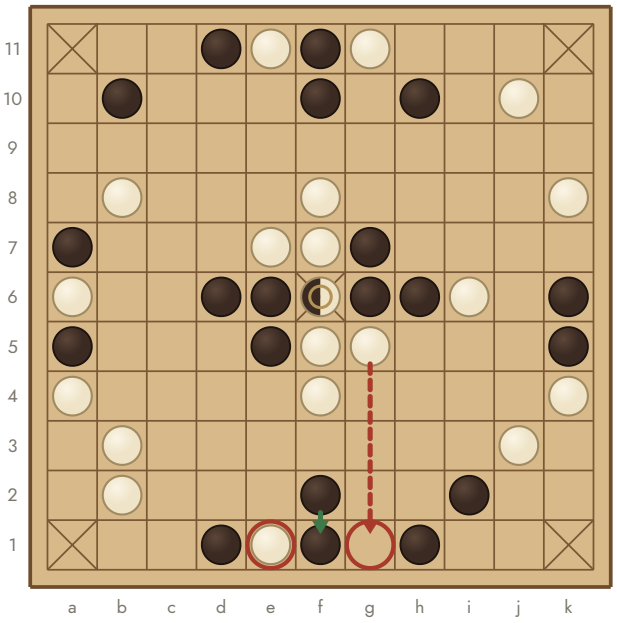
5.6 Black to play and capture a pawn.

6 Poisoned captures

LEVEL 2 · INTERMEDIATE

Not every capture is good. The capturing pawn lands on a new square, and it leaves the one it was guarding. Before capturing, ask yourself three questions.

- **Will your pawn be captured in turn?** Look at both sides of the arrival square: an enemy anvil on one side and an empty square on the other are enough.
- **What was your pawn protecting?** By leaving, it may open a line to a corner, or remove the anvil that protected another pawn.
- **Is the reply stronger?** A recapture that takes two pieces at once turns the gain into a loss.

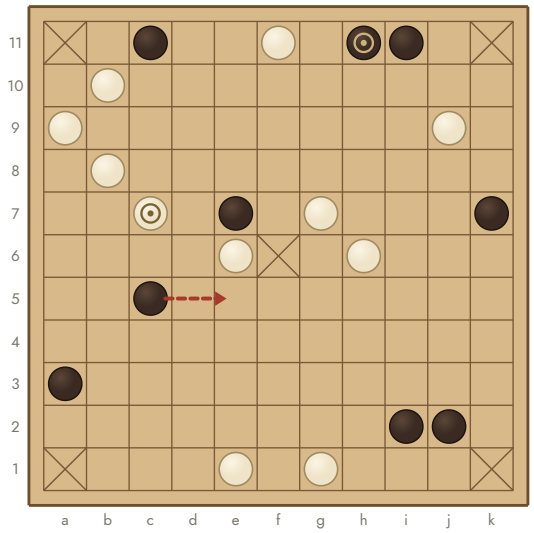


White can capture f1 with g5-g1, but Black answers f2-f1 and captures both e1 and g1 in one move.

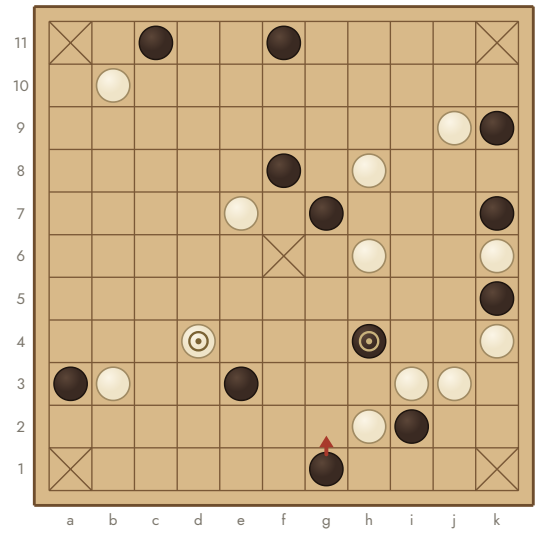
REMEMBER

Calculate a capture to the end: count the pieces after your opponent's reply, not before.

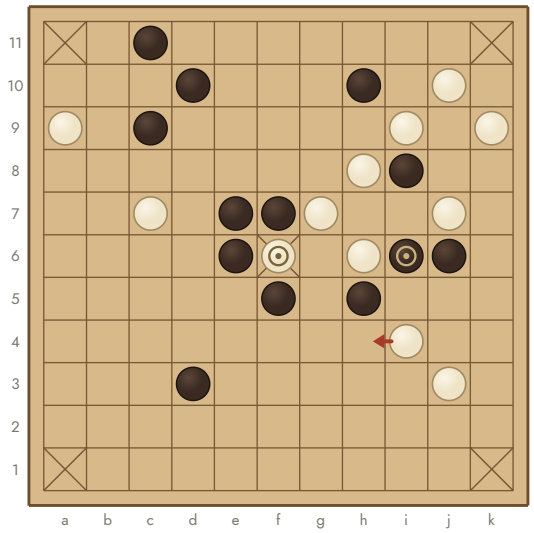
Exercises



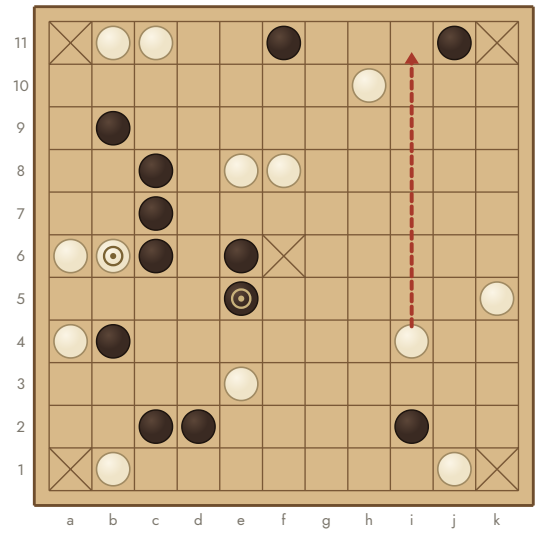
6.1 Black can capture with c5-e5. Is it a good idea?



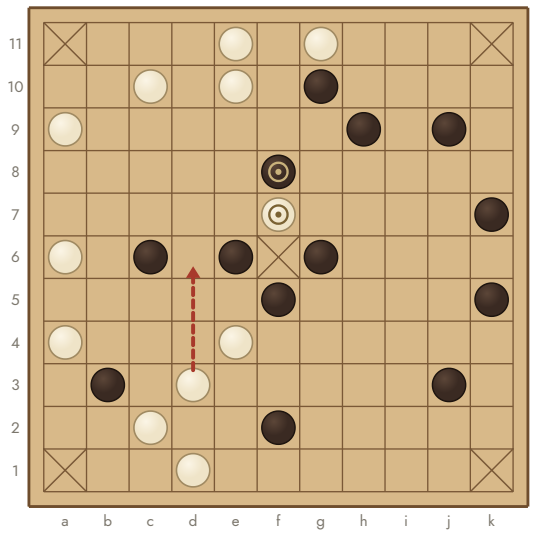
6.2 Black can capture with g1-g2. Is it a good idea?



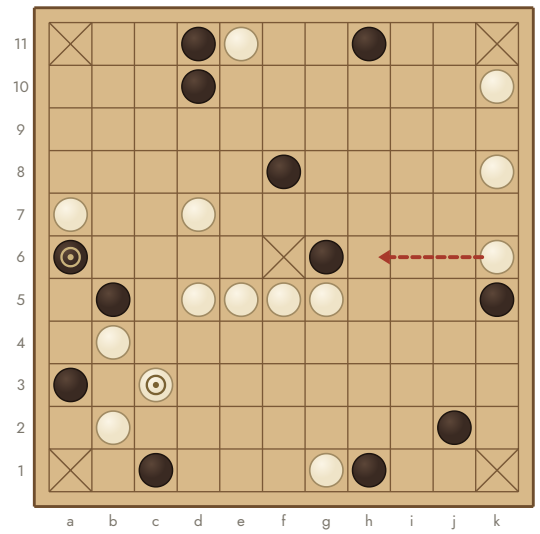
6.3 White can capture with i4-h4. Is it a good idea?



6.4 White can capture with i4-i11. Is it a good idea?



6.5 White can capture with d3-d6. Is it a good idea?



6.6 White can capture with k6-h6. Is it a good idea?

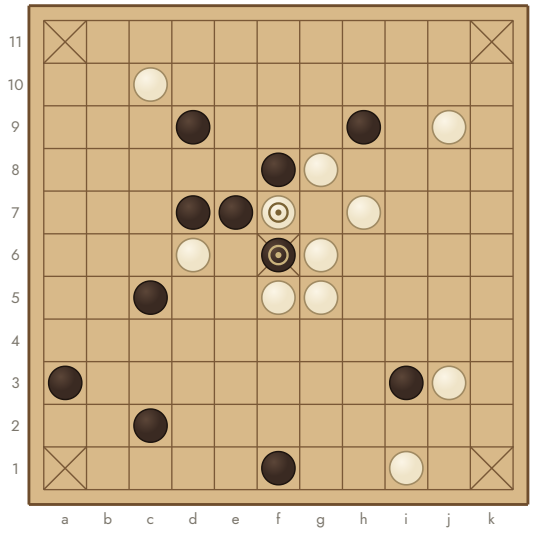
7 Combinations

When no capture is available at once, you can **prepare** one. A combination is a short sequence of moves that forces your opponent to give up material.

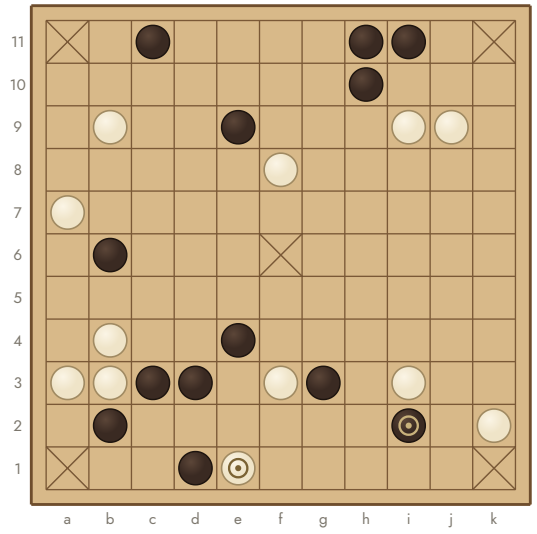
- **Place the anvil first**, then bring the hammer: your opponent cannot always save the target.
- **Threaten two things at once**: a capture and a line to a corner, or two captures.
- **Force the reply**: a winning threat forces your opponent to defend, and gives you the time for your next move.
- **Deflect a defender**: force an enemy piece to leave the square it was guarding.

> **REMEMBER**
>
> The best combinations start with a forcing move. When your opponent has only a few allowed moves, you can calculate far ahead.

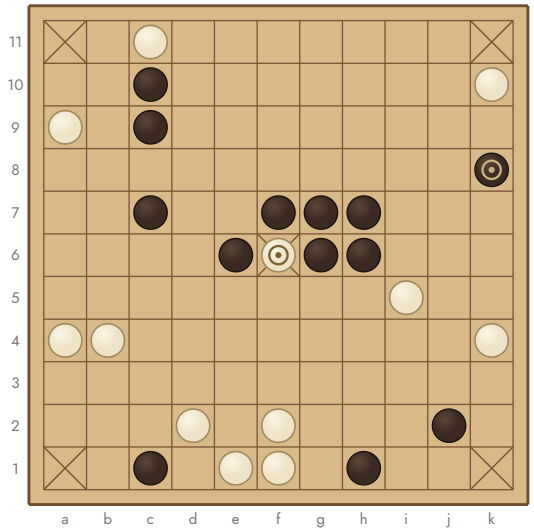
Exercises



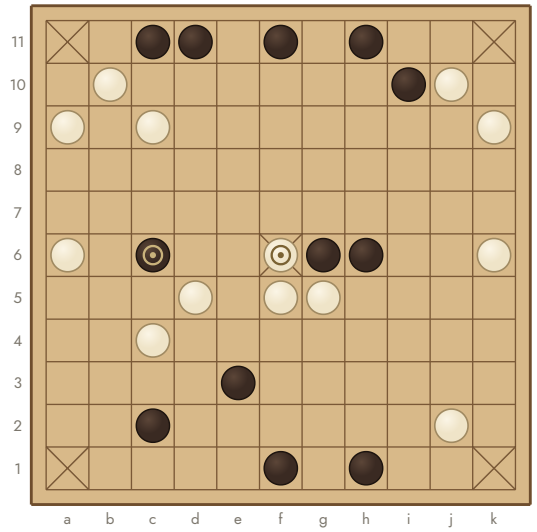
7.1 Black to play and win a pawn.



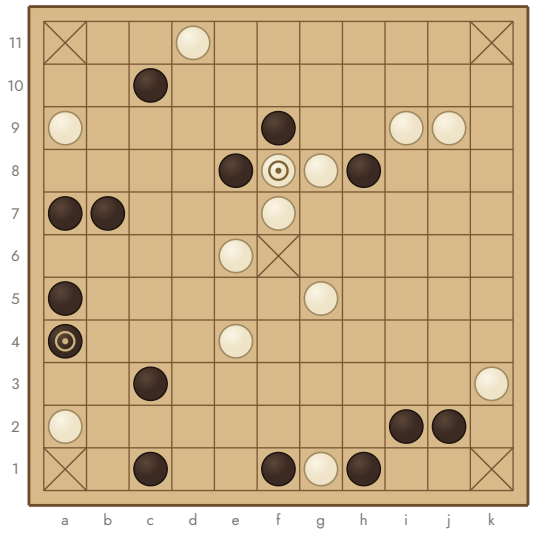
7.2 Black to play and win a pawn.



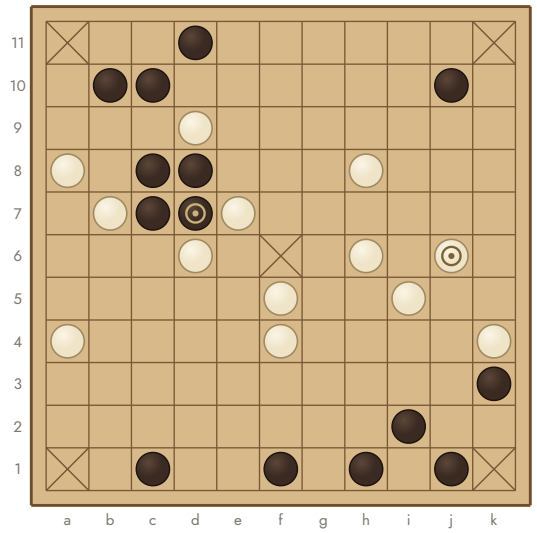
7.3 Black to play and win a pawn.



7.4 White to play and win a pawn.



7.5 Black to play and win a pawn.



7.6 White to play and win a pawn.

8 Defending and preventing

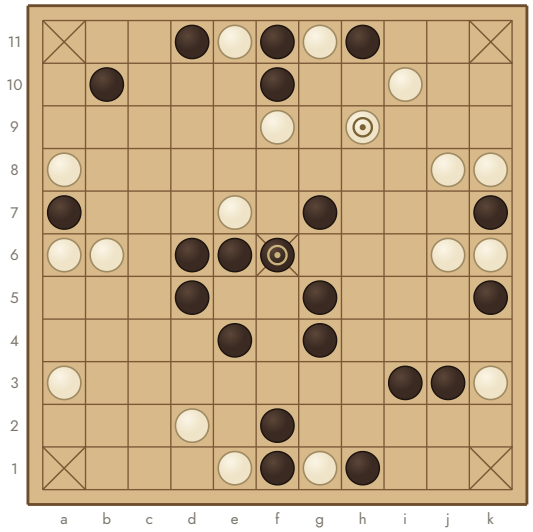
Good defence means seeing the threat **before** it becomes unstoppable. At every move, ask yourself: "If you passed, what would your opponent play?" If he has a move that creates an unstoppable threat, your move must prevent it.

Ways to prevent

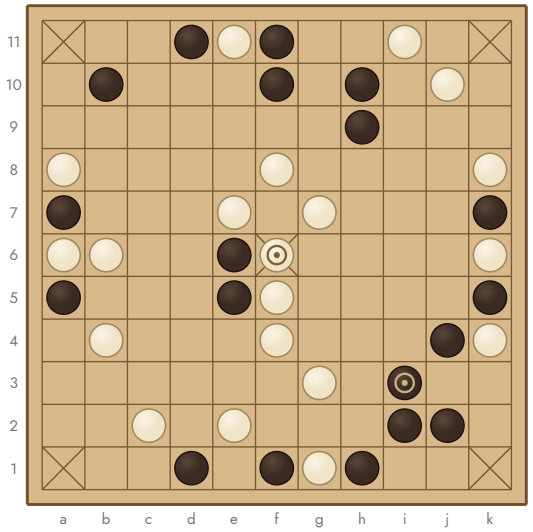
- Put a piece on the line the enemy king wants to use, or on its arrival square.
- Prepare a piece able to reach that line next move: the threat will no longer be unstoppable.
- Move your king away from a pincer being prepared.
- Capture the key piece of the attack.

> **REMEMBER**
>
> A threat you can stop is not deadly. The danger is the threat none of your pieces will be able to stop: count your defenders before it is too late.

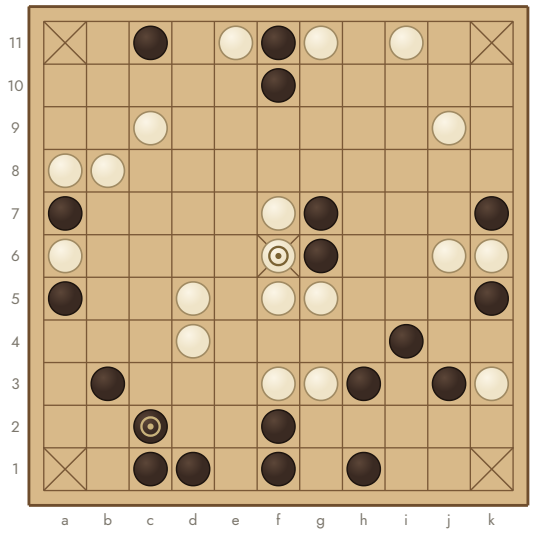
Exercises



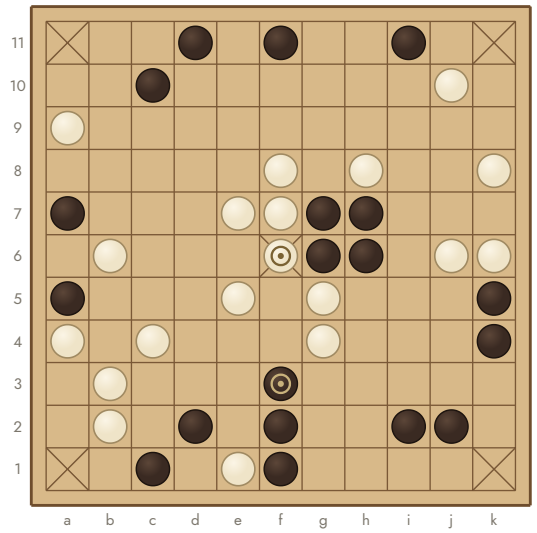
8.1 White is preparing an unstoppable threat. Black to play: find the defence.



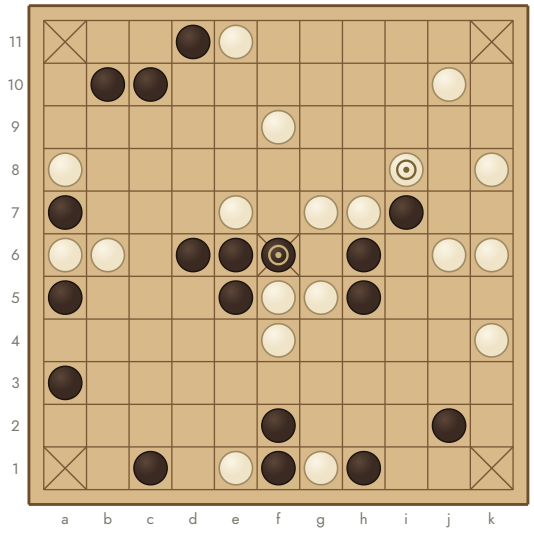
8.2 Black is preparing an unstoppable threat. White to play: find the defence.



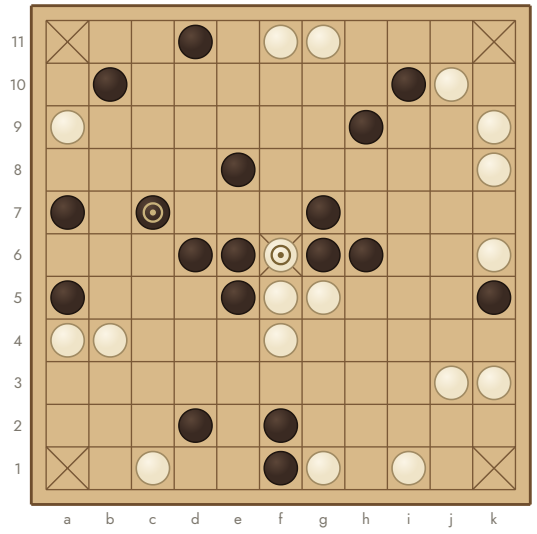
8.3 Black is preparing an unstoppable threat. White to play: find the defence.



8.4 Black is preparing an unstoppable threat. White to play: find the defence.



8.5 White is preparing an unstoppable threat. Black to play: find the defence.

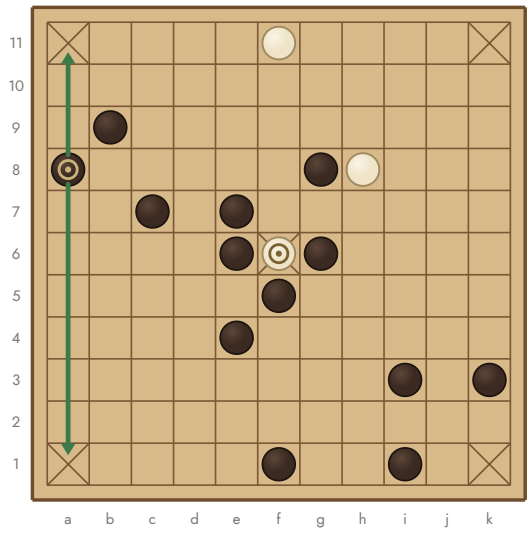


8.6 Black is preparing an unstoppable threat. White to play: find the defence.

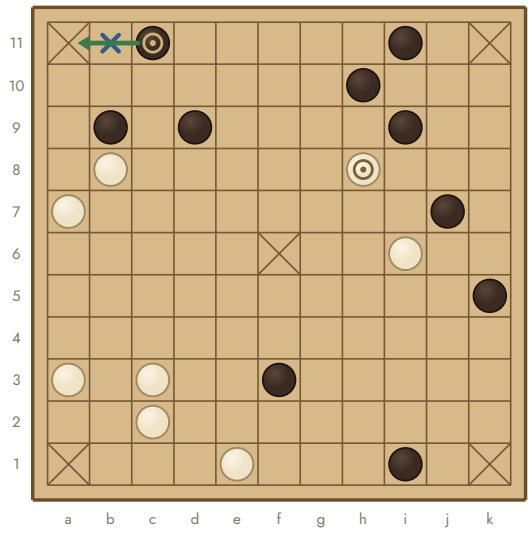
9 The unstoppable threat

A threat is unstoppable when none of the four defences works. Three shapes keep coming back.

- **The open line**: the king threatens a corner, no enemy piece can reach the line leading to it, and the king cannot be captured.
- **The corner fork**: the king threatens two corners at once; your opponent can close only one line.
- **The double pincer**: two different moves capture the enemy king, and a single defence is not enough.



Fork: the black king on a8 threatens a11 and a1.



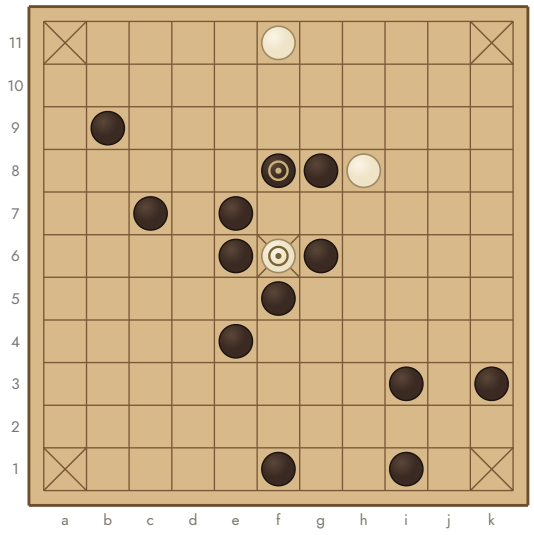
Open line: the black king threatens a11 through b11, a square no white piece can reach.

The method

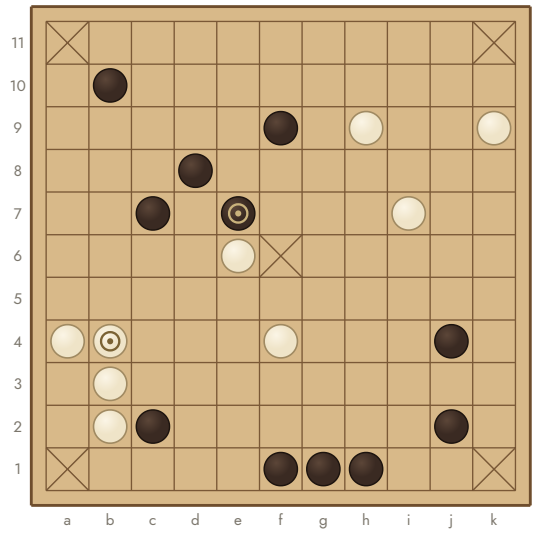
For each line between a king and a corner, list the empty squares and, for each one, the enemy pieces able to reach it in one move. Also check the squares from which an enemy pawn would capture the king. If nothing can intervene, the threat is unstoppable.

> **REMEMBER**
>
> The winning king often stays inside the board, then slips onto the edge at the last moment, on a line nobody can close any more.

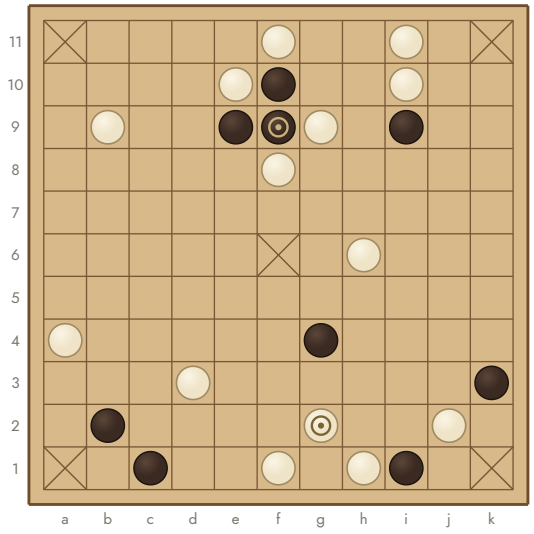
Exercises



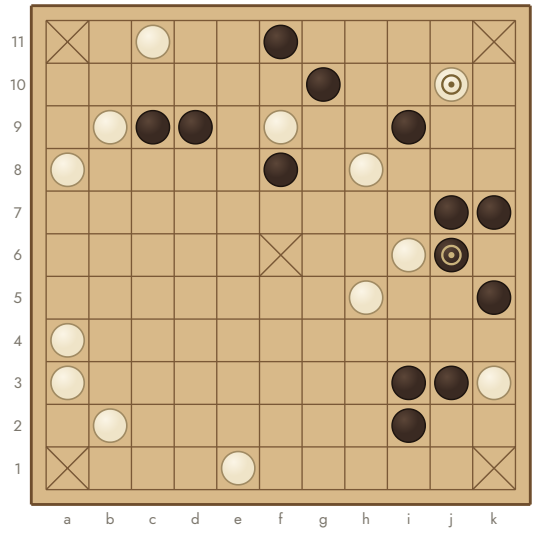
9.1 Black to play and win.



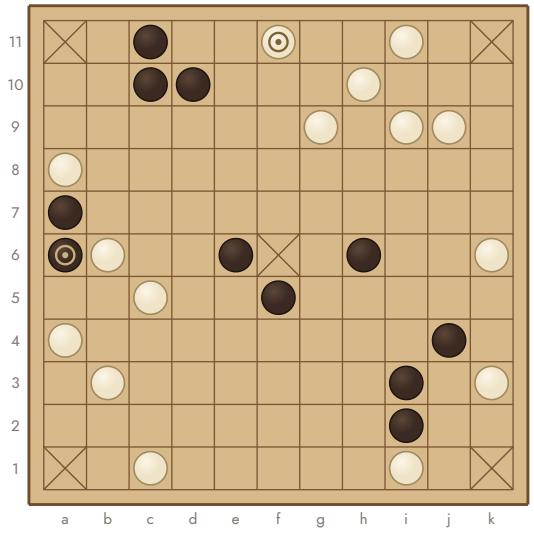
9.2 Black to play and win.



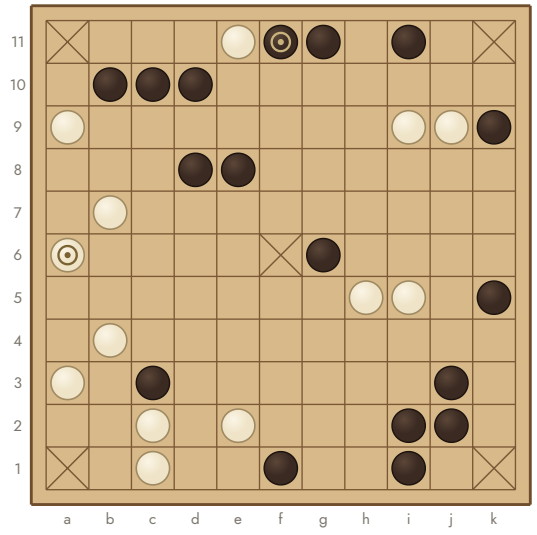
9.3 White to play and win.



9.4 Black to play and win.



9.5 White to play and win.



9.6 White to play and win.

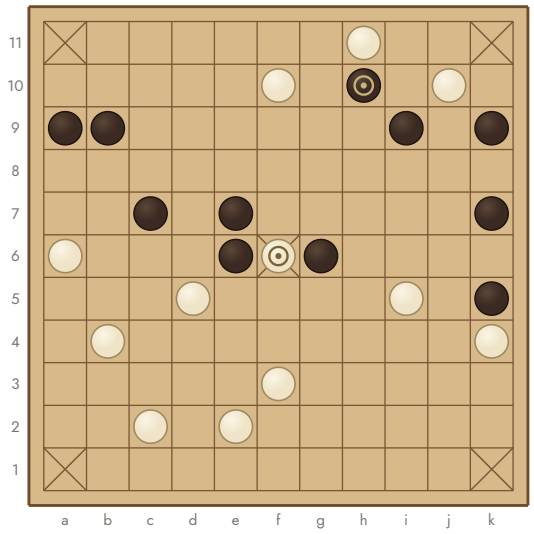
10 Winning in two moves

A win in two moves often begins with a **quiet** move: it threatens nothing visible, but it prepares an unstoppable threat your opponent can no longer prevent. At other times, the first move is a threat that forces a precise defence, after which the second threat becomes unstoppable.

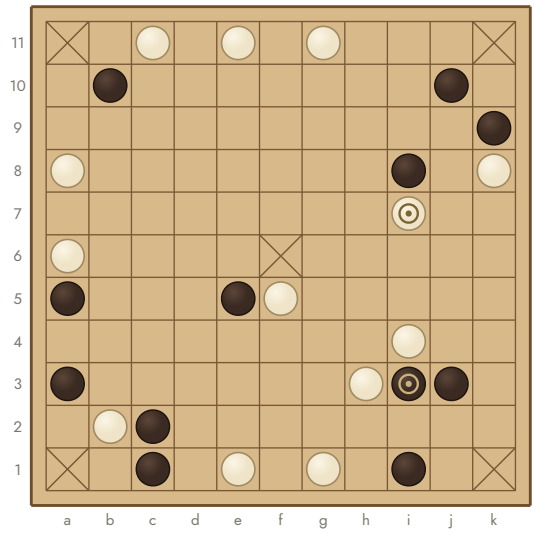
- First look at moves that threaten: they cut your opponent's replies down to a handful.
- Then look at moves that take an enemy piece out of the defence, or that open a second road for your king.
- Check every reply: a single defence is enough to make the combination fail.

> **REMEMBER**
>
> Thanks to the threat rule, your opponent has few allowed moves after a threat: this is what makes combinations possible to calculate.

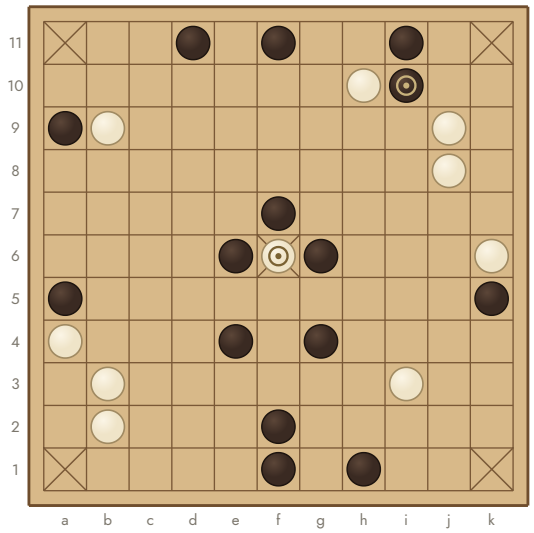
Exercises



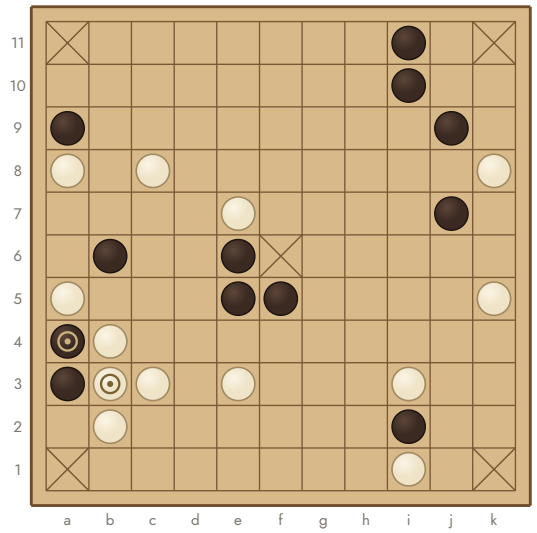
10.1 White to play and win in two moves.



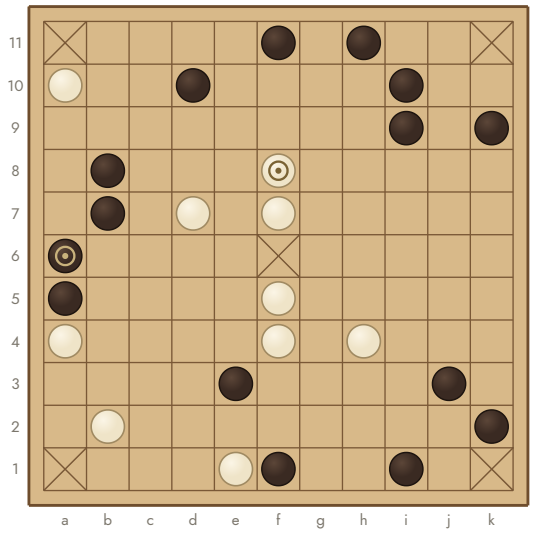
10.2 Black to play and win in two moves.



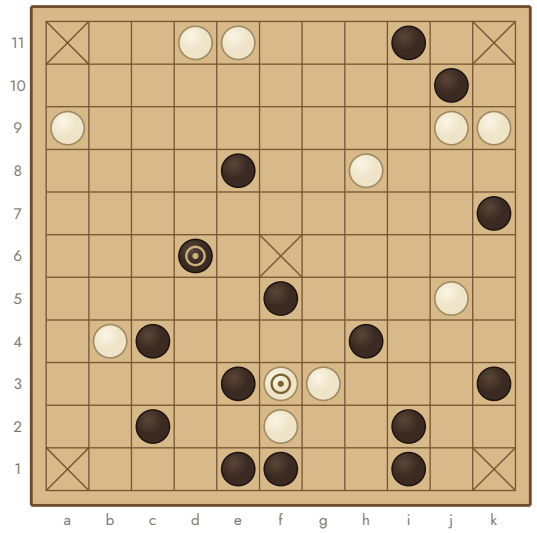
10.3 Black to play and win in two moves.



10.4 White to play and win in two moves.



10.5 Black to play and win in two moves.



10.6 Black to play and win in two moves.

III

LEVEL 3 · ADVANCED

The opening

The NORMAN TAFL opening follows a precise logic: the battle for the ends of the edge chains. This part gives its principles, the main lines explored in depth, and the traps you need to know.

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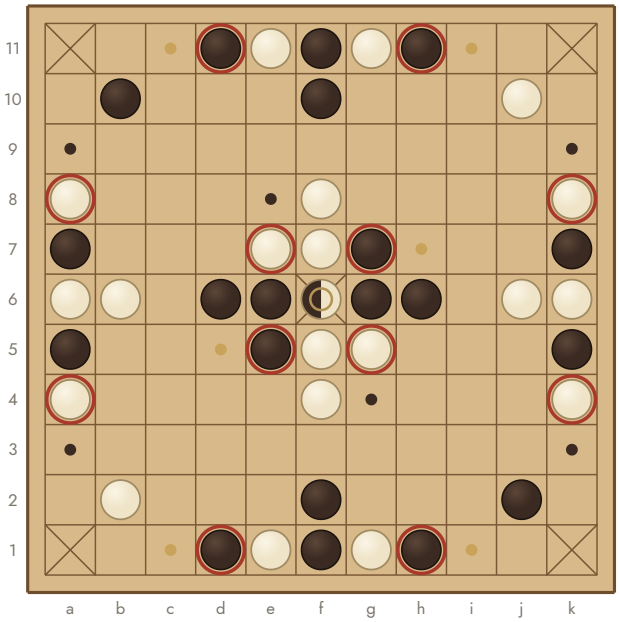
11 Opening principles

12 The main lines

13 Opening traps

11 Opening principles

The starting position hides an important feature: on each edge, the pawns of both sides **alternate**. On the a-file, for example, there is a white pawn on a4, a black one on a5, a white one on a6, a black one on a7 and a white one on a8. A pawn arriving at the end of one of these chains captures the last enemy pawn.

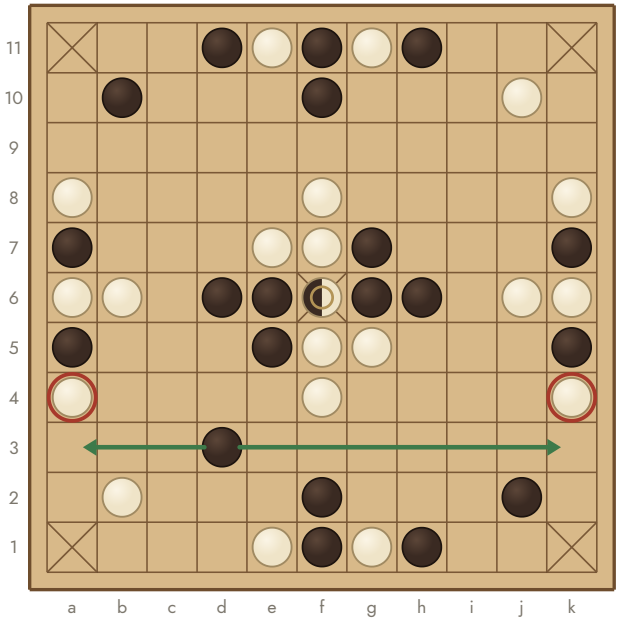


The capture squares at the start. Black aims at a3, a9, k3, k9, g4 and e8 (dark dots); White aims at c1, i1, c11, i11, d5 and h7 (light dots).

Black attacks the edge files, White attacks the edge ranks. None of these squares can be reached on the first move: a pawn must first be brought onto the right line. That is what the first moves are about.

The opening double threat

The move **d1-d3** puts a black pawn on the third rank, from where it threatens both **d3-a3**, capturing a4, and **d3-k3**, capturing k4. White cannot stop both threats: he must prepare a recapture. A sound continuation is to accept an **exchange** on the edge.

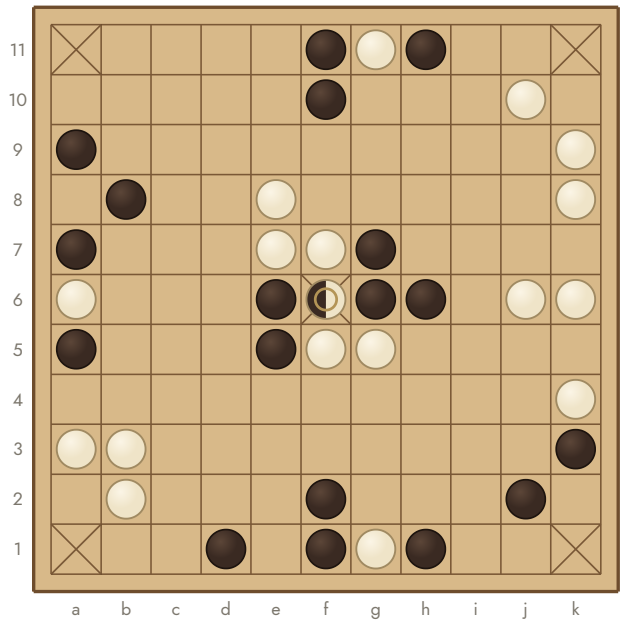


After d1-d3, Black threatens a3 and k3.

A model line

Here is the beginning of a high-level game. It shows the typical rhythm of the opening: each side captures on an edge, the other recaptures at once, and the exchanges follow one another until one side comes out of the series a pawn up.

1.	**d1-d3**	**b6-b3**	White closes a3, but not k3.
2.	**d3-k3** takes k4	**f4-k4** takes k5	The exchange is immediate.
3.	**d11-d9**	**k8-k9**	Same idea at the top: Black threatens a9 and k9.
4.	**d9-a9** takes a8	**a4-a3**	
5.	**b10-b8**	**e11-e8**	
6.	**d6-d1** takes e1	**f8-k8** takes k7	Black comes out of the series a pawn up.



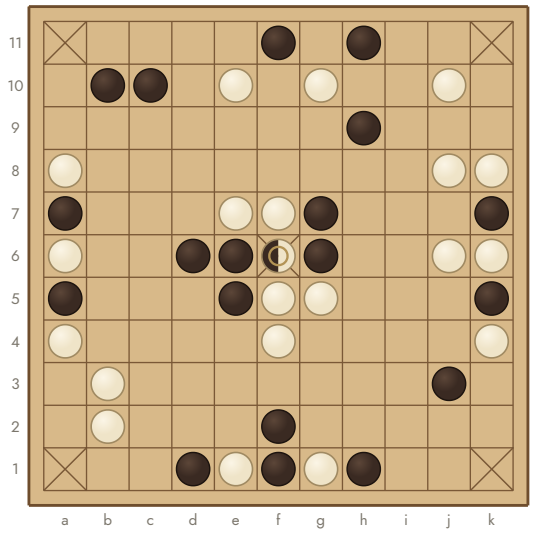
The position after White's sixth move. The edges have thinned out; the central cross is intact.

Opening principles

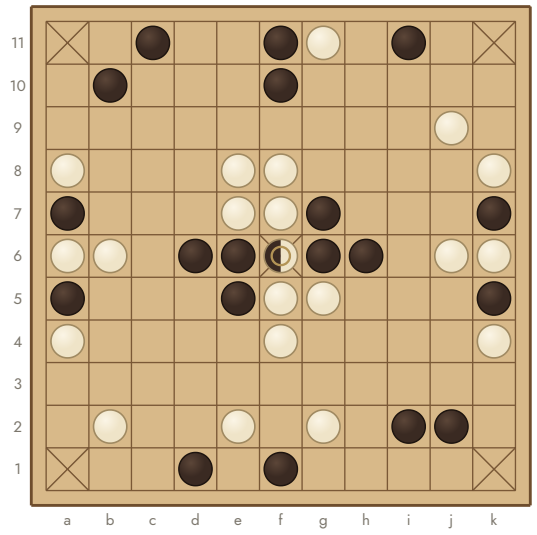
- Bring a pawn onto a line joining two capture squares: the third or ninth rank for Black (a3 and k3, a9 and k9), the c- or i-file for White (c1 and c11, i1 and i11). From there it threatens two captures at once.
- Before each move, check your opponent's capture squares: a pawn leaving a chain may open its end.
- In an exchange, count who captures last. A series of exchanges is won or lost by one pawn.
- Do not touch the central cross (e6, g6 for Black; f5, f7 for White) without a reason: it holds the pair and protects your neighbouring pawns.
- Do not put a pawn next to an empty corner: a single enemy pawn is enough to capture it.

> **REMEMBER**
>
> The opening is a battle for the ends of the chains. Create double threats, prepare your recaptures, and have the last word in the exchanges.

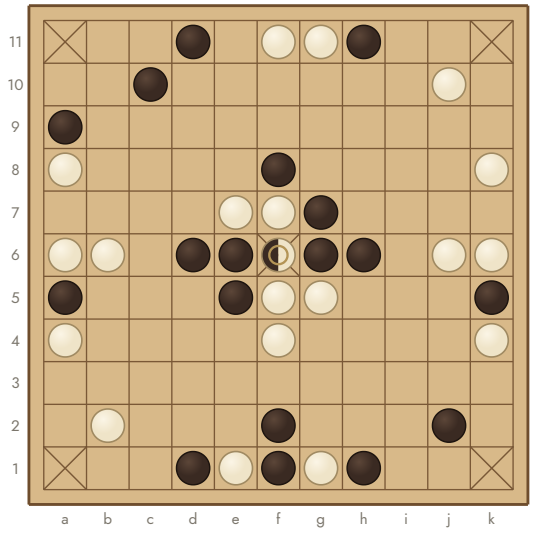
Exercises



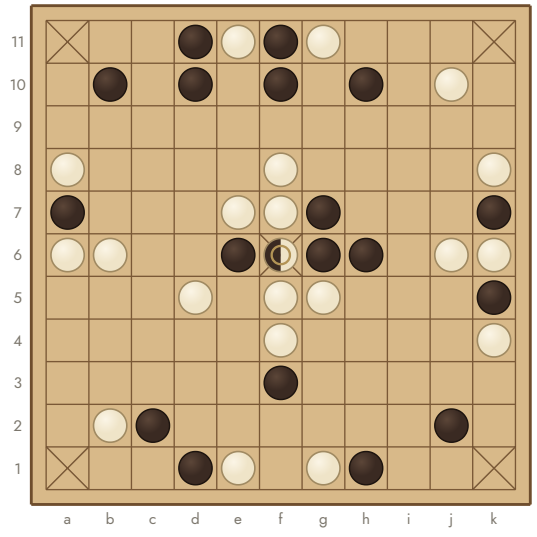
11.1 Black to play and win a pawn.



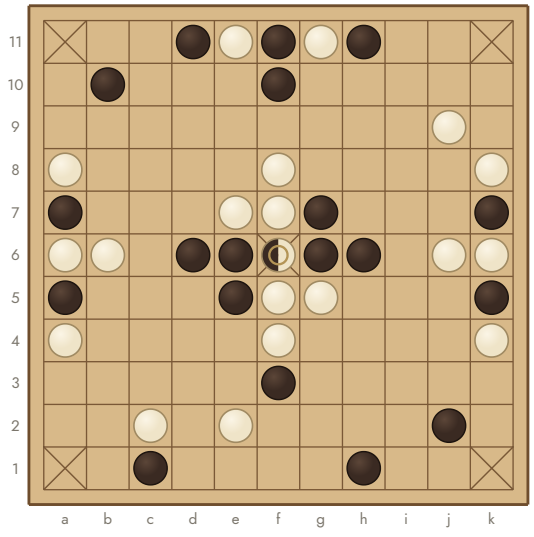
11.2 Black to play and capture a pawn.



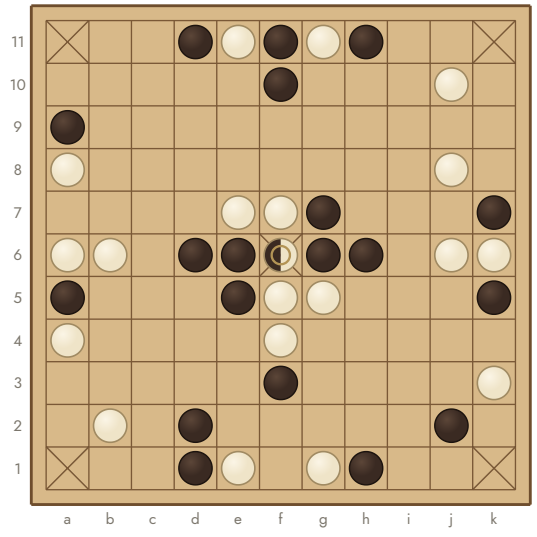
11.3 Black to play and capture a pawn.



11.4 Black to play and capture a pawn.



11.5 Black to play and capture a pawn.



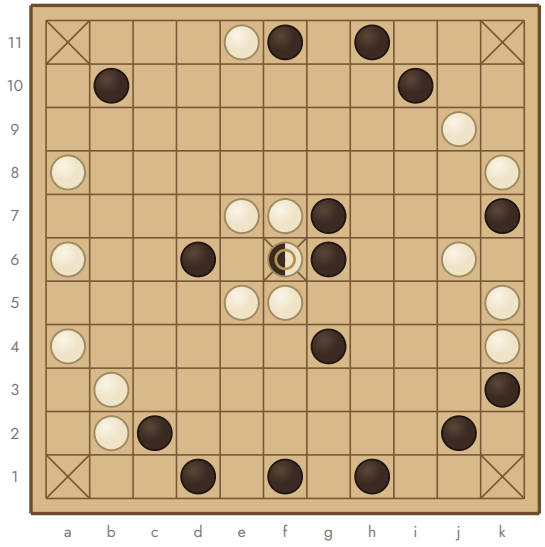
11.6 Black to play and capture a pawn.

12 The main lines

First system: 1. d1-d3, the double threat

The most direct first move. The pawn on d3 threatens both a3 and k3; White closes one side and accepts the exchange on the other. A long series of captures and recaptures on all four edges follows: each side brings its inner pawns back to the chain ends that the exchanges have freed.

1.	**d1-d3**	**b6-b3**	White closes a3. Closing k3 with j6-j3 amounts to the same.
2.	**d3-k3** takes k4	**f4-k4** takes k5	Exchange on the k-file.
3.	**d6-d1** takes e1	**j10-j9**	The d6 pawn drops to the edge along the d-file, freed by the departure of the d1 pawn, and takes e1.
4.	**h1-h4**	**a8-d8**	Both sides aim at the central cross.
5.	**h4-g4** takes g5	**d8-d5** takes e5	Capture and recapture in the centre.
6.	**f10-i10**	**a6-b6**	
7.	**d11-d6**	**d5-e5** takes e6	
8.	**h6-h1** takes g1	**b6-a6** takes a5	Black takes on the first rank, White on the a-file.
9.	**h11-h5**	**k6-k5**	
10.	**h5-h11** takes g11	**f8-a8** takes a7	The h5 pawn returns to the edge and takes g11.
11.	**f2-c2**		



The position after 11. f2-c2.

Material is equal and the pair has not moved. The edges are very thin: the middlegame will turn on the remaining chains and on when the kings come out.

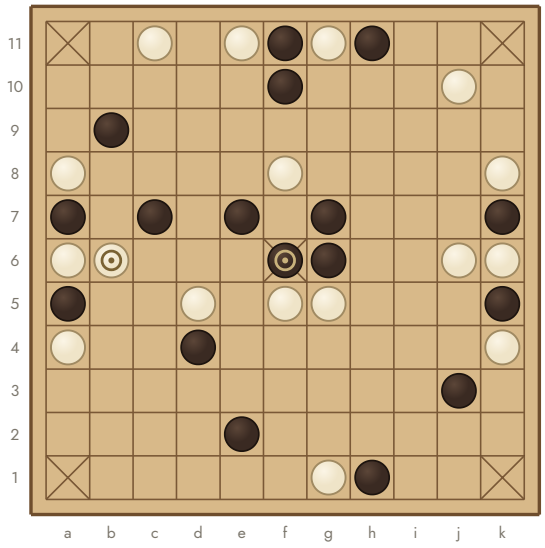
Variations

- **1... j6-j3**: White closes k3 instead of a3; Black then takes on the other side, on the a-file. 2. d11-d9 j10-j9 3. k5-j5 f8-i8 4. d3-a3 takes a4 i8-i1 takes h1 5. d9-a9 takes a8 k8-a8 takes a7. Balanced position.
- **2. h11-h9**: Black delays the capture to prepare a second double threat at the top of the board. 2... k8-i8 3. f2-i2 a8-c8 4. d11-c11 k4-k3 5. k7-k9 b2-d2 6. k9-k7 takes k6. Balanced position.
- **2... k8-i8**: White does not recapture at once and prepares a capture on the i-file. 3. f10-i10 a6-b6 4. k7-k10 takes j10 f4-k4 takes k5 5. a5-a6 i8-i1 takes h1 6. d6-c6 takes b6 a8-d8. Balanced position.
- **3. d11-d9**: Black creates the same double threat on the ninth rank instead of taking on e1. 3... a4-i4 4. d9-a9 takes a8 i4-i1 takes h1 5. b10-b4 k8-k9 6. b4-g4 takes g5 b2-c2 7. g6-g5 takes f5. Black is clearly better.

Second system: 1. d1-c1, occupying the capture square

Black puts his pawn on c1, the square from which White would take d1. The pawn will come back to d1 to take e1. The exchanges open the d-file and the centre very quickly, and White uses this to bring a king out early.

1.	**d1-c1**	**b2-d2**	
2.	**c1-d1** takes e1	**d2-d5** takes e5	The pawn comes back and takes at the end of the chain; White recaptures in the centre.
3.	**d1-d4** takes d5	**f5-e5** takes e6	Another exchange in the centre.
4.	**f1-c1**	**f6-e6**	White brings a king out: the open centre already gives it roads.
5.	**b10-e10**	**e5-d5**	
6.	**j2-j3**	**e7-d7** takes d6	
7.	**h6-h9**	**b6-b11**	
8.	**e10-e7** takes f7	**b11-c11** takes d11	Each side takes another pawn.
9.	**c1-c7** takes d7	**f4-f5**	
10.	**f2-e2**	**e6-b6**	The white king heads west.
11.	**h9-b9**		



The position after 11. h9-b9.

Material is equal, but the white king is out and active: White has slightly better chances. Against this system, bringing the king out early is a weapon: the king does not stop in the centre, it heads west at once along the road the exchanges have opened.

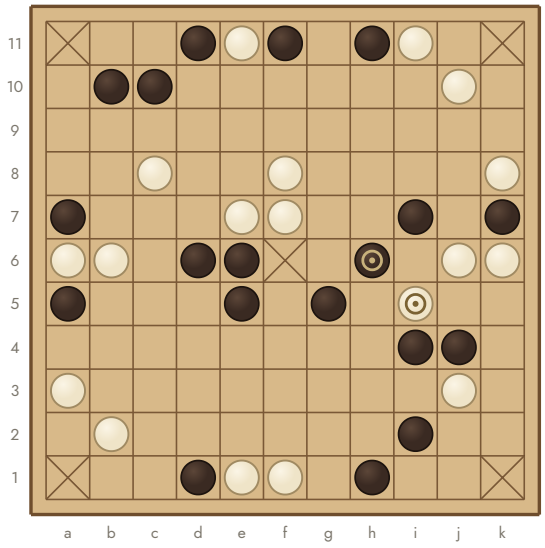
Variations

- **1... a8-a9**: White occupies a9, one of Black's capture squares, instead of answering on d2. 2. h1-h3 j6-j3 3. h3-h1 takes g1 f8-a8 takes a7 4. k7-j7 a4-c4 5. d11-d8 g5-g1 takes f1. Balanced position.
- **2. h1-h3**: Black delays the return to d1 by one move. 2... d2-d5 takes e5 3. c1-d1 takes e1 a4-d4 4. f10-c10 f5-e5 takes e6 5. f11-f10 e7-d7 takes d6 6. f10-f11 takes g11, e11. Balanced position.
- **2... a4-d4**: White brings a pawn to d4, from where he will take e5. 3. h1-h3 d4-d5 takes e5 4. h3-d3 takes d2 d5-e5 takes e6 5. h6-h1 takes g1 f6-e6 6. a7-d7 f4-a4 takes a5. Balanced position.
- **3. h11-h9**: Black launches a capture on the k-file and leaves the centre to White. 3... d5-e5 takes e6 4. h9-k9 takes k8 f6-e6 5. k9-a9 takes a8 e5-d5 6. j2-j3 e7-d7 takes d6 7. b10-c10. Balanced position.

Third system: 1. f10-c10, the reply on the i-file

Black plays a quiet move. White at once takes the initiative with k4-i4, his own double threat: i4-i1 takes h1 and i4-i11 takes h11. Black closes i1 and recaptures at the top. What follows shows the white king coming out at the wrong moment.

1.	**f10-c10**	**k4-i4**	White's double threat on the i-file.
2.	**f2-i2**	**i4-i11** takes h11	Black closes i1, White takes h11.
3.	**h6-h11** takes g11	**a4-a3**	Black recaptures at once.
4.	**f1-f2**	**g1-f1**	
5.	**k5-k4**	**f4-f3** takes f2	White takes f2; Black answers with two captures in the centre.
6.	**k4-g4** takes g5	**a8-c8**	
7.	**g6-g5** takes f5	**f6-i6**	White brings his king out to the east, where the black pawns are waiting.
8.	**g7-i7**	**i6-i4**	The best retreat: i6-h6 would lose in three moves (exercise 12.2).
9.	**j2-j4**	**i4-i5**	
10.	**f6-h6**	**f3-j3**	The black king comes out in turn to cut the white king's roads.
11.	**g4-i4**		



The position after 11. g4-i4.

The white king is surrounded by black pawns on the east wing: Black is clearly better. The lesson is clear: a king that goes out on the side where the opponent is strongest gets hunted.

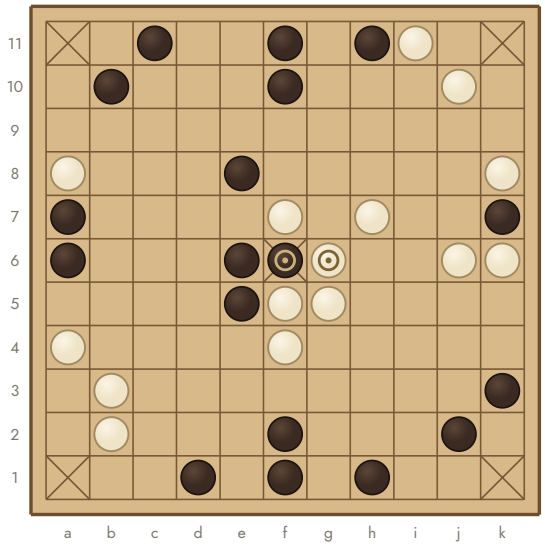
Variations

- **1... a4-a3**: White occupies a3, Black's capture square, instead of launching his double threat. 2. d6-d9 k8-i8 3. d1-d3 j6-j3 4. d9-a9 takes a8 f4-a4 takes a5 5. e5-a5 takes a6 i8-i11 takes h11. White is slightly better.
- **2. k5-k3**: Black leaves i1 open and plays on the k-file. 2... i4-i1 takes h1 3. h11-i11 k8-h8 4. k7-h7 a8-a9 5. i11-h11 takes g11 f8-a8 takes a7 6. h11-h9 takes h8. Balanced position.
- **2... i4-k4 takes k5**: White takes on the k-file instead of going to i11. 3. d1-c1 a8-a9 4. d6-d1 takes e1 j6-j9 5. d1-d8 a9-a8 takes a7 6. j2-j3 a4-d4. Balanced position.
- **3. j2-j4**: Black does not recapture on h11 and attacks the central cross via g4. 3... b6-b9 4. j4-g4 takes g5 a6-c6 5. k5-g5 takes f5 k6-k3 6. f6-f5 c6-a6 takes a7, a5 7. e5-e4 takes f4. Black is clearly better.

Fourth system: 1. d11-c11, the race for pawns

The same idea as d1-c1, at the top of the board. White again answers with the double threat k4-i4, and Black closes i1 by occupying it. Black then wins several pawns on the edges, while White keeps the pair on the throne.

1.	**d11-c11**	**k4-i4**	White's double threat on the i-file.
2.	**h1-i1**	**i4-i11** takes h11	Black closes i1 by occupying it; White takes h11.
3.	**h6-h11** takes g11	**f8-h8**	
4.	**k5-k3**	**b6-b3**	
5.	**i1-h1** takes g1	**a6-b6**	The i1 pawn comes back and takes g1.
6.	**a5-a6**	**b6-b9**	
7.	**d6-b6**	**h8-h7** takes g7	Black clears the d-file; White takes g7.
8.	**d1-d11** takes e11	**f7-g7** takes g6	Black takes e11 along the open file, White takes g6.
9.	**b6-b8** takes b9	**g7-f7**	Black takes b9.
10.	**b8-e8** takes e7	**f6-g6**	White finally brings a king out.
11.	**d11-d1** takes e1		Black takes again on the edge.



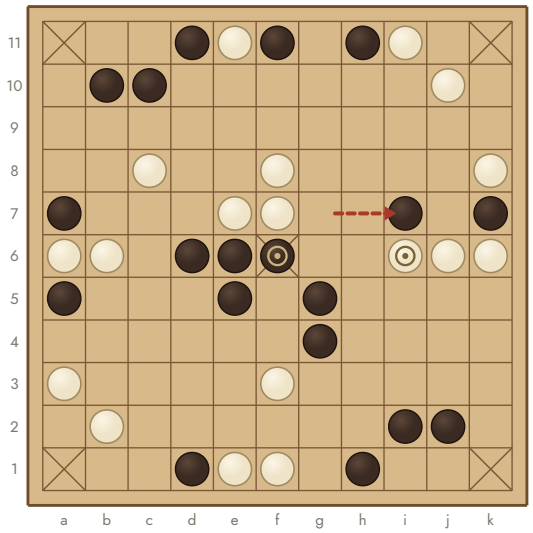
The position after 11. d11-d1.

Black has several extra pawns and keeps the advantage, but the white king has just come out into a very open centre: Black must now close its roads before thinking of attacking. Pawns won on the edge are only worth something if you keep the tempi to defend.

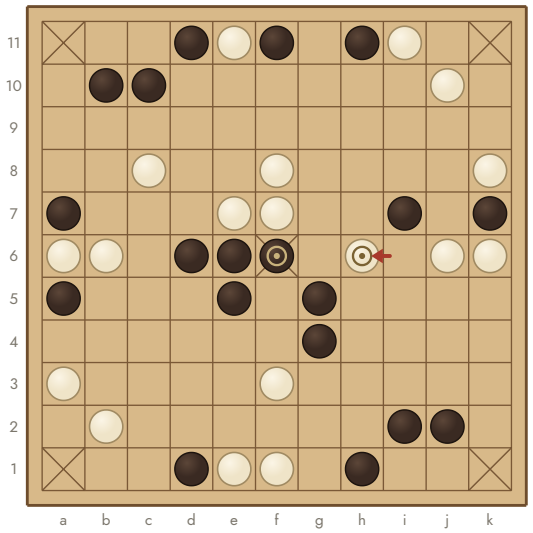
Variations

- **1... k4-k3**: White occupies k3 instead of launching the double threat. 2. d6-d11 takes e11 f4-k4 takes k5 3. h11-h9 a4-d4 4. h9-k9 takes k8 d4-d5 takes e5 5. f2-c2 k4-a4 takes a5. White is slightly better.
- **2. k5-k3**: Black leaves i1 open and plays on the k-file. 2... a4-c4 3. d6-d11 takes e11 c4-c1 takes d1 4. a5-a3 b2-d2 5. h11-h9 d2-d5 takes e5 6. h9-a9 takes a8. Balanced position.
- **2... f8-h8**: White cuts the h-file to prevent the recapture h6-h11. 3. j2-j4 i4-i11 takes h11 4. j4-g4 takes g5 b6-b3 5. k5-g5 takes f5 a6-b6 6. e5-e4 takes f4 b6-a6 takes a7, a5. Balanced position.
- **3. h6-h1 takes g1**: Black recaptures at the bottom instead of the top. 3... f8-h8 4. f1-g1 f4-g4 5. g1-f1 takes e1 h8-h7 takes g7 6. k5-i5 g11-g7 takes g6 7. d6-d11 takes e11. White is slightly better.

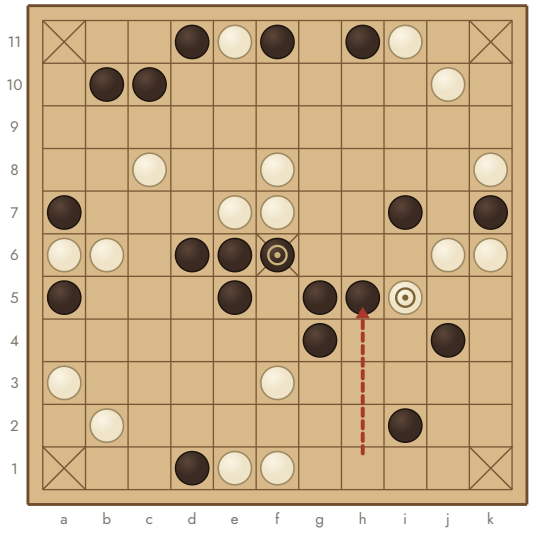
Exercises



12.1 Third system, after 8. g7-i7. Black threatens to capture the white king with i2-i5 or g5-i5. White has only three moves: i6-h6, i6-i5 or i6-i4. Which one?



12.2 Same line, but White has played 8... i6-h6, a mistake. Black to play and win in three moves.



12.3 Third system: Black plays 10. h1-h5 instead of 10. f6-h6. White to play and win a pawn.

13 Opening traps

LEVEL 3 · ADVANCED

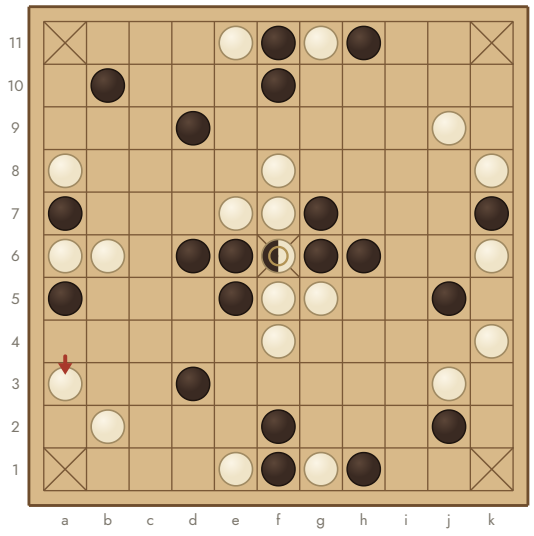
In the opening, every move can open the end of a chain or leave a pawn hanging. Typical mistakes look alike: you soon recognise them, and punish them even faster.

Typical mistakes

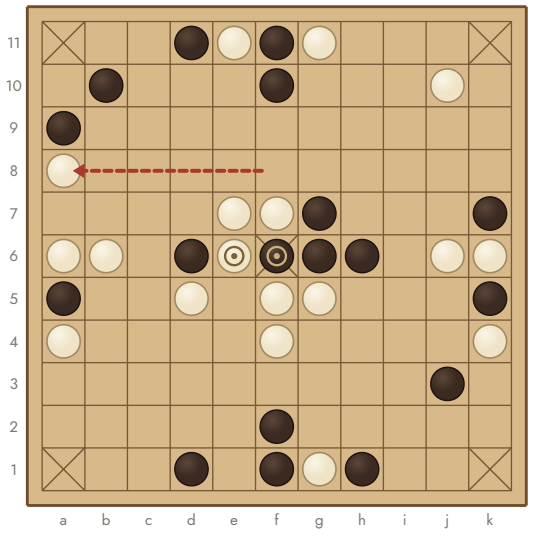
- **Leaving the end of a chain.** A pawn leaving an edge may free a capture square for your opponent.
- **Moving a pawn of the central cross.** The pawns on e6, g6, f5 and f7 hold each other; removing one often allows an immediate capture.
- **Forgetting the recapture.** After a capture on the edge, your opponent can often recapture at once; if you cannot recapture in turn, the exchange goes wrong.
- **Moving next to an empty corner.** A single enemy pawn is then enough to capture.

> **REMEMBER**
>
> Before every opening move, ask which capture square your move frees, and who will capture last in the exchange that follows.

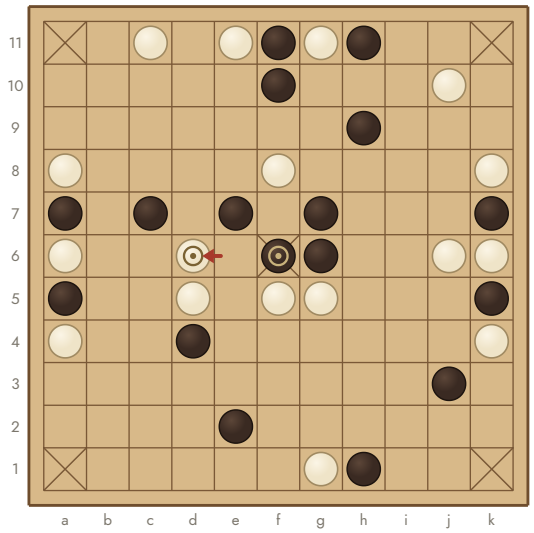
Exercises



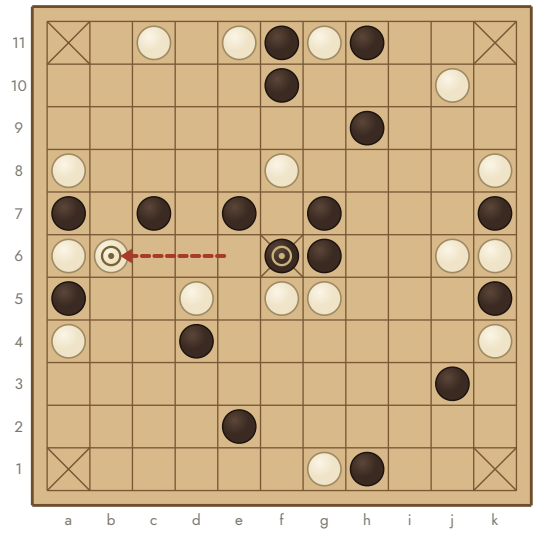
13.1 First system, variation 1... j6-j3 2. d11-d9 j10-j9 3. k5-j5. White plays 3... a4-a3 instead of f8-i8. Black to play and win a pawn.



13.2 Second system, variation 3. h11-h9, up to 6. j2-j3. White plays 6... f8-a8 (which takes a7) instead of e7-d7. Black to play and win a pawn.



13.3 Second system, main line up to 10. f2-e2. White plays 10... e6-d6 instead of e6-b6. Black to play and win a pawn.



13.4 Second system, after 10... e6-b6. Black can capture in five ways: j3-k3, j3-a3, d4-g4, h9-k9 or h9-a9. What should he play?

IV

LEVEL 3 · ADVANCED

The middlegame

The middlegame rests on two notions: the control and the occupation of squares. This part derives from them a simple way of counting the value of pawns, the functions of pawn structures, then the art of sacrificing, moving the pair out and choosing a plan.

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14 Control and occupation

15 The value of pawns

16 The functions of pawn structures

17 Sacrifices and initiative

18 Moving the pair out

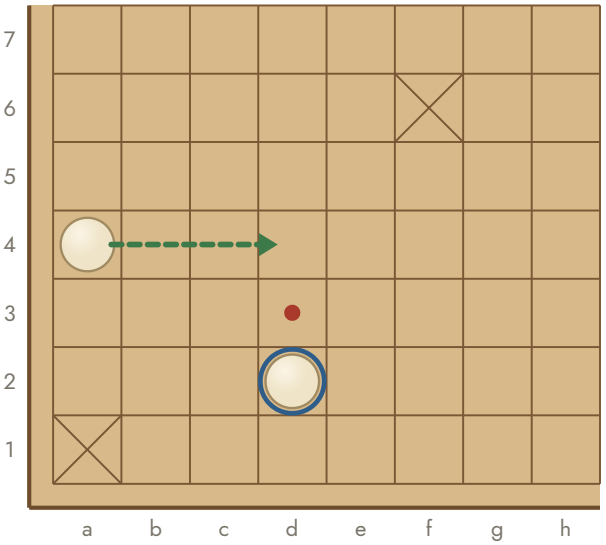
19 Game plans

14 Control and occupation

A pawn does only two things on the board: it **controls** squares and it **occupies** one. Everything else follows from them: locks, barrages, nets around a king, solid or fragile structures. These two notions are the basis of the whole middlegame.

Control

A side **controls** an empty square when an enemy piece stopping there could be captured next move. Two things are needed: an **anvil** on one side of the square (one of your pawns, an empty corner or the empty throne) and a pawn able to **arrive** on the other side. Control works against pawns as well as against the king, except for a king left on the throne; on an edge, it works only along the edge.



White controls d3: a black pawn or the black king stopping there would be captured by a4-d4 against the anvil d2. The white king, on the other hand, can stop there safely.

Control is **offensive** and above all **asymmetric**: it denies a square to your opponent without closing it to your own pieces. It is the only way to stop the enemy king from stopping on a square while leaving it free for yours.

One detail matters a great deal: the capturing pawn must be able to **arrive** on its square. A pawn already standing on that square controls nothing on that side; moved back one square, it controls it again.

Occupation

A piece standing on a square **occupies** it. Occupation is **defensive** and has three effects: it bars the line to every piece, to both kings as well as to pawns; it denies the square to your opponent, who can no longer place his anvil there or bring his pawn there; and it provides your pawns with an anvil on the neighbouring squares. Occupation is therefore also the raw material of control.

Its price: a pawn occupying a square can itself be controlled, that is, **hanging**. Its hidden advantage: a barrage closes the line to both kings, but you decide when to open it, by moving your pawn at the right moment.

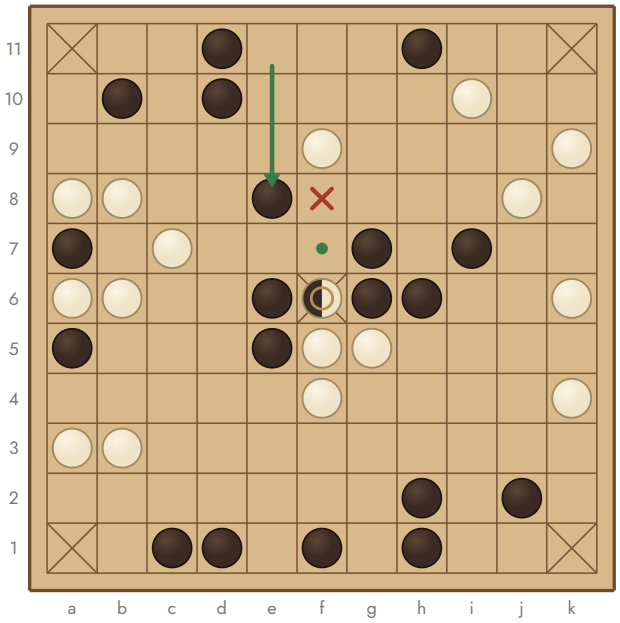
Reach

A pawn's **reach** is the set of squares it can reach in one move. It serves both notions: a pawn controls from afar the squares where it can arrive to capture, and it can occupy in one move a line it reaches. A pawn that sees far is often worth more than a pawn glued to its neighbours.

Before the pair comes out: the landing squares

As long as the pair is on the throne, both kings start from the same square and use the same lines. The only difference between them is the **landing squares**: those where the king can stop without being captured. The king that stays behind occupies the throne: during the exit, the throne is neither empty, nor an anvil, nor a passage. The side that controls the squares near the throne can come out first in safety, while the opponent, on the same lines, finds only traps.

The typical technique combines both notions: you **occupy** one line to close it, and you **control** the other landing squares against the enemy king only.

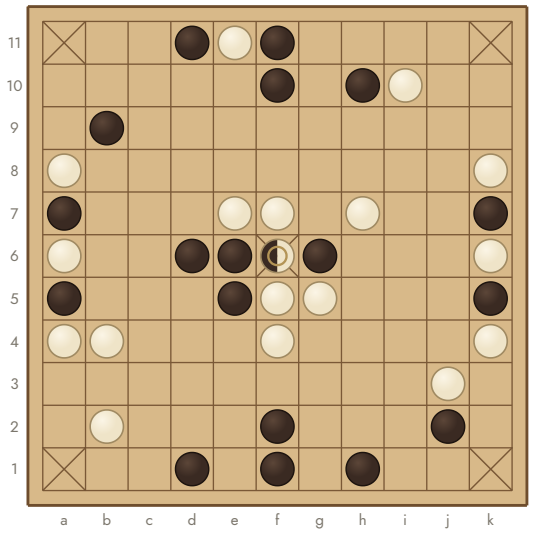


White to play, after e11-e8. Both exits, f7 and f8, are controlled against the white king (by e6, e8 and g7). The green dot marks f7, which remains safe for the black king; the cross marks f8, a trap for both kings. Only Black can move the pair out without risk.

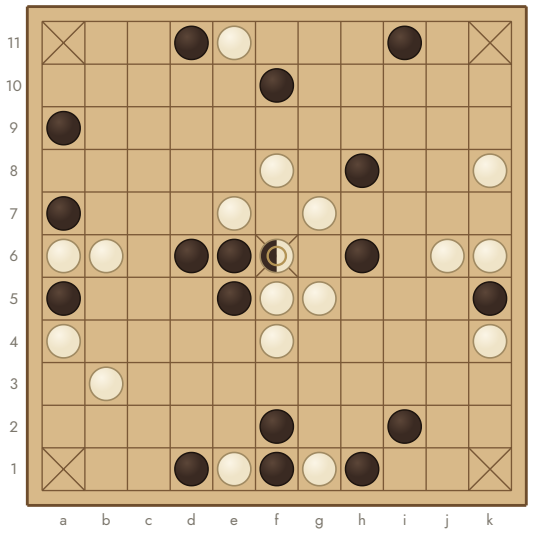
The pawns that best control the landing squares are not those of the central cross, which mainly serve as a barrage: they are the pawns placed two or three squares from the throne, off the cross, which can arrive on several lines at once.

> **REMEMBER**
>
> Control to forbid, occupy to bar. Before the pair comes out, the side that controls the landing squares around the throne will be the first to come out.

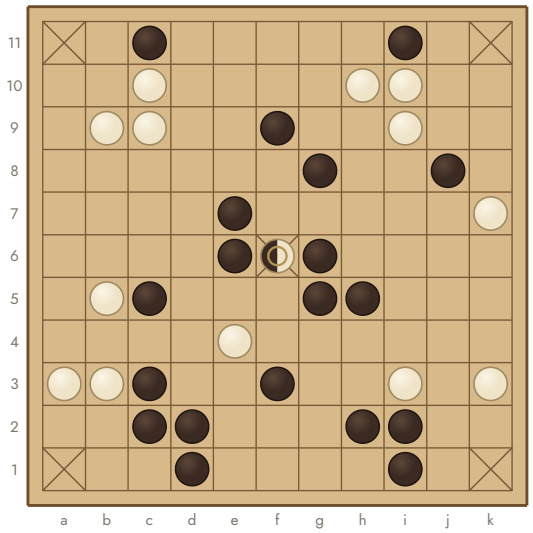
Exercises



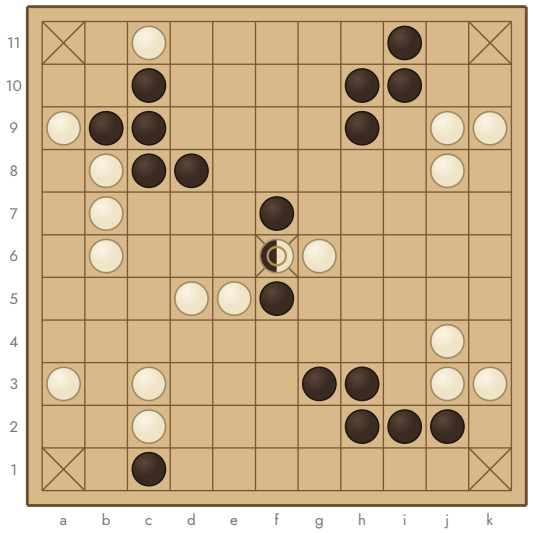
14.1 Black to play. Which pawns are hanging, on each side, and by which move?



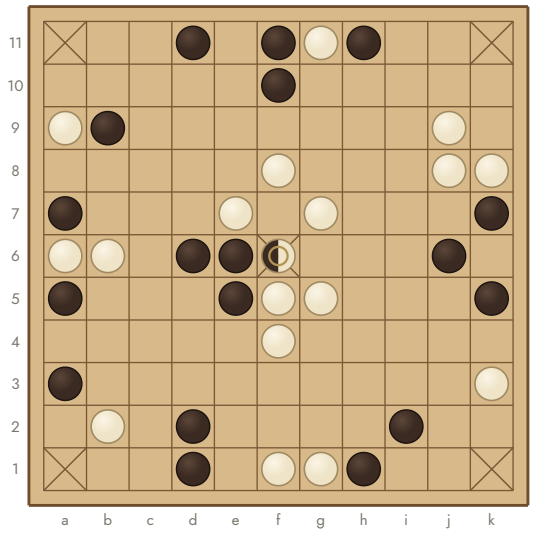
14.2 Black to play. Which pawns are hanging, on each side, and by which move?



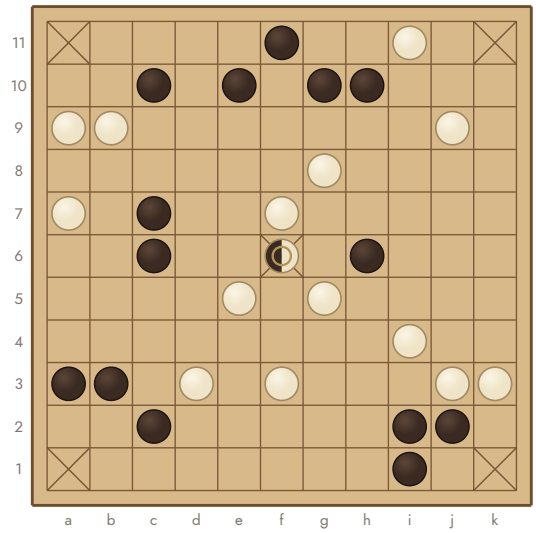
14.3 The pair is on the throne. On which squares can each king come out without being captured next move?



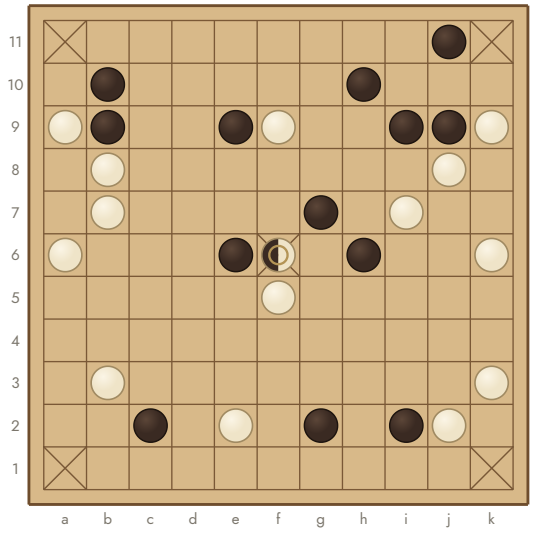
14.4 The pair is on the throne. On which squares can each king come out without being captured next move?



14.5 The pair is on the throne. White to play: find the move that leaves the black king no safe exit, without closing those of the white king.



14.6 The pair is on the throne. White to play: find the move that leaves the black king no safe exit, without closing those of the white king.



14.7 The pair is on the throne. Black to play: find the move that leaves the white king no safe exit, without closing those of the black king.

15 The value of pawns

Not all pawns are worth the same. A pawn draws its value first from its **square**: this is its strategic value. It then draws value from what it **does** in relation to the other pieces: this is its tactical value. After the pair comes out, the **king's assets** are added. This chapter gives a simple scale, in quarters of a pawn, to weigh all of this together.

Strategic value: the square

	Pair on the throne	After the pair comes out
Edge pawn	three quarters of a pawn	three quarters of a pawn
Central pawn (in the square from d4 to h8)	one and a quarter pawns	one pawn
Corner guard (b2, b3, c2, c3 and their mirror squares)	one pawn	one and a half pawns
Other pawn	one pawn	one pawn

Before the pair comes out, central pawns are worth more because they control the landing squares around the throne. After it comes out, the **corner guards**, near the corners, gain value: they lock the corners and bar the last lines. Edge pawns are worth less in both phases: solid, they are also passive. The pawns of the central cross count as central pawns, but their value comes mostly from their role as a barrage.

Tactical value: what the pawn does

What the pawn does	Pair on the throne	After the pair comes out
Barrage: first obstacle on a line of the enemy king (or of the throne)	half a pawn more	half a pawn more
Net: helps control a square next to the enemy king	—	half a pawn more
Lock: helps control a square next to a corner	—	a quarter of a pawn more
Hanging	a quarter of a pawn less	a quarter of a pawn less
Glued to a friendly pawn	—	a quarter of a pawn less

There are at most four barrages per side: on each of the four lines leaving the enemy king (or the throne), the first piece met, if it is yours. A pawn is glued when it touches a friendly pawn along a side, and each of the two counts. The next chapter describes these functions in detail. Remember already that a pawn barring a road or squeezing the enemy king is worth clearly more than an idle pawn, and that two pawns glued together are worth less than two pawns each covering their own lines.

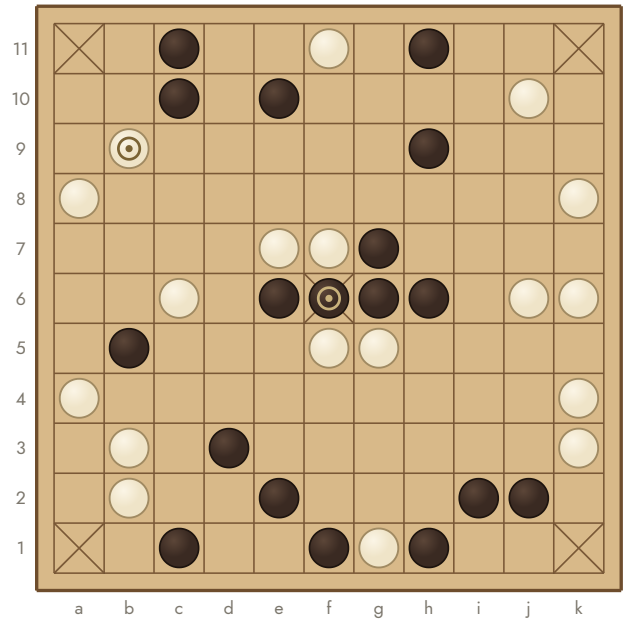
The king's assets and the move

Asset	Value
After the pair comes out, each square your king can go to without being captured	a quarter of a pawn
After the pair comes out, each corner threat your king can create without being captured	half a pawn
The move	a quarter of a pawn before the pair comes out, half a pawn after

A free king is soon worth several pawns: a king with eight safe squares and one corner threat is worth, on its own, two and a half pawns more than a shut-in king.

Doing the count

First count the material, then only the **differences** between the two sides: there is no need to add everything up. To go faster, ignore the quarters of a pawn: they only separate very close positions. An example:



Black to play. Material is level. The white king, out on b9, has eleven safe squares (two and three-quarter pawns) and two corner threats (one pawn); the black king, still on the throne, has neither. On the other hand, White has three more edge pawns and three more glued pawns (one and a half pawns against him), and Black has the move (half a pawn). In total, White is better by about two pawns.

Intermediate goals

- **Pair on the throne.** Win material; place pawns in the centre, two or three squares from the throne; keep your barrages around the pair, which are the key to your exit; leave no pawn hanging.
- **After the pair comes out.** Give your king freedom and look for corner threats; place your corner guards near the corners the enemy king can reach; bar its lines and squeeze it in a net; separate pawns that are needlessly grouped.

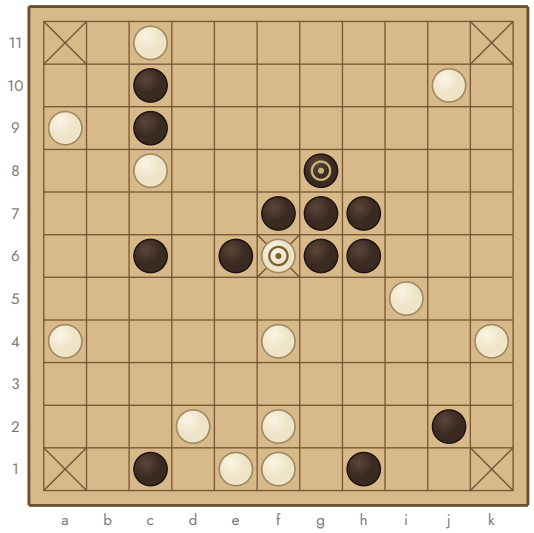
What to sacrifice?

The scale also tells you what to give. Sacrifice first the pawns that are worth least: edge pawns, glued pawns, hanging or idle pawns. Give up a corner guard or a barrage only for a clear gain. A sacrifice is good when what it brings exceeds the value of the pawn given: an edge pawn that opens two safe squares for your king and gives it a corner threat brings one pawn for three quarters of a pawn.

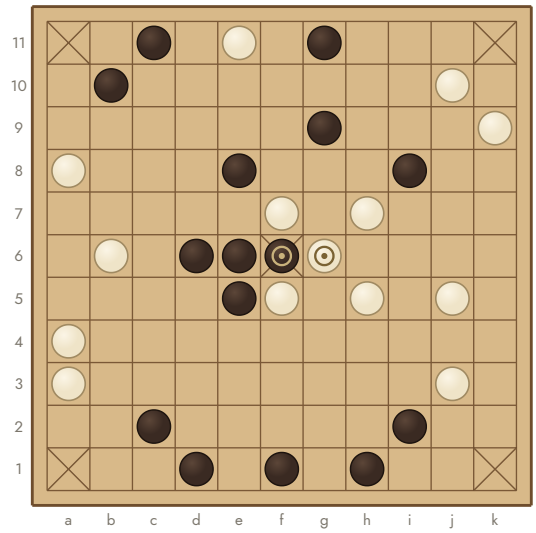
The scale is a guide, not an absolute rule: a combination comes before any count. It weighs most after the pair comes out; before that, material and the centre decide almost alone.

> **REMEMBER**
>
> A pawn is worth something by its square and by what it does. Before the pair comes out, the centre; after, corner guards, barrages, nets and the king's freedom.

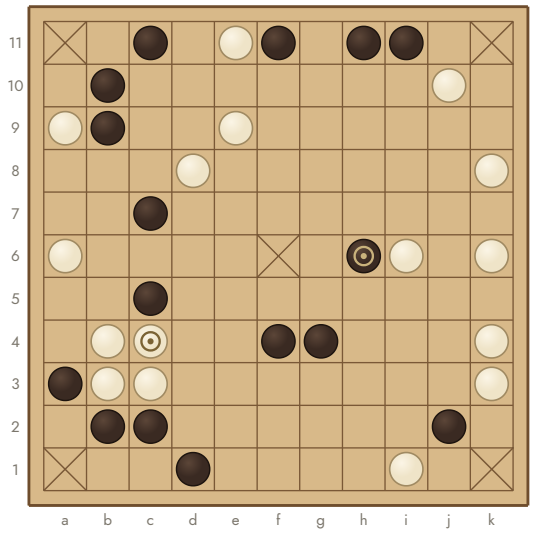
Exercises



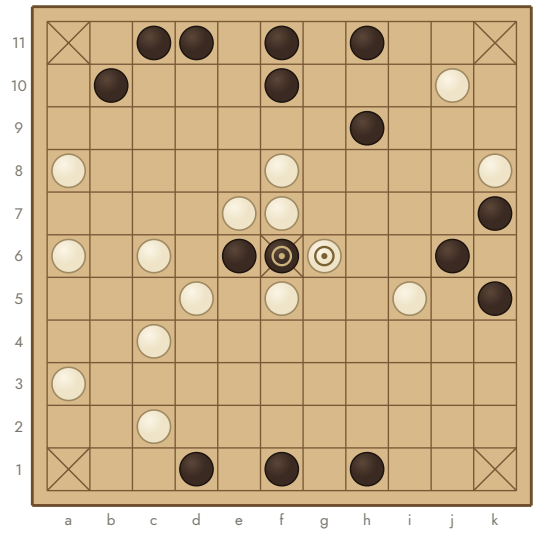
15.1 Black to play. Do the count: which side is better, and by about how much?



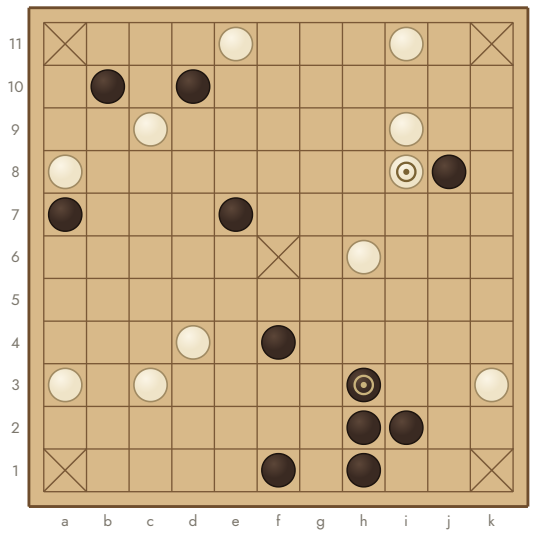
15.2 Black to play. Do the count: which side is better, and by about how much?



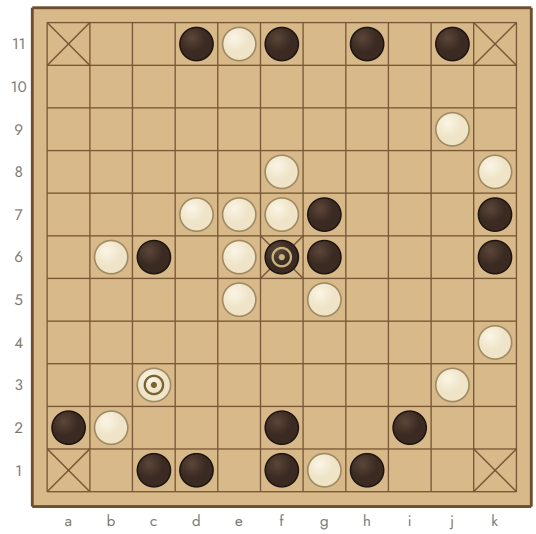
15.3 Black to play. Do the count: which side is better, and by about how much?



15.4 Black to play. Do the count: which side is better, and by about how much?



15.5 Black to play. Do the count: which side is better, and by about how much?



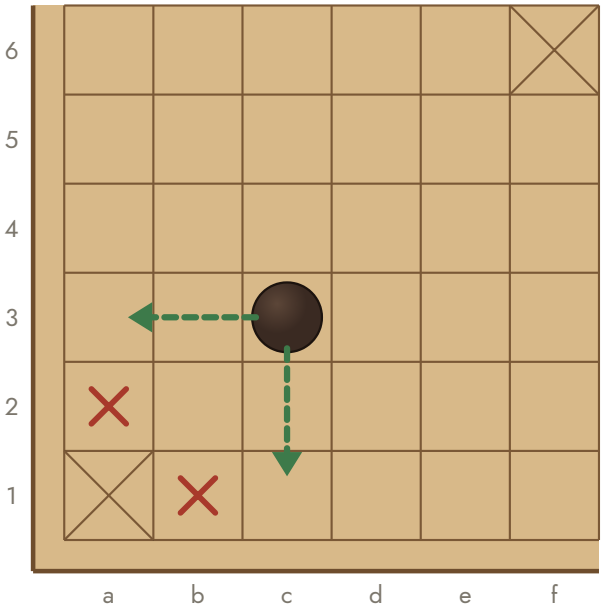
15.6 Black to play. Do the count: which side is better, and by about how much?

16 The functions of pawn structures

From control and occupation come a few precise functions. A good pawn structure is not a pretty shape: it is a set of pawns fulfilling these functions with as few pieces as possible.

The lock

The empty corner acts as an anvil. A pawn able to arrive on the second edge square after the corner (c1 or a3 for corner a1) therefore forbids the enemy king from stopping next to that corner. The best spot is c3, on the corner's diagonal: from there, a single pawn closes both approaches at once.

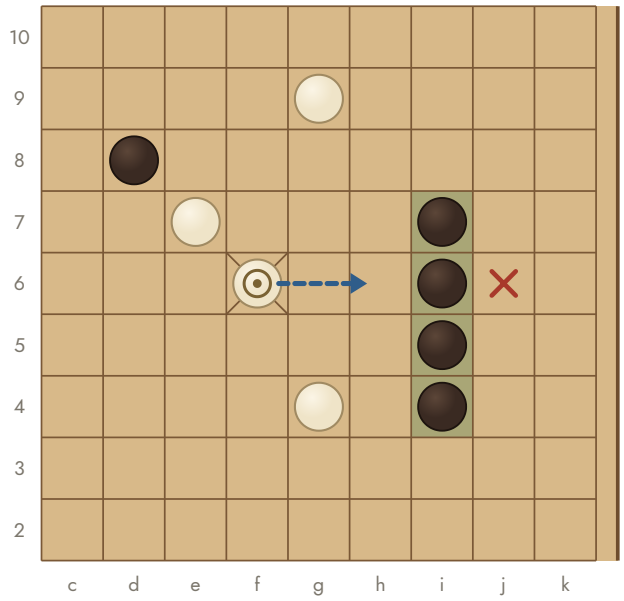


The black pawn on c3 can arrive on c1 or a3: the white king can stop neither on b1 nor on a2.

Two frequent mistakes: placing the pawn **on** the square next to the corner, where it locks nothing and can be captured against the empty corner; and placing it **on** the capture square itself, from which it cannot arrive there. Edge pawns standing three squares from the corner, such as a4 or d1, make excellent locks: this is what redeems their passivity.

The barrage

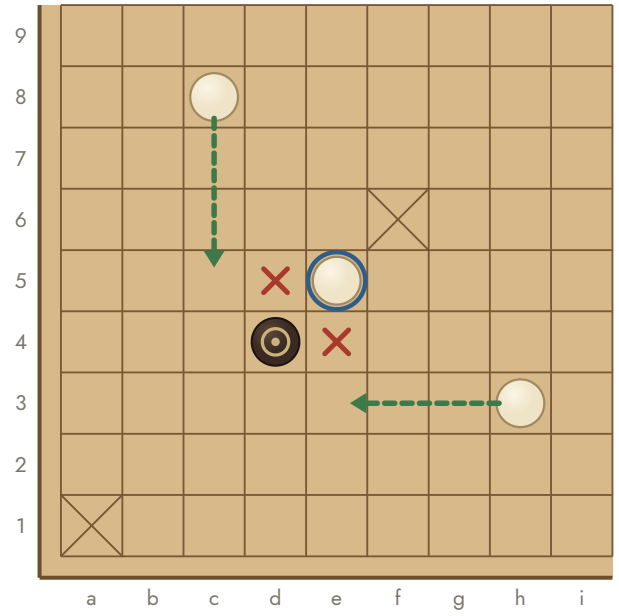
A pawn occupying a king's line closes it: this is a **barrage**. Right next to the enemy king, a single piece thus cuts a whole line. The **wall**, at least three adjacent pawns in a line, is a long barrage that cannot be captured along its line. Before the pair comes out, the pawns of the central cross are barrages for both kings: whoever owns them chooses when to move them.



The black wall i4-i7 cuts the white king off from the east: it cannot cross it, and White cannot capture any of its pawns along the file.

The net

The **net** is the control of the squares next to the enemy king. The most efficient pawn stands diagonally from the king: it serves as an anvil on two of its neighbouring squares at once. A complete net, with one or two barrages, leaves the king no safe square.



The pawn on e5, diagonally from the black king, is the anvil for d5 and for e4: the king would be captured on d5 by c8-c5, on e4 by h3-e3.

The screen and the interception

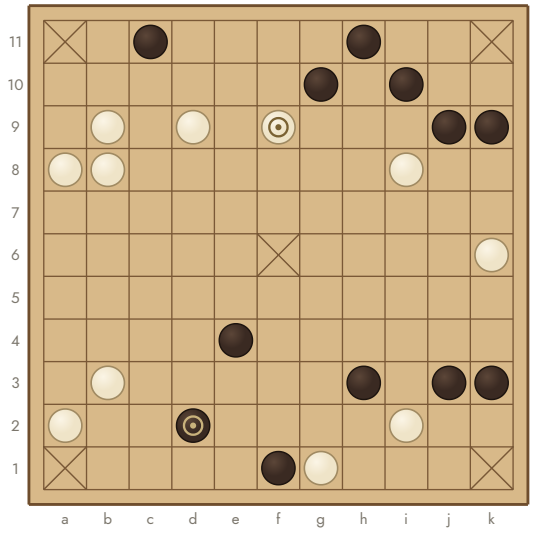
The **screen** is a defensive occupation: you place a pawn on the square your opponent needs, for his anvil or for his pawn's arrival, and his control disappears. The **interception** is an occupation held in reserve: a pawn able to cut, in one move, the enemy king's line to a corner. Keep at least one on every edge the enemy king can reach.

Support and glued pawns

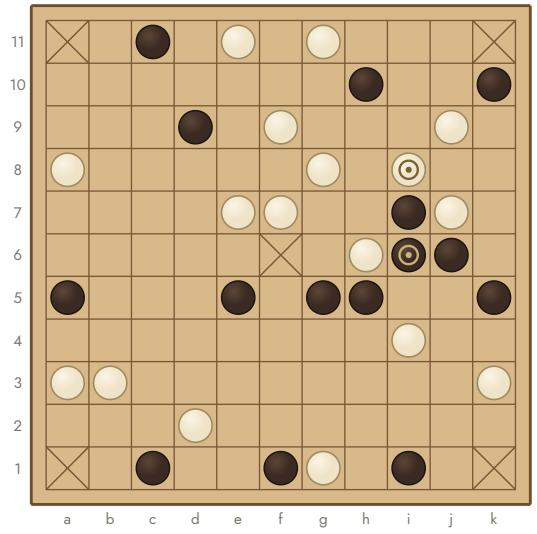
Two pawns touching each other cannot be captured along the line joining them: they support each other. But they control fewer squares than two separate pawns, and each one blocks one of the other's lines. An isolated pawn is not weak in itself; a pawn glued without reason is a pawn working at half strength. Group your pawns where they bar a road, and separate them everywhere else.

> **REMEMBER**
>
> Lock from afar, bar from close up, squeeze diagonally. A pawn that locks, bars and squeezes nothing is the first to move or to sacrifice.

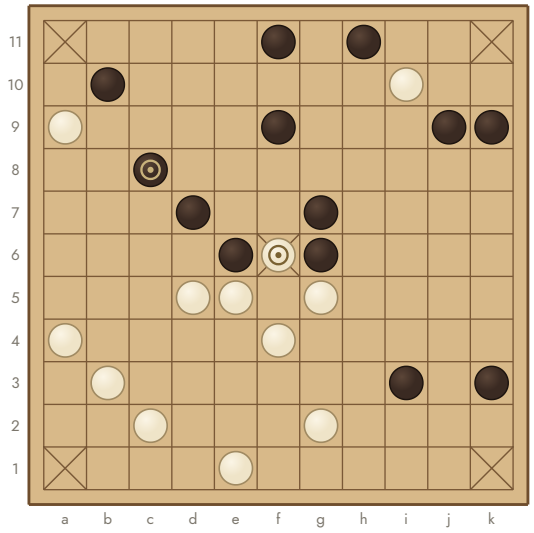
Exercises



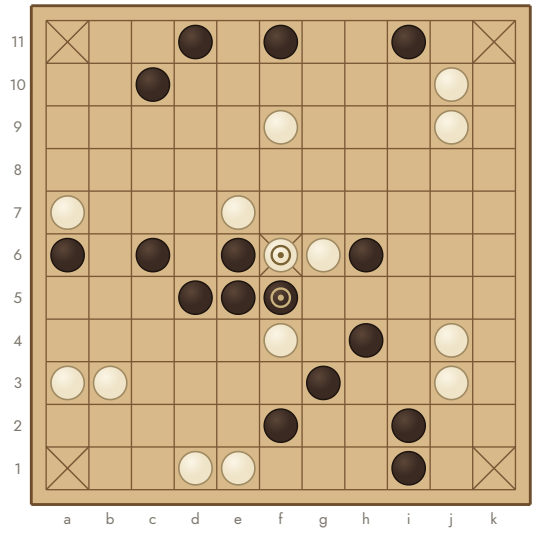
16.1 White to play. Find the move that closes both squares next to the corner a1 to the black king.



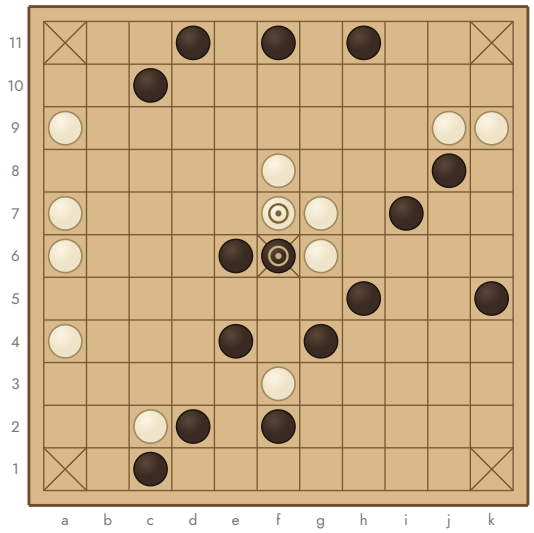
16.2 Black to play. Find the move that closes both squares next to the corner k11 to the white king.



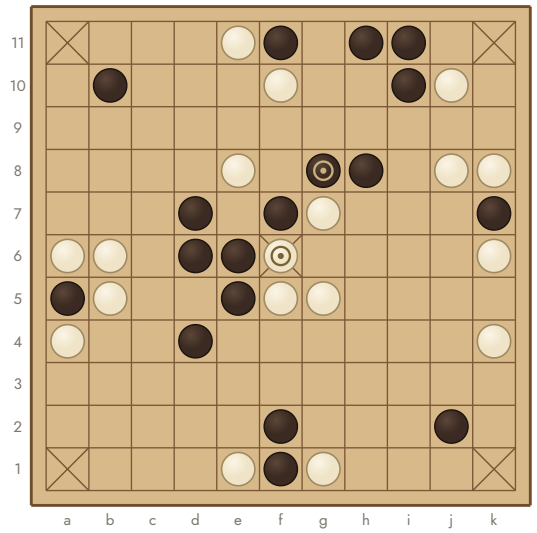
16.3 White to play. Find the move that closes both squares next to the corner a11 to the black king.



16.4 White to play. Find the move that leaves the black king no safe square.



16.5 Black to play. Find the move that leaves the white king no safe square.



16.6 White to play. Find the move that leaves the black king no safe square.

17 Sacrifices and initiative

LEVEL 3 · ADVANCED

The best players sometimes give a pawn away. A sacrifice is justified when it brings in more than the pawn given: an open road for the king, a threat that forces the opponent's replies, or a pawn won back a little later in better conditions.

The three reasons to sacrifice

- **Time.** The pawn you give away deflects an enemy piece or costs it a move; meanwhile, your attack moves forward.
- **The line.** The captured pawn leaves an empty square: if that square opens a road to a corner, it can be worth far more than the pawn.
- **The recapture.** The enemy pawn that captures often ends up badly placed, ready to be captured in turn.

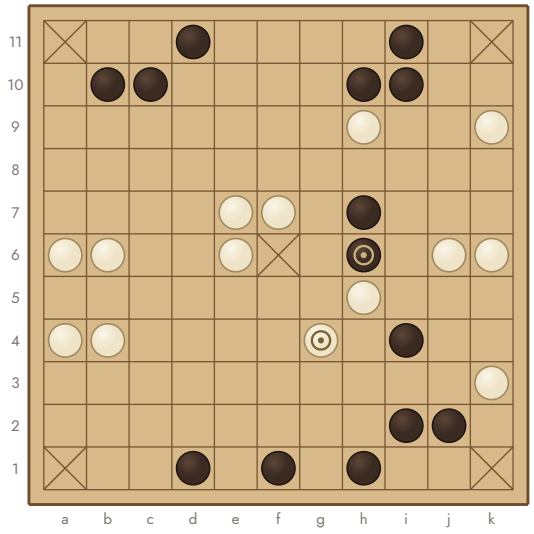
How to recognise it

A sacrifice is calculated like a combination: list your opponent's replies, starting with the capture, and check that each one leaves you something better than a pawn. If a single quiet reply leaves you simply a pawn down, the sacrifice is bad.

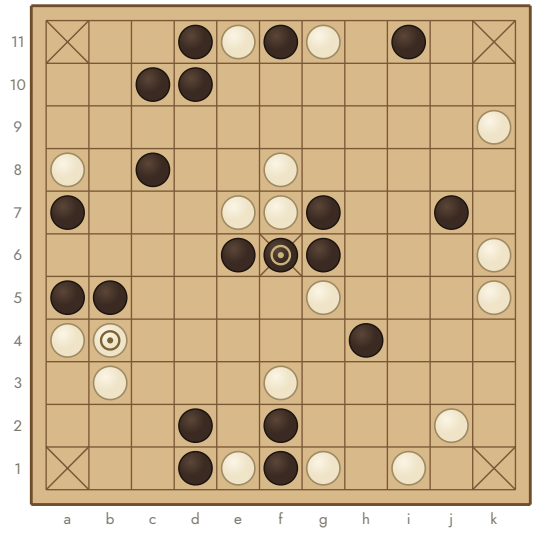
The scale of chapter 15 helps you judge a quiet sacrifice: add up what you gain (the move, king's freedom, corner threats, barrages, net) and compare it with the value of the pawn given. Preferably give an edge pawn, a glued pawn or a pawn that is already hanging.

> **REMEMBER**
>
> A pawn is given for time, a line or a recapture. Before sacrificing, look for your opponent's quiet reply: it is the judge of the sacrifice.

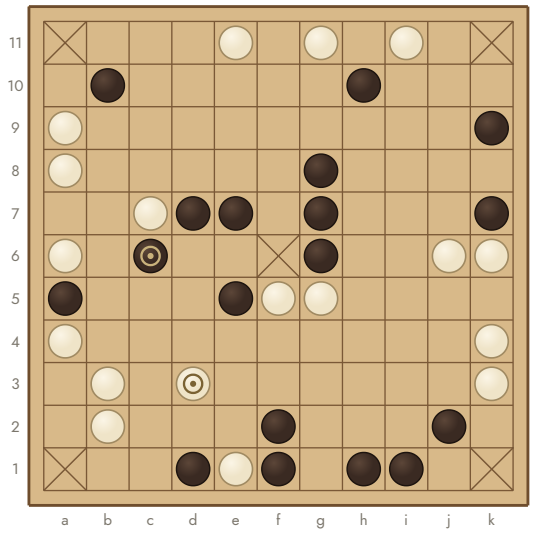
Exercises



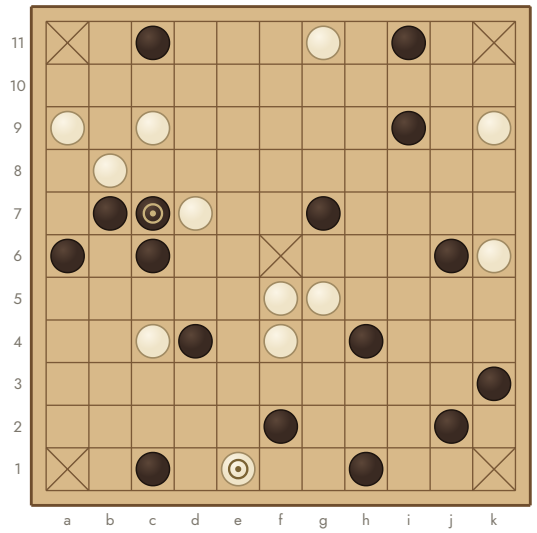
17.1 White to play and win in two moves, even though Black takes a pawn on the way.



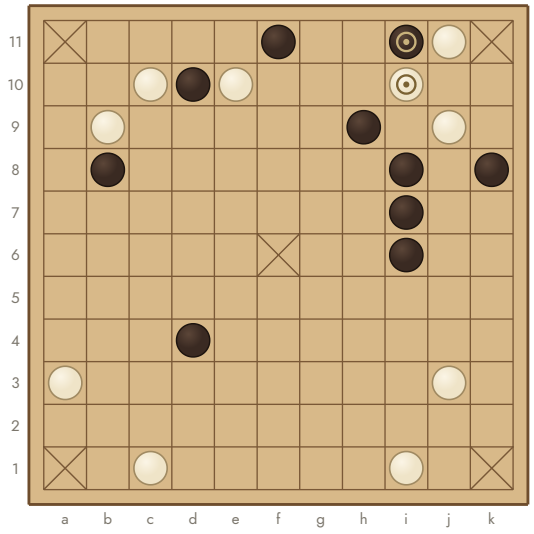
17.2 White to play and win in three moves, even though Black takes a pawn on the way.



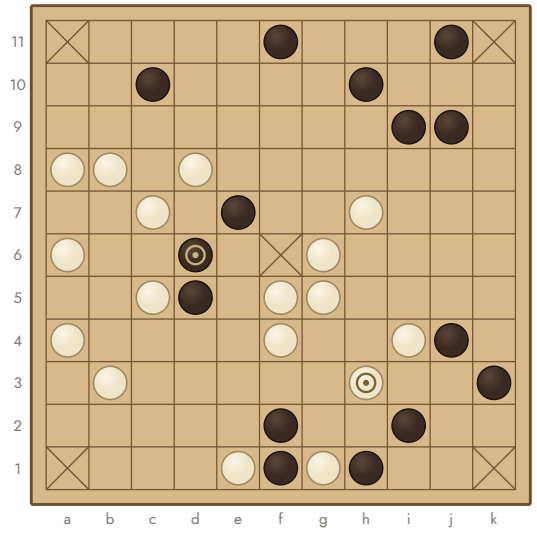
17.3 White to play and win in three moves, even though Black takes a pawn on the way.



17.4 Black to play and win in three moves, even though White takes a pawn on the way.



17.5 White to play and win in three moves, even though Black takes a pawn on the way.



17.6 White to play and win in three moves, even though Black takes a pawn on the way.

18 Moving the pair out

LEVEL 3 · ADVANCED

Moving the pair out is the main strategic decision of the game. The player who leaves takes his king along and can seize the initiative, but his king becomes a target. The player who stays keeps an untouchable king on the throne, but lets his opponent choose when the race begins.

When to leave?

- **With more pawns, leave.** Your opponent lacks the pieces to close all your king's roads, and a race to a corner is a good way to cash in the advantage.
- **With equal material, choose your moment.** Leave when your king can reach an edge or an open road, not before.
- **With fewer pawns, stay.** Leaving on a gamble usually fails: your king becomes a target and you lack the pieces to protect it. Wait for the enemy king to venture into the middle, or for a safe road to open for yours.

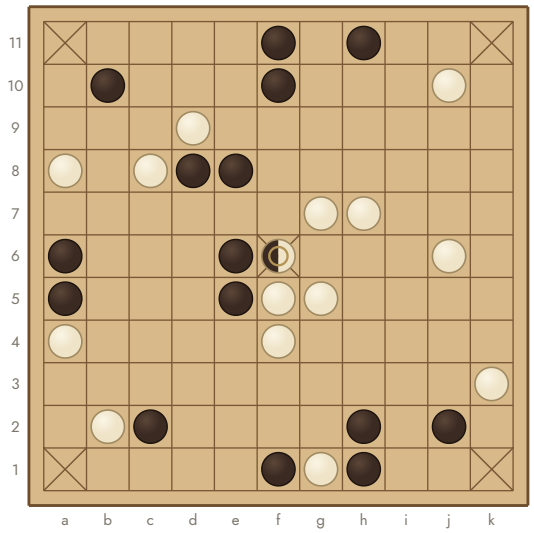
Before coming out, look at the landing squares (chapter 14): come out when yours are safer than your opponent's. Coming out while he controls the squares around the throne better than you gives you a bad start.

Where to go?

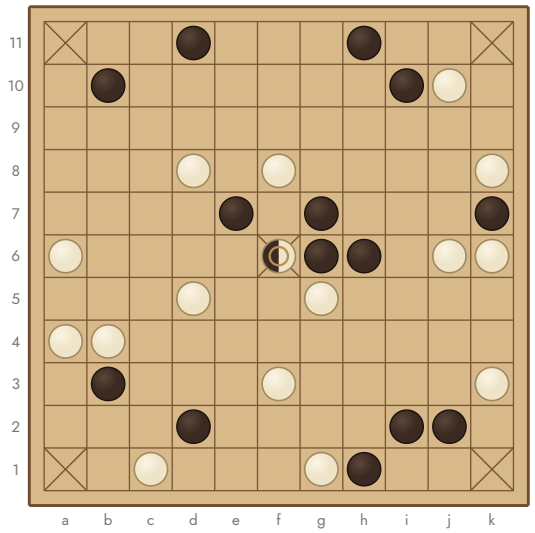
The pair can travel far in a single move. Aim for a square from which your king will have **two roads** to corners, out of reach of pincers. The edge is a safe refuge, but a king that gets there too early soon has its road blocked. Above all, avoid stopping in the middle, surrounded by enemy pawns able to form a pincer.

> **REMEMBER**
>
> Leave with an advantage, towards a double road. Otherwise, keep your king on the throne and let your opponent take the risks.

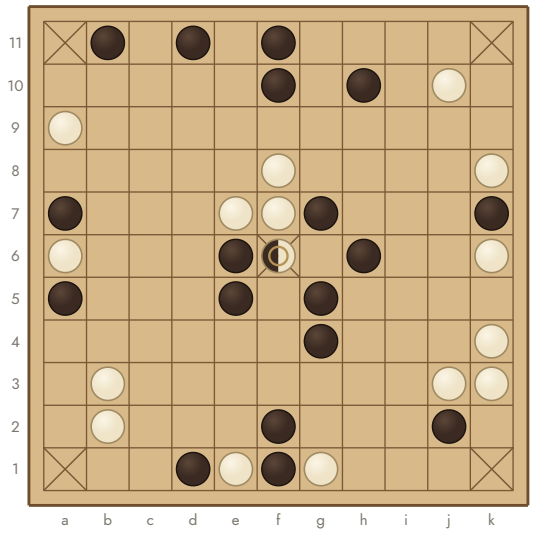
Exercises



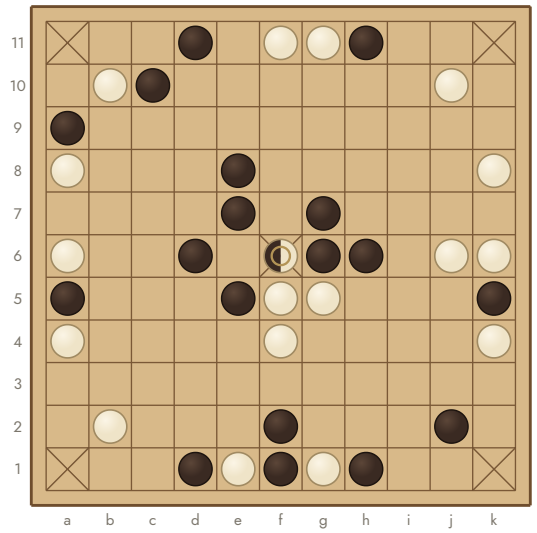
18.1 Black to play. Should the pair move out? If so, where?



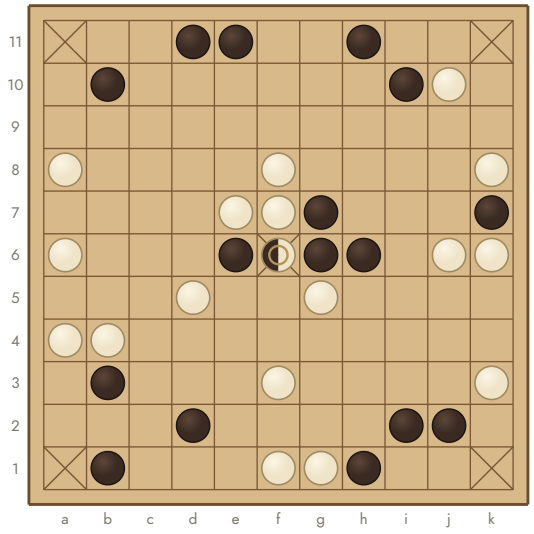
18.2 White to play. Should the pair move out? If so, where?



18.3 White to play. Should the pair move out? If so, where?



18.4 White to play. Should the pair move out? If so, where?



18.5 White to play. Should the pair move out? If so, where?

19 Game plans

Plan A · The race to a corner

This is the most frequent winning plan. Move the pair out, keep your king inside, away from pincers, and prepare a line to an edge that your opponent will no longer be able to close. The winning king moves little; the final strike is sudden. It can even succeed with fewer pawns, if your opponent lets a road open: an unstoppable line is worth more than an extra pawn.

Plan B · Hunting the king

Keep your king on the throne, win material, then capture the enemy king as it crosses the middle of the board. The hunter is generally the player who stayed on f6, and he is normally ahead in pawns. The key: two pawns aiming at the same square next to the enemy king, with an anvil on the other side.

Plan C · Suffocation

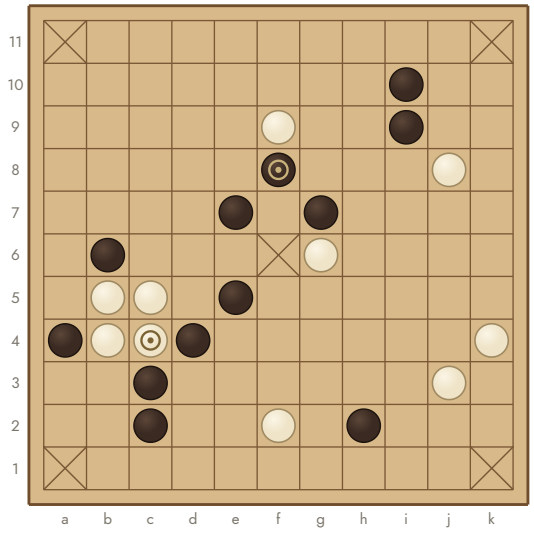
Immobilising every enemy piece is not a realistic plan: your opponent almost always has a free pawn move left.

Choosing and switching plans

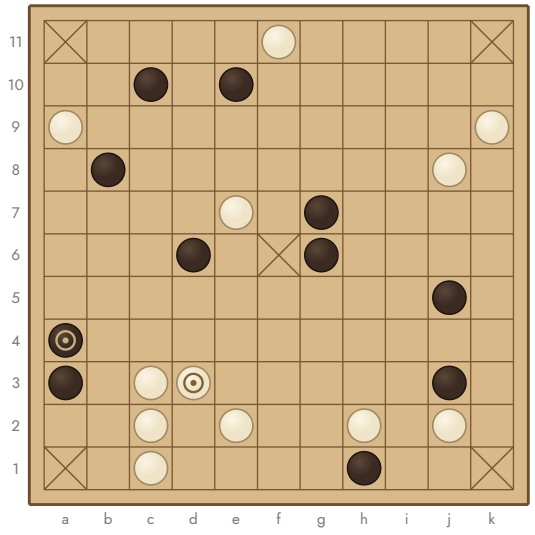
- When ahead in pawns, leave and race: the race is a good way to cash in the lead.
- When behind, stay and hunt: the king that leaves first often ends up crossing the middle.
- Plan B can turn into plan A: once material is won, leave in turn, with the advantage.
- Plan A requires watching your back: a king that leaves too early becomes the target of your opponent's plan B.

> **REMEMBER**
>
> Two plans win: the race to a corner and the king hunt. Material decides which one to choose.

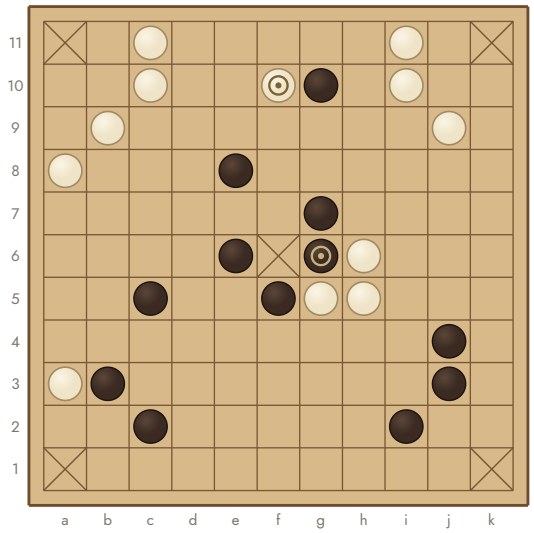
Exercises



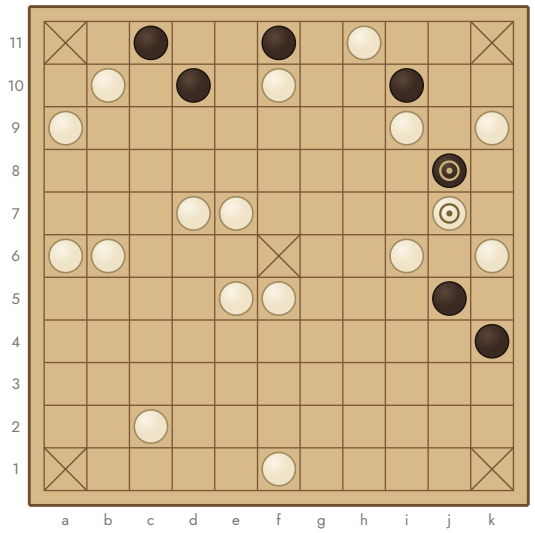
19.1 Black to play and win in three moves.



19.2 Black to play and win in three moves.



19.3 White to play and win in three moves.



19.4 White to play and win in three moves.

V

LEVEL 4 · STRONG PLAYER

Advanced tactics

Combinations of several moves decide high-level games. This part shows how to build and calculate them, with hard problems.

---

20 Long combinations

21 Calculating

20 Long combinations

At the highest level, games are often decided by a combination of three, four or five moves, often starting with a quiet move. They look long, but they can be calculated thanks to the threat rule: every threat cuts your opponent's replies to a handful, and the sequence becomes a single line to follow.

The three building blocks

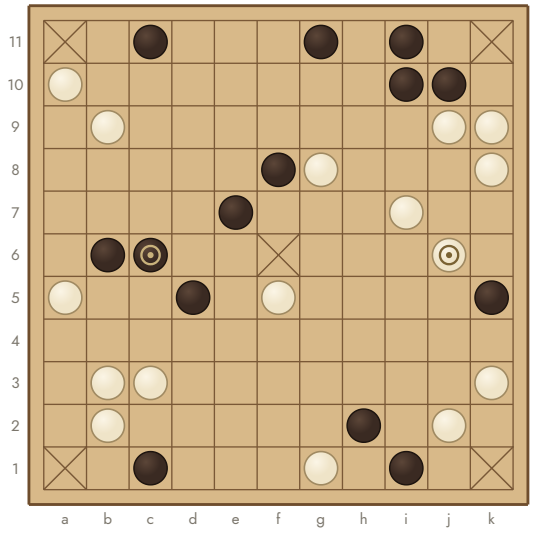
- **The forcing threat**: it forces a precise defence, which often moves a useful piece elsewhere.
- **The quiet move**: it threatens nothing visible, but prepares two threats your opponent cannot stop together.
- **The double threat**: two corners, two pincers, or a corner and a pincer; it is almost always the last move of the combination.

Building a combination

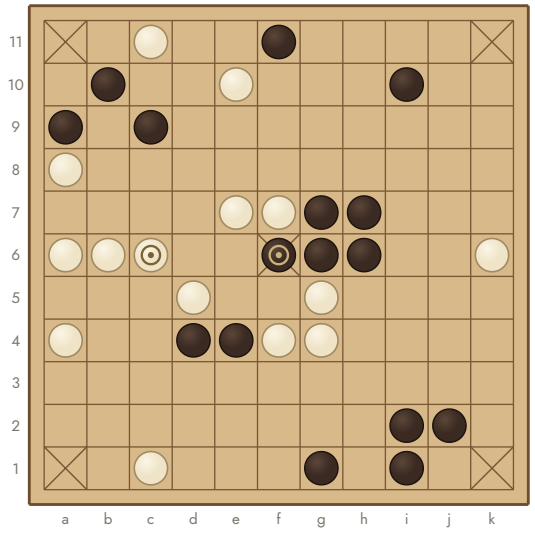
Start from the end: look for a position where your king would have two threats at once. Then ask which enemy pieces prevent it, and how to deflect them with threats. A long combination is almost always a sequence of threats that remove the defenders one by one.

> **REMEMBER**
>
> Start from the final double threat and work backwards: every move of the combination must remove a defender or gain a tempo.

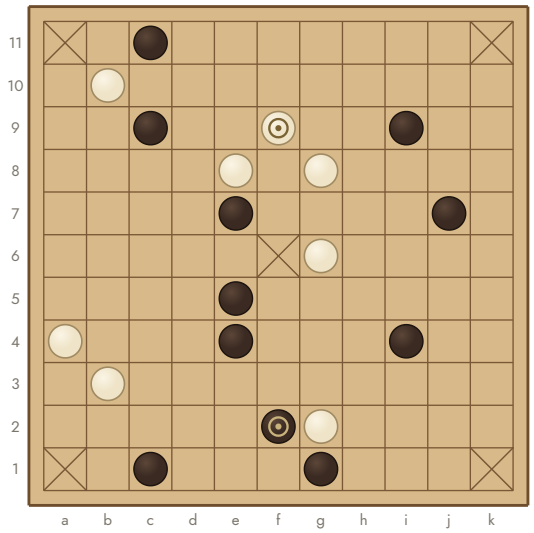
Exercises



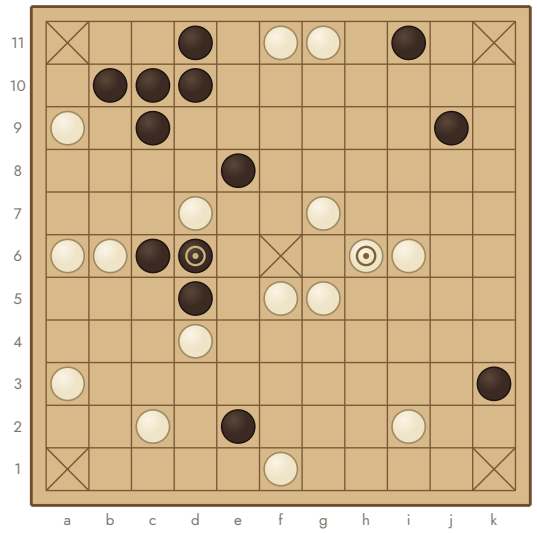
20.1 White to play and win in three moves.



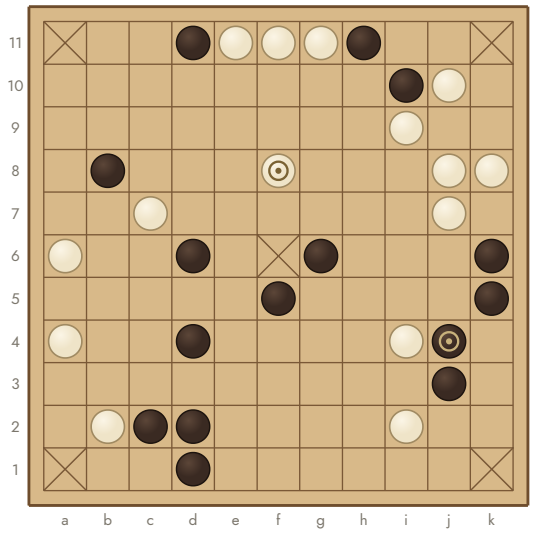
20.2 White to play and win in three moves.



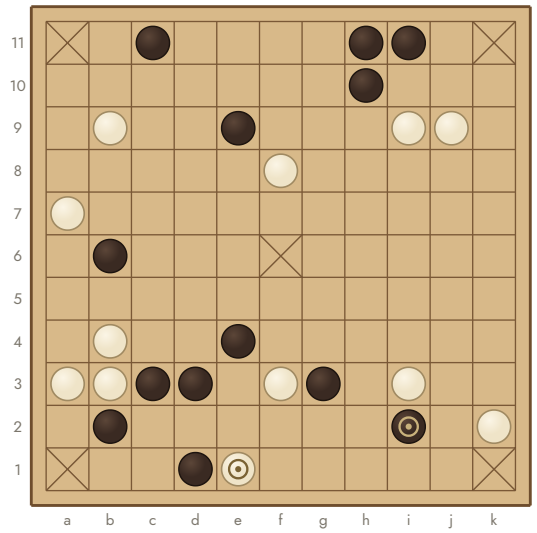
20.3 Black to play and win in three moves.



20.4 White to play and win in three moves.



20.5 Black to play and win in three moves.



20.6 Black to play and win in four moves.

21 Calculating

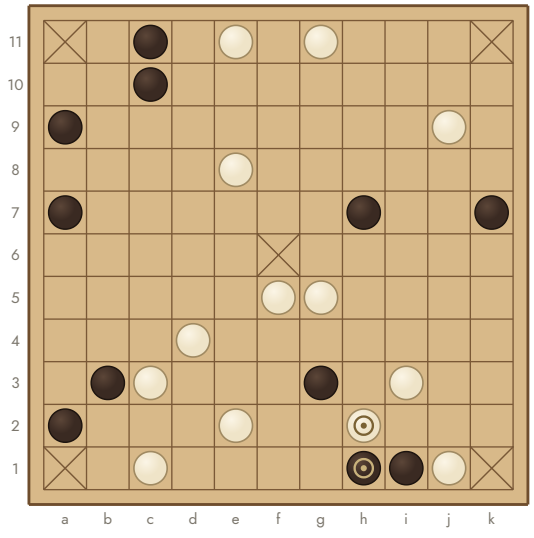
The threat rule makes calculation easier than it looks: after a threat, your opponent often has only a handful of allowed moves, sometimes a single one. A sequence of threats can therefore be calculated far ahead. The game rewards calculation: a move that looks natural can lose a pawn two or three moves later.

A method

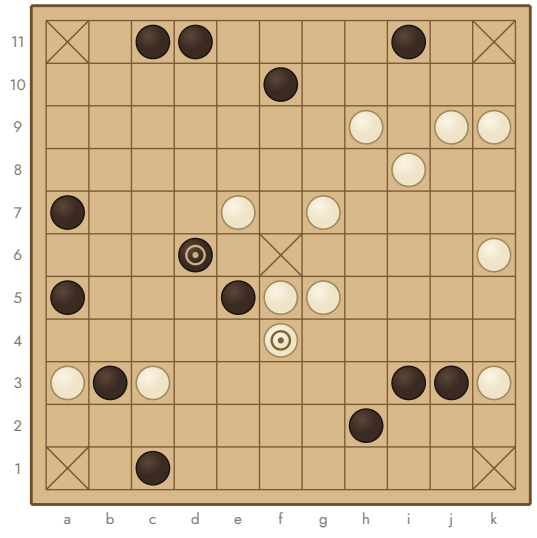
1. **List the forcing moves**: winning threats first, then captures, then capture threats.
2. **For each one, list the allowed replies.** After a threat there are only a few: write them all down.
3. **Look for the most stubborn reply**, the one that delays your win the most, and start again from there.
4. **Stop at a quiet position** and compare: material, king placement, remaining threats.
5. **Check the counter-attack.** Before playing, ask whether your opponent has an unstoppable threat faster than yours.

> **REMEMBER**
>
> Calculate the forcing moves first: there are few of them, and most wins hide among them. If they lead nowhere, look for the quiet move that prepares a double threat.

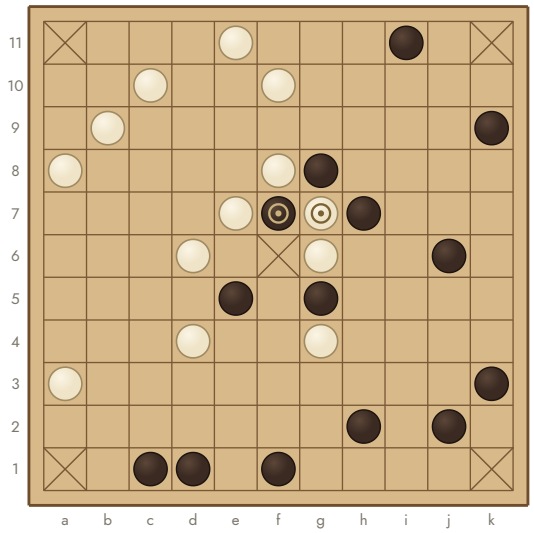
Exercises



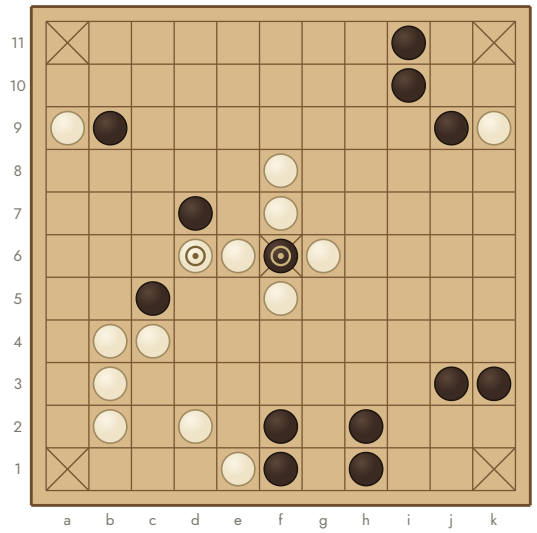
21.1 White to play and win in three moves.



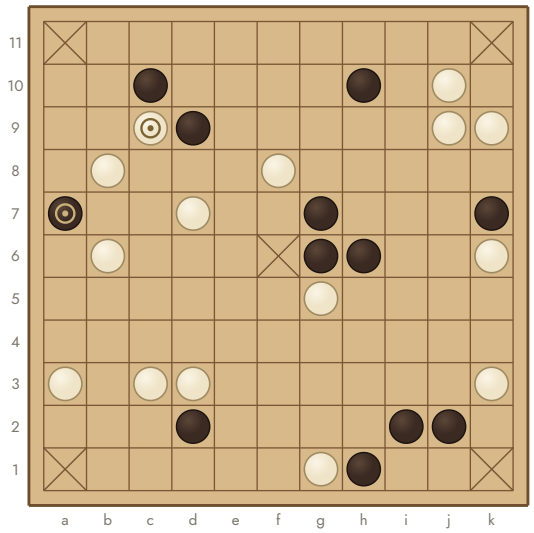
21.2 Black to play and win in three moves.



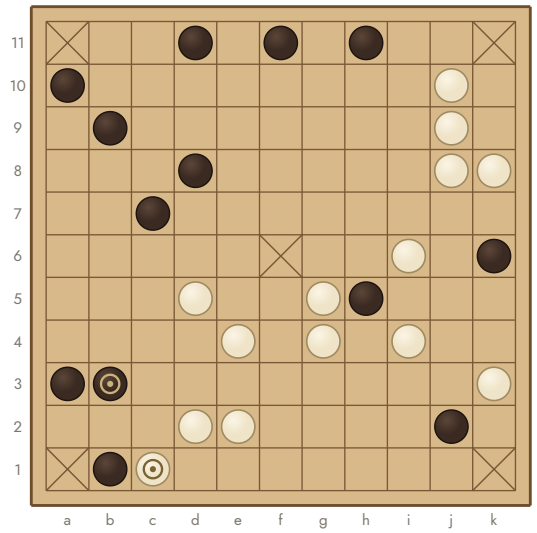
21.3 Black to play and win in three moves.



21.4 White to play and win in three moves.



21.5 White to play and win in three moves.



21.6 Black to play and win in three moves.

VI

LEVEL 4 · STRONG PLAYER

The endgame

When both kings are out, the game becomes a race. This part gives the theory of that phase: counting tempi, corner locks and the art of defending.

---

22 The final phase

23 Counting tempi

24 Races, locks and defence

22 The final phase

Most games are decided while many pawns are still on the board: the "endgame" is not an empty board, it is the moment when **both kings are out** and fight for the corners. The game becomes a race in which every tempo counts.

The rules of the race

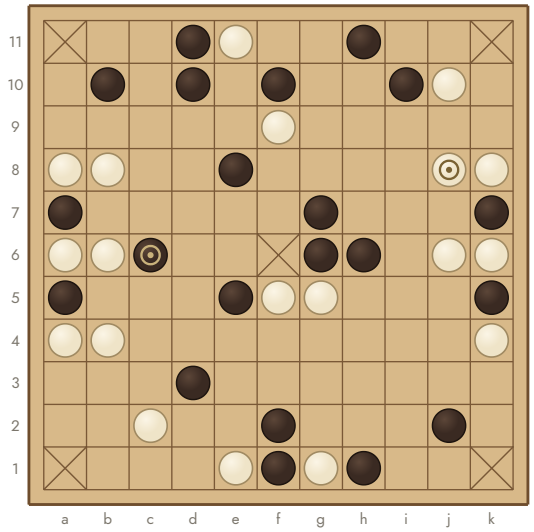
- **Count tempi.** How many moves does your king need to create an unstoppable threat? And your opponent's? The side with a tempo in hand can ignore slow threats.
- **A threat gains a tempo.** Every threat forces your opponent to defend instead of advancing: chain them.
- **Do not close your own lines.** A defence that blocks your king's road costs double.
- **Keep a piece on every dangerous edge.** A single piece able to reach a line is enough to make a threat stoppable.
- **A king on an edge is strong, next to a corner it is fragile**: a single pawn is enough to capture it there.

When few pawns are left

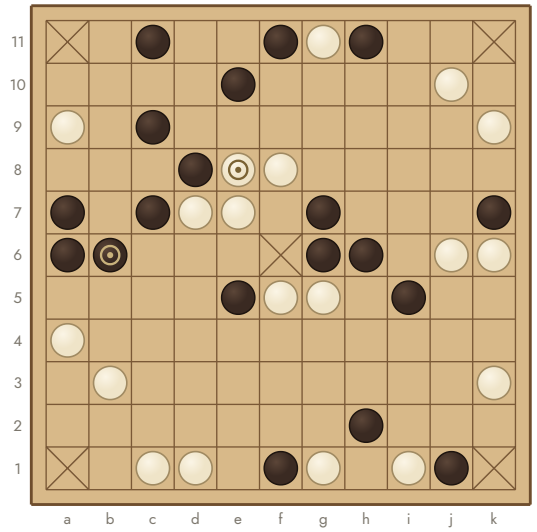
The fewer pawns remain, the freer the kings and the easier corner forks become. The side with more pieces able to reach the edges keeps the advantage: before exchanging, ask yourself who will still be able to close the lines.

> **REMEMBER**
>
> In the final phase, a tempo can be worth more than a pawn. Threaten, count, and never leave an edge without a defender.

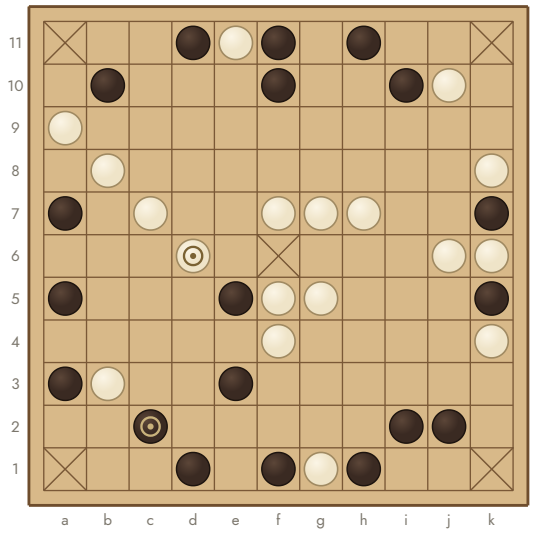
Exercises



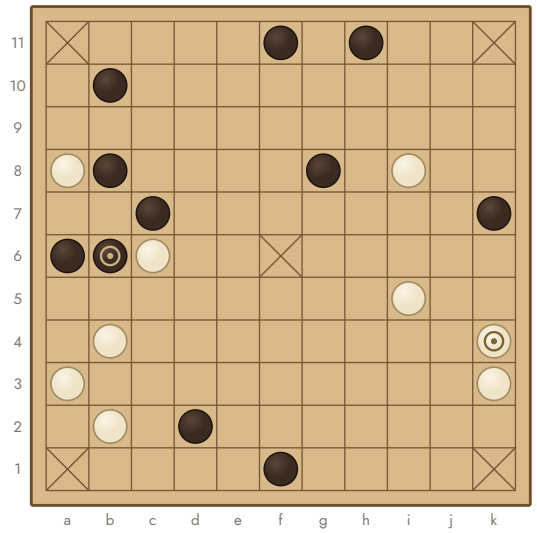
22.1 Black is preparing an unstoppable threat. White to play: find the defence.



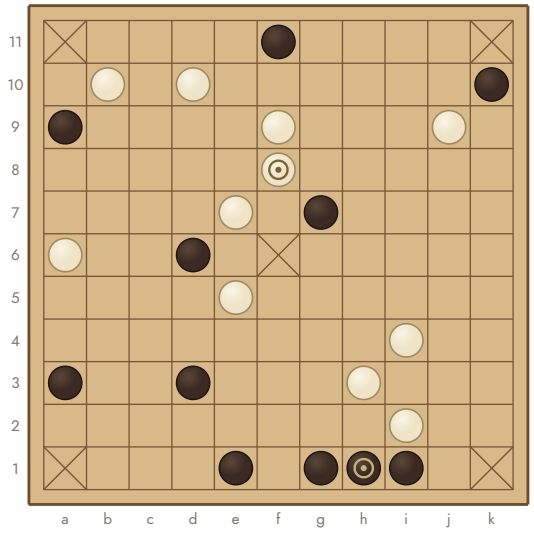
22.2 Black is preparing an unstoppable threat. White to play: find the defence.



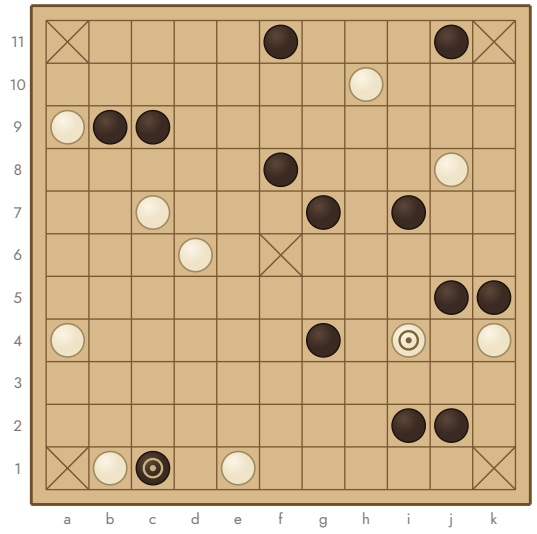
22.3 Black is preparing an unstoppable threat. White to play: find the defence.



22.4 White to play and win in two moves.



22.5 White to play and win in two moves.



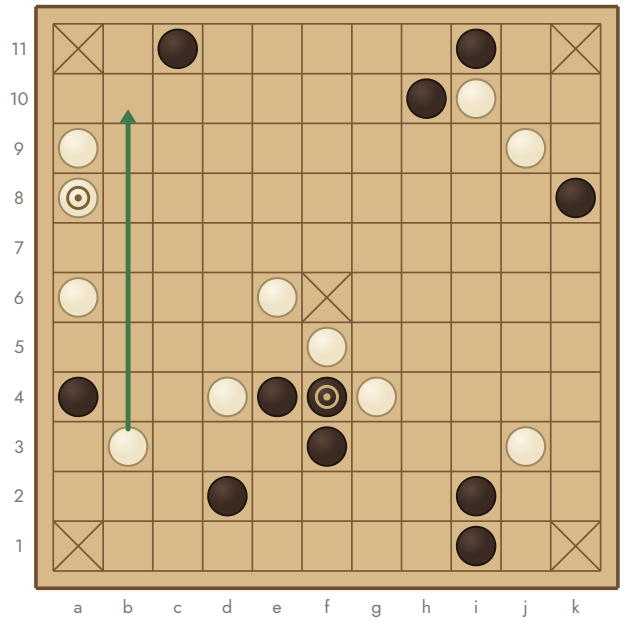
22.6 Black to play and win in two moves.

23 Counting tempi

When both kings are out, the game becomes a race. To judge it, strong players **count tempi**: for each king, how many moves would it need, if the opponent did nothing, to create an unstoppable threat?

The rule of the race

- **Compare your tempi.** The side that needs fewer moves is favoured, and a tempo in hand usually weighs more than a pawn.
- **With equal tempi, the move decides.** The side to play has the advantage: it gets there first.
- **If your opponent is one tempo away**, that is, he threatens to create an unstoppable threat next move, you must stop him first: your own race will wait.
- **Every threat gains a tempo**, since it forces your opponent to defend instead of advancing. A chain of threats can overturn an unfavourable count.



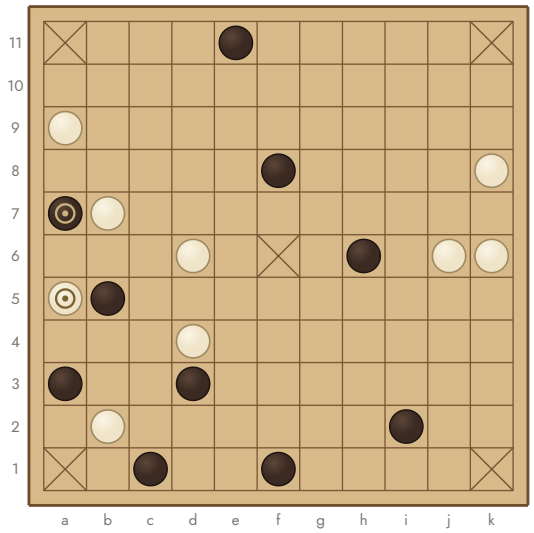
White to play. The white king needs two moves to create an unstoppable threat, the black king three. With this tempo in hand, White wins: b3-b10! prepares a9-b9, which will open the a-file to a11 for the king, and Black can do nothing about it.

How to count

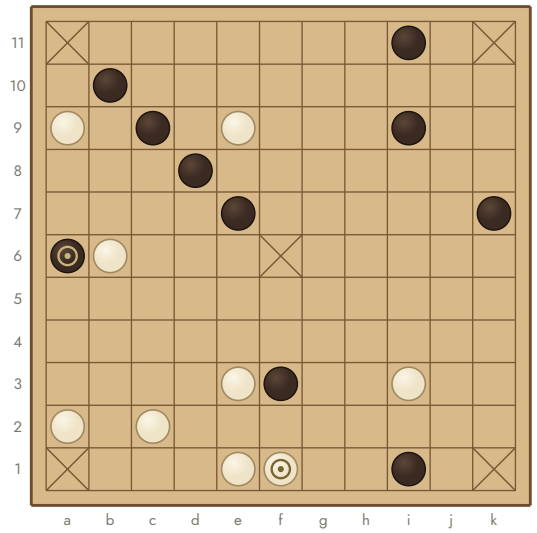
For a given king, first look for a move that creates an unstoppable threat: if there is one, the count is one. Otherwise, look for a move that prepares such a threat for the following move, and so on. Enemy pieces matter: a defender able to reach a line in one move lengthens the count by at least one tempo.

> **REMEMBER**
>
> In the race, count both kings' tempi before every move. A tempo in hand is often worth more than a pawn; with equal counts, the move decides.

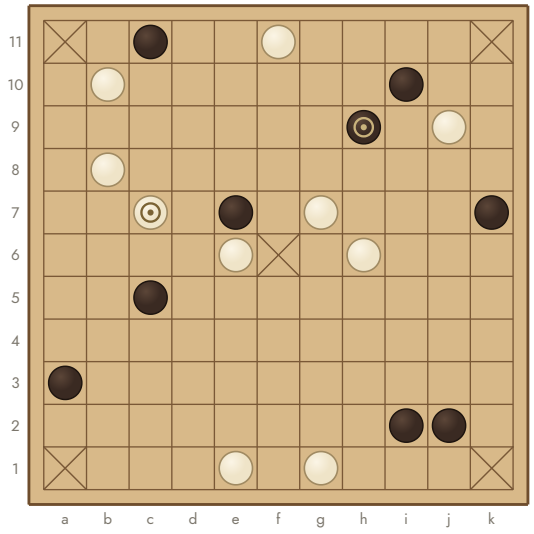
Exercises



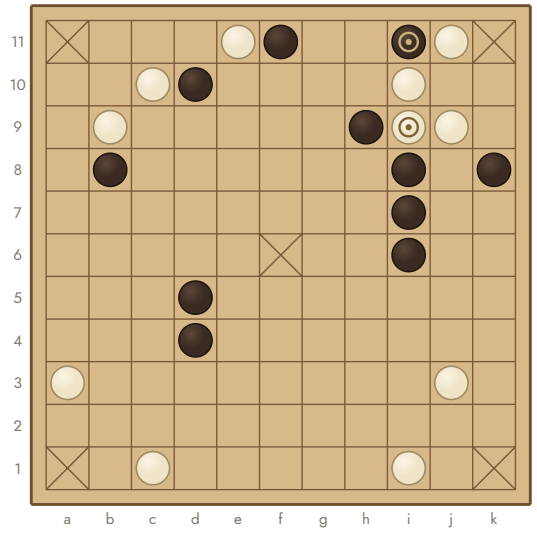
23.1 White to play. Count the tempi: how many moves would each side need, if the other did nothing, to create an unstoppable threat?



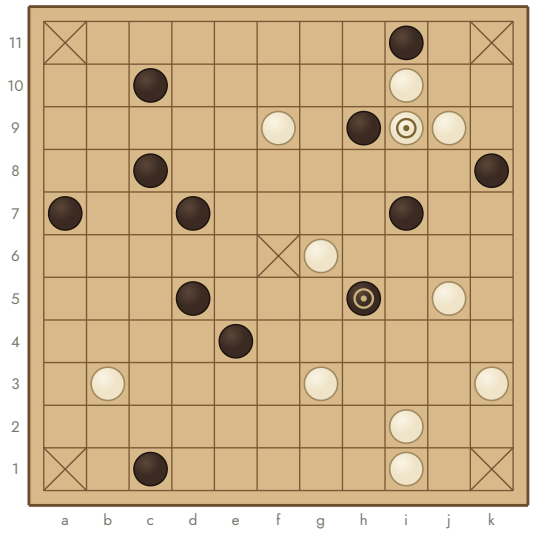
23.2 Black to play. Count the tempi: how many moves would each side need, if the other did nothing, to create an unstoppable threat?



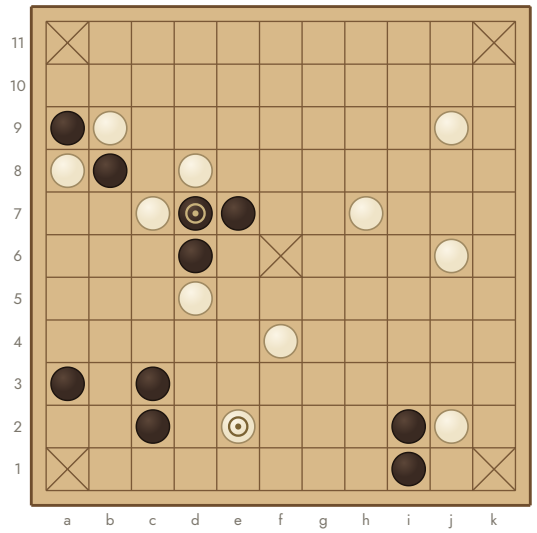
23.3 Black to play. Count the tempi: how many moves would each side need, if the other did nothing, to create an unstoppable threat?



23.4 White to play. Count the tempi: how many moves would each side need, if the other did nothing, to create an unstoppable threat?



23.5 Black to play. Count the tempi: how many moves would each side need, if the other did nothing, to create an unstoppable threat?



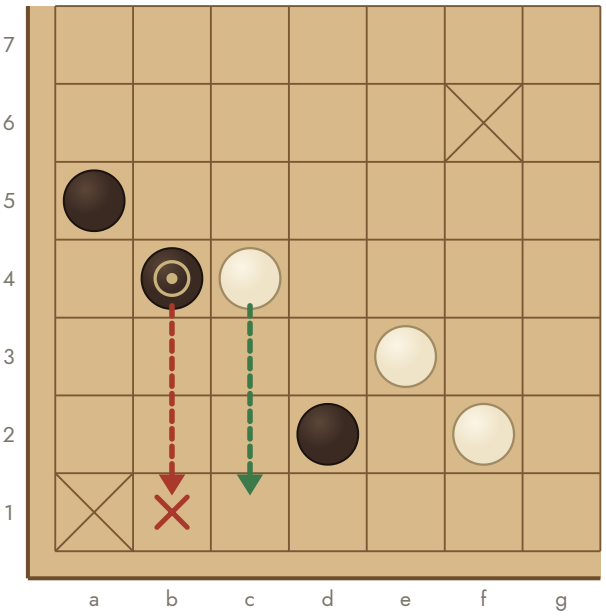
23.6 Black to play. Count the tempi: how many moves would each side need, if the other did nothing, to create an unstoppable threat?

24 Races, locks and defence

Defending a final phase means lengthening the enemy king's tempo count. Three tools are usually enough.

The corner lock

A pawn able to reach the edge square two squares from a corner (c1 for corner a1, for example) forbids the enemy king from stopping next to that corner: it would be captured there against the empty corner, and the threat rule makes that move illegal. A single well-placed pawn thus holds a whole corner.



The white pawn on c4 can reach c1: the black king cannot stop on b1: this approach to corner a1 is locked.

Chapter 16 shows how to build these locks from the middlegame on; in the endgame, they become the main defensive weapon.

One defender per edge

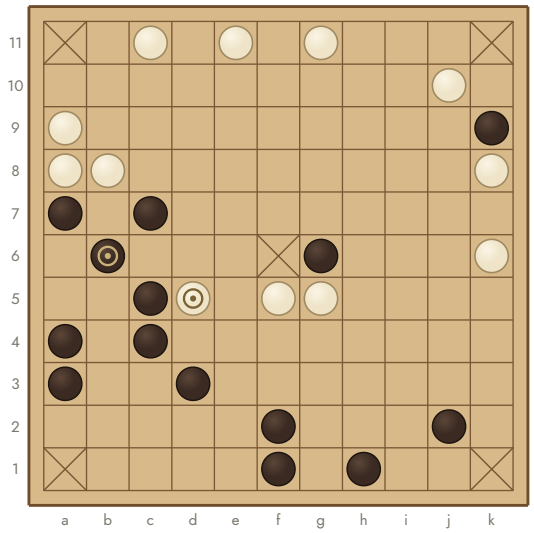
A threat is unstoppable only if no piece can reach the king's line. So keep, on every edge the enemy king can reach, at least one piece able to cut the line in one move. Corner forks are always born on an unguarded edge.

The counter-threat

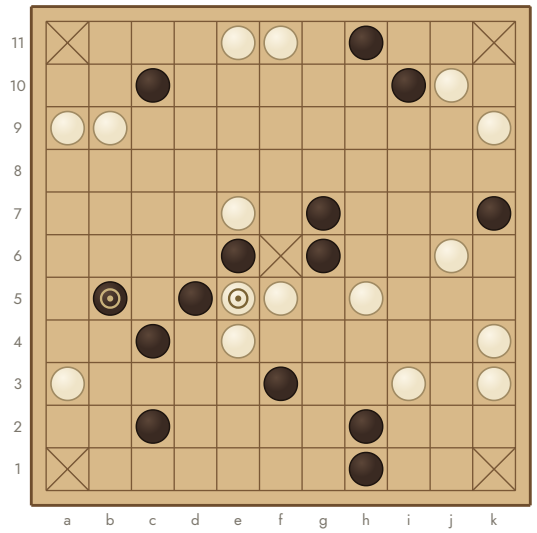
The best defence is often a threat: if your defence itself creates a threat, your opponent must answer, and you gain a tempo. Always look, among your defences, for the one that threatens something.

> **REMEMBER**
>
> Lock the corners, keep a defender on every threatened edge, and prefer defences that threaten.

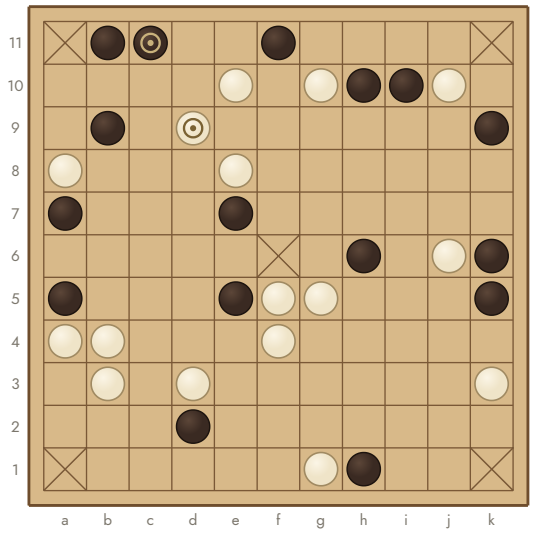
Exercises



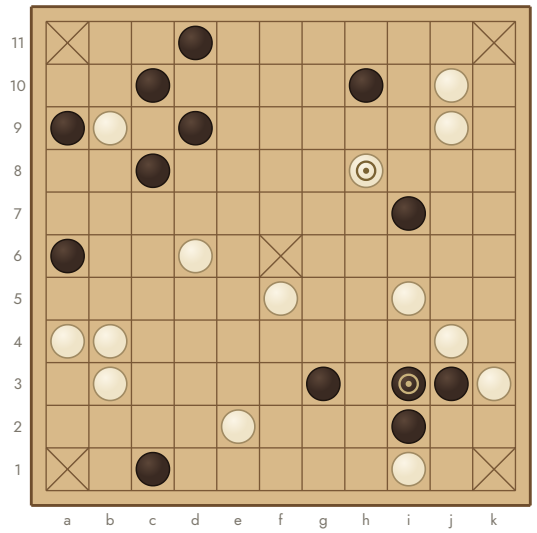
24.1 Black is preparing an unstoppable threat. White to play: find the defence.



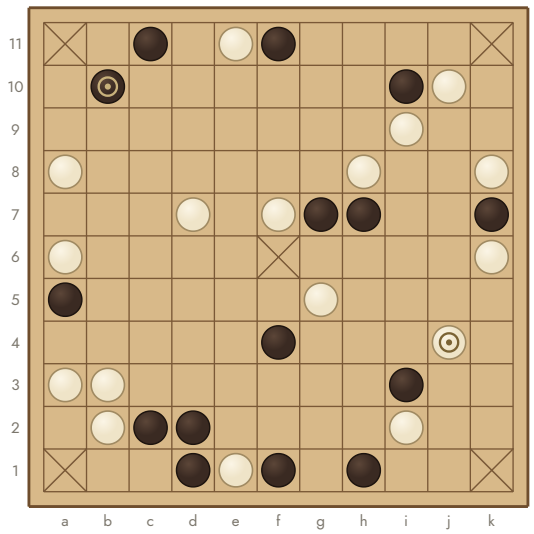
24.2 Black is preparing an unstoppable threat. White to play: find the defence.



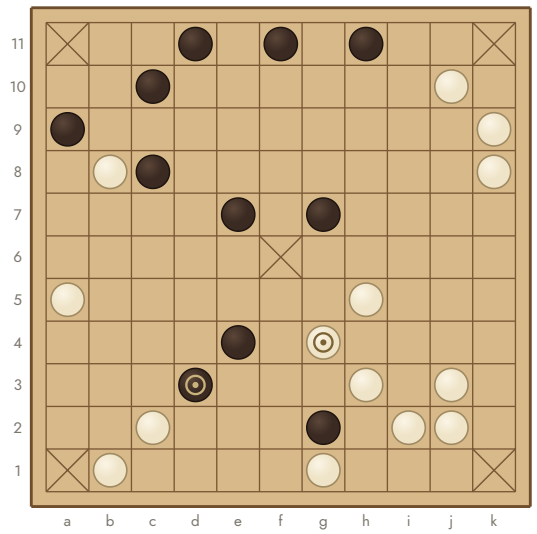
24.3 Black is preparing an unstoppable threat. White to play: find the defence.



24.4 White is preparing an unstoppable threat. Black to play: find the defence.



24.5 Black is preparing an unstoppable threat. White to play: find the defence.



24.6 White is preparing an unstoppable threat. Black to play: find the defence.

VII

LEVEL 4 · STRONG PLAYER

Annotated games

To finish, high-level games annotated at the key moments: they bring together the opening, structures, moving the pair out and the final race.

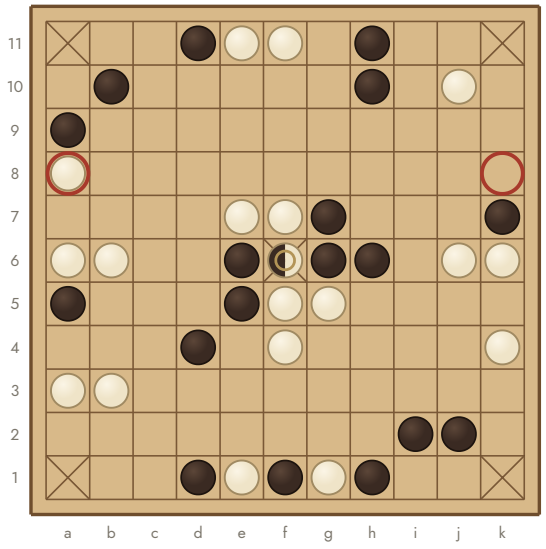
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25 Annotated games

Game 1 · Racing to the corner with fewer pawns

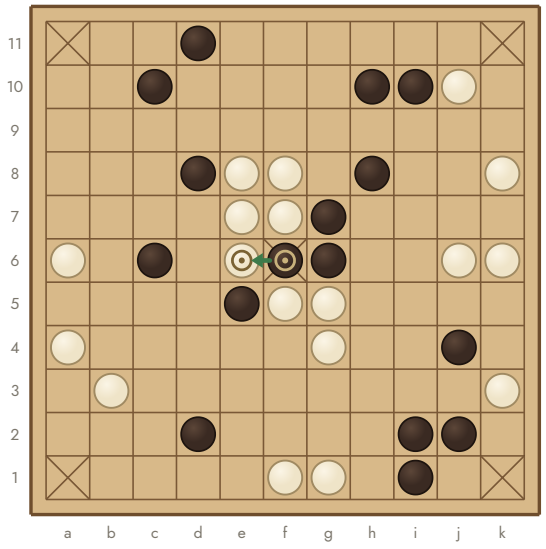
A high-level game. Black, behind in material, keeps his king on the throne for a long time; he takes it out only when a road opens, and the race to the corner beats the pawns.

1. f10-h10 b2-b3
2. f2-i2 a4-a3
3. d6-d4 k4-j4
4. f11-f9 g11-f11
5. f9-k9×k8 j4-k4×k5
6. k9-a9×a8 f8-a8×a7
7. f1-f3 e1-f1
8. h6-h8 e11-e8
9. f3-j3 k4-k3
10. b10-c10 a3-a4×a5
11. h10-i10 f11-f10
12. h11-h10 b6-b10
13. a9-a10×b10 a8-a9×a10
14. j3-j4 f4-g4
15. d1-d2 a9-k9
16. d4-d8 f10-f8
17. h1-i1 k9-k8×k7
18. e6-c6 f6-e6
19. d11-i11 a4-a5
20. d8-d5 e8-b8
21. h10-e10 b3-j3
22. j4-i4 j6-j4
23. i4-i5 k6-k7
24. g6-k6 e7-e9
25. f6-i6 e6-e7
26. c6-c7 b8-b9
27. i6-d6 k7-h7×g7
28. d6-d11 b9-b11
29. c10-c11×b11 g4-b4
30. k6-b6 e9-h9×h8
31. c11-c10



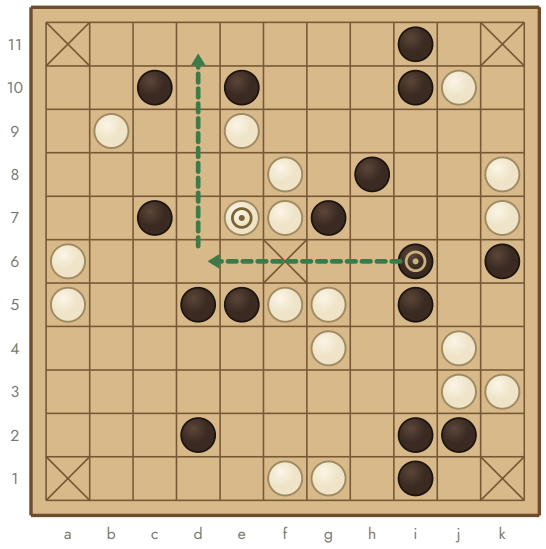
After 6. ... f8-a8

The first moves follow the opening pattern: Black captures on k8 then on a8, White recaptures at once on k5 then on a7. Material stays even and the edges thin out.



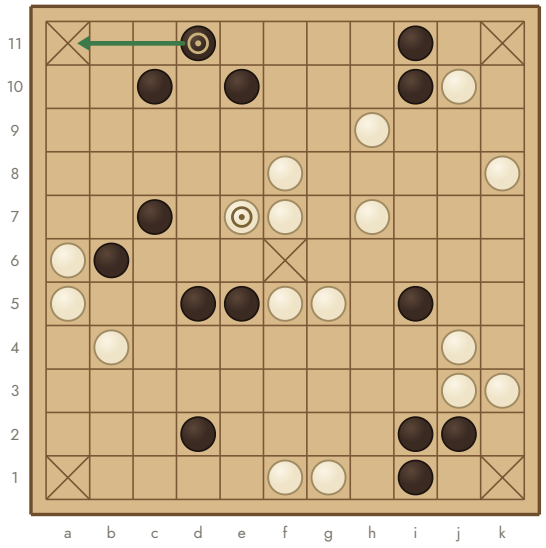
After 18. ... f6-e6

During the middlegame, White has won two pawns. He now moves the pair out with f6-e6, as chapter 18 advises when ahead in pawns; the black king stays alone on the throne.



After 26. ... b8-b9

White has just played b8-b9?, a waiting move. He should have played k7-h7, capturing g7 and keeping the initiative. Black seizes the chance at once: his king heads for the edge with i6-d6, then d6-d11.



After 31. c11-c10

After d6-d11, the black king threatens a11; White defends with b9-b11, but c10-c11 captures b11 against the empty corner. Finally c11-c10! clears the line: the king threatens a11 through c11 and b11, and no white piece can reach them any more. Black wins three pawns down.

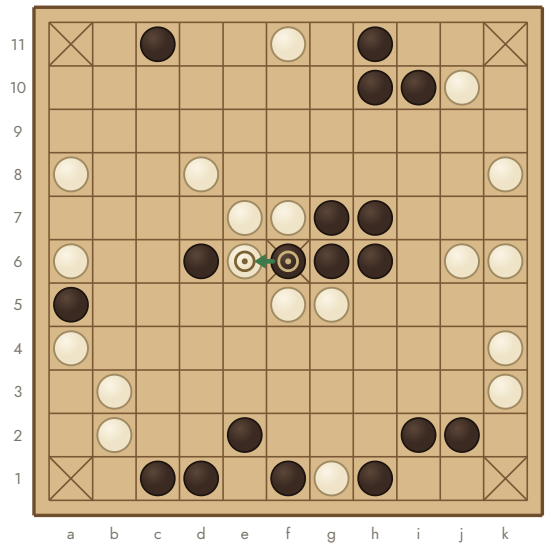
THE LESSON

Two extra pawns do not protect you from an active king. Once the pair has left, every waiting move gives time to the enemy king: White had to keep his pieces on the roads to the corners.

Game 2 · One capture too many, then the king hunt

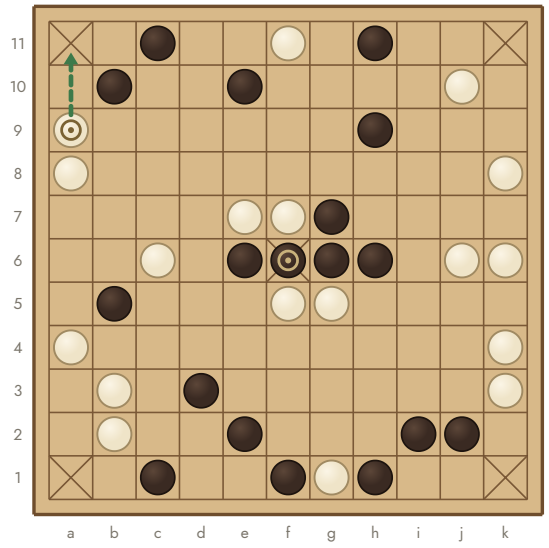
A second high-level game. The white king takes the initiative, a capture that overlooks the opponent's threat tips the balance, and the black king, leaving too late, ends up trapped.

1. d11-c11 a8-d8
2. f2-i2 d8-a8×a7
3. f10-i10 b6-b3
4. d6-d11×e11 e7-e11×f11
5. b10-f10 g11-f11
6. f10-h10 f8-h8
7. k7-h7 k4-k3
8. d1-c1 f4-k4×k5
9. d11-d1×e1 h8-d8
10. e5-e2 e11-e7
11. e6-d6 f6-e6
12. h10-e10 e6-e3
13. d1-d3 e3-e5
14. e10-c10 a6-c6
15. d6-e6 e5-d5
16. a5-b5 d8-d10
17. i10-e10×d10 d5-d9
18. h7-h9 d9-b9
19. c10-b10 b9-a9
20. b10-a10 a9-b9
21. e10-b10 b9-b7
22. a10-a9 j6-j3
23. g7-k7 k8-k9
24. e6-e4 k4-f4
25. e4-c4 j10-j9
26. h6-h4 a4-b4
27. b5-a5 f4-d4×c4
28. a5-a7×a8 b7-d7
29. h4-j4×j3 b2-d2×d3
30. a7-a3 f5-f2×e2
31. c1-c3×b3 e7-e1×f1
32. g6-g8 d2-c2
33. c3-d3 f2-d2×d3
34. h1-h7 k6-i6
35. f6-f3 g5-g3
36. f3-f2 g1-i1
37. g8-f8×f7 d4-f4
38. h7-e7 d7-d5
39. h9-h1 g3-g2
40. e7-e2 k3-e3×e2



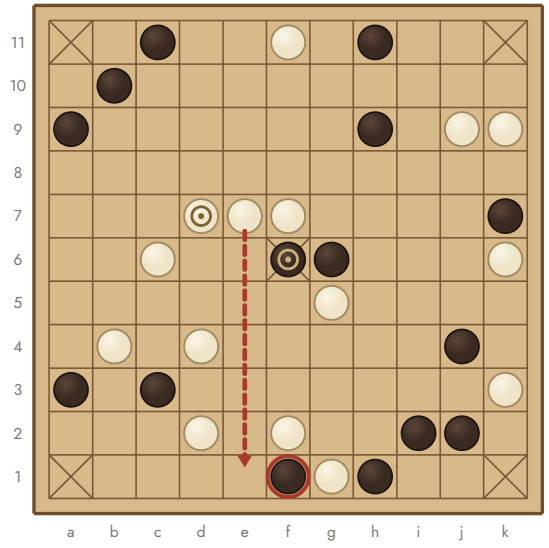
After 11. ... f6-e6

White is a pawn up and moves the pair out with f6-e6. The black king stays on the throne, untouchable.



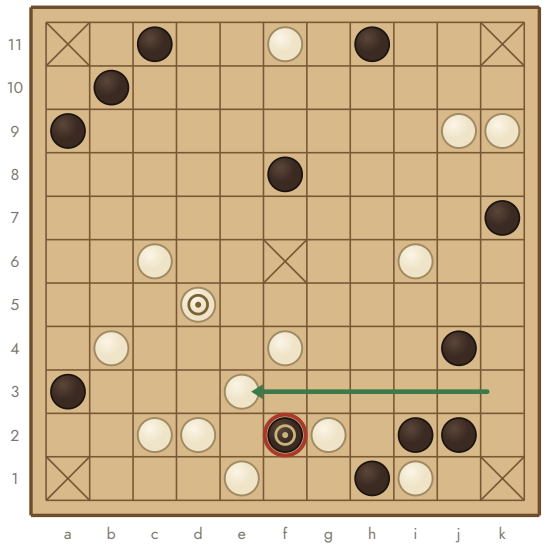
After 19. ... b9-a9

Black has restored the balance, but the white king has reached the edge: b9-a9 threatens a11, and Black must close the line with b10-a10. For many moves, the white king shuttles along this edge and ties several black pieces to the defence.



After 31. c1-c3

Black has just played c1-c3?, capturing b3, but overlooks White's threat: the e-file has been open since the capture on e2, and e7-e1 captures f1. Black should have played f1-e1, closing the file. Soon after, f2-d2 also captures d3, and White is two pawns up.



After 40. ... k3-e3

In trouble, Black tried his luck with f6-f3, then took refuge on f2, among the white pawns. k3-e3 captures e2 and threatens to capture the king in three ways (e1-e2, d2-e2 or e3-e2): there is no defence.

THE LESSON

Before capturing, look at what your opponent is threatening. And a king that leaves the throne when behind in material becomes the target of the opponent's plan B.

Glossary

Anvil — The piece, or the empty special square, holding the other side of a capture.

Barrage — A pawn occupying a king's line and closing it.

Capture square — A square from which a pawn would at once capture an enemy pawn.

Chain — A line of alternating pawns on an edge; the pawn that reaches its end takes the last enemy pawn.

Control — A side controls an empty square when an enemy piece stopping there could be captured next move.

Corner fork — A king threatening two corners at once: only one line can be closed.

Corner guard — A pawn standing near a corner, on b2, b3, c2, c3 or a mirror square.

Counter-threat — A defence that itself creates a threat: the opponent must answer instead of pressing on.

Defence — A move that prevents an immediate win by the opponent.

Edge — Files a and k, ranks 1 and 11. The edge is never an anvil.

Glued pawn — A pawn touching a friendly pawn along a side: it cannot be captured along the line joining them, but it controls fewer squares.

Hammer — The arriving pawn that makes the capture.

Hanging pawn — A pawn the opponent can capture next move.

Interception — A pawn able to cut, in one move, the enemy king's line to a corner.

Lock — A pawn able to arrive on the capture square of a square next to a corner: it forbids the enemy king from stopping there.

Net — The control of the squares next to the enemy king.

Occupation — The presence of a piece on a square: it bars the line to every piece.

Pair — The two kings stacked on the throne at the start of the game.

Pincer — The position of a piece between the hammer and the anvil, on the same line.

Reach — The squares a pawn can reach in one move.

Road — A sequence of empty squares leading a king to a corner.

Screen — A pawn placed on the square your opponent needs for a capture, cancelling his control.

Special square — One of the four corners or the throne f6. When empty, it acts as an anvil.

Strategic value — What a pawn is worth according to its square.

Tactical value — What a pawn is worth according to what it does in relation to the other pieces.

Tempo — A move. Counting tempi means counting the moves each king needs to create an unstoppable threat.

The move — The right to play: the side "to move" is the one whose turn it is.

Throne — The central square f6, where the pair starts. No piece may stop there; the king left on it cannot be captured.

Unstoppable threat — A winning threat against which no defence exists: the game is won.

Wall — At least three adjacent pawns in a line: a long barrage, none of whose pawns can be captured along that line.

Solutions

Chapter 1 · The board and the pieces

1.1 Six: d3 e3 b3 c4 c5 c6.

1.2 Six: e6 c6 d7 d8 d5 d4.

1.3 Six: c1 d1 e1 b2 b3 b4.

1.4 Nine: d9 e9 f9 b9 a9 c10 c11 c8 c7.

Chapter 2 · Capturing

2.1 e9-e5.

2.2 f1-c1.

2.3 e8-e5 captures two pieces at once.

2.4 d8-f8 (or f10-f8).

2.5 d8-d6.

2.6 g4-f4.

Chapter 3 · The king and the pair

3.1 i7-i11.

3.2 h8-f8.

3.3 e7-e4.

3.4 j1-i1.

3.5 e1-c1.

3.6 j5-j8.

Chapter 4 · Threats and winning

4.1 d5-k5. White was threatening a5-b5, an unstoppable threat: the king threatens a1, and no black piece can reach a5, a4, a3 or a2; it also threatens to capture the black king with e3-e5. Any other move loses.

4.2 h8-k8. Black was threatening c8-c11, an unstoppable threat: the king threatens a11, and no white piece can reach b11. Any other move loses.

4.3 i11-i3. White was threatening g3-k3, an unstoppable threat: the king threatens k1, and no black piece can reach k2. Any other move loses.

4.4 b5-b1! The king threatens the neighbouring corner a1, and it cannot be captured, since no white pawn can reach c1.

4.5 c8-c11! The king threatens a11, and no white piece can reach b11.

4.6 b2-b1! The king threatens the neighbouring corner a1.

Chapter 5 · Winning material

5.1 d7-d5 captures e5 against f5. The capturing pawn cannot be taken back.

5.2 d5-d11 captures e11 against f11. The capturing pawn cannot be taken back.

5.3 h9-h6 captures g6 against the empty throne. The capturing pawn cannot be taken back.

5.4 i11-i3 captures i2 against i1. The capturing pawn cannot be taken back.

5.5 i5-i3 captures h3 against g3. The capturing pawn cannot be taken back.

5.6 e6-e4 captures f4 against g4. The capturing pawn cannot be taken back.

Chapter 6 · Poisoned captures

6.1 No. After c5-e5, which captures e6 against e7, White answers c7-c1: the king threatens a1, and no black piece can reach b1. The game is lost.

6.2 No. After g1-g2, which captures h2 against i2, White answers d4-d1: the king threatens k1 and a1 at once: Black can close only one line. The game is lost.

6.3 No. After i4-h4, which captures h5 against h6, Black answers i6-i1: the king threatens k1 and a1 at once: White can close only one line. The game is lost.

6.4 No. After i4-i11, which captures j11 against the empty corner k11, Black answers c7-b7 and threatens to capture the white king with b4-b5. The game is lost.

6.5 No. After d3-d6, which captures e6 against the empty throne, Black answers f8-k8: the king threatens k11, and no white piece can reach k9 or k10. The game is lost.

6.6 No. After k6-h6, which captures g6 against the empty throne, Black answers a3-b3 and captures b4 against b5. The balance is negative: one pawn down.

Chapter 7 · Combinations

7.1 c5-d5 captures d6 against d7.

7.2 i2-i1: the king threatens k1. Against the best defence, j9-j1, Black plays d3-e3 and captures f3 against g3.

7.3 c9-k9 captures k10 against the empty corner k11. Against the best defence, d2-d10, Black plays e6-e10 and captures d10 against c10.

7.4 c9-c7: threatens to capture the black king in two ways (c4-c5 or d5-c5). Against the best defence, c6-e6, White plays c7-g7 and captures g6 against g5.

7.5 c3-a3 captures a2 against the empty corner a1.

7.6 j6-k6: the king threatens k11. Against the best defence, j10-k10, White plays d9-k9 and captures k10 against the empty corner k11.

Chapter 8 · Defending and preventing

8.1 i3-i9. White was threatening h9-k9, an unstoppable threat: the king threatens k11, and no black piece can reach k10. Any other move loses.

8.2 k4-k3 or **k4-k2**: these are the only two defences. Black was threatening i3-k3, an unstoppable threat: the king threatens k1, and no white piece can reach k2. Any other move loses.

8.3 d4-a4. Black was threatening c2-a2, an unstoppable threat: the king threatens the neighbouring corner a1, and it cannot be captured, since no white pawn can reach a3. Any other move loses.

8.4 g4-g3 or **j6-j3**: these are the only two defences. Black was threatening f3-k3, an unstoppable threat: the king threatens k1, and no white piece can reach k2. Any other move loses.

8.5 c10-i10. White was threatening i8-i11, an unstoppable threat: the king threatens k11, and no black piece can reach j11. Any other move loses.

8.6 a9-c9. Black was threatening c7-c11, an unstoppable threat: the king threatens a11, and no white piece can reach b11. Any other move loses.

Chapter 9 · The unstoppable threat

9.1 f8-a8! The king threatens a11 and a1 at once: White can close only one line.

9.2 e7-e11! The king threatens k11 and a11 at once: White can close only one line.

9.3 i10-g10! Captures f10 against e10; threatens to capture the black king in three ways (e10-f10, g10-f10 or f11-f10).

9.4 k7-k10! Threatens to capture the white king in two ways (i9-i10 or g10-i10).

9.5 i11-i10! The king threatens k11, and no black piece can reach g11, h11, i11 or j11.

9.6 a3-b3! The king threatens a1, and no black piece can reach a5, a4, a3 or a2.

Chapter 10 · Winning in two moves

10.1 f3-h3! Threatens to capture the black king with h3-h9 — for example g6-h6 — f6-f1: the king threatens k1 and a1 at once: Black can close only one line.

10.2 j3-j6! Threatens to capture the white king with j6-i6 — for example i4-i6 — i3-k3: the king threatens k1, and no white piece can reach k2.

10.3 i10-j10! A quiet move that prepares what follows — for example b2-c2 — j10-j11: the king threatens the neighbouring corner k11.

10.4 b2-f2! A quiet move that prepares what follows — for example i2-j2 — b3-b1: the king threatens the neighbouring corner a1, and it cannot be captured, since no black pawn can reach c1.

10.5 i9-a9! Captures a10 against the empty corner a11 — for example d7-d9 — a9-b9: the king threatens a11, and no white piece can reach a7, a8, a9 or a10.

10.6 c2-b2! A quiet move that prepares what follows — for example g3-h3 — d6-d1: the king threatens a1, and no white piece can reach c1 or b1.

Chapter 11 · Opening principles

11.1 h9-k9 captures k8 against k7. Against the best defence, e1-e2, Black plays d1-d2 and captures e2 against f2.

11.2 i11-h11 captures g11 against f11. The capturing pawn cannot be taken back.

11.3 f8-e8 captures e7 against e6. The capturing pawn cannot be taken back.

11.4 f3-f1 captures e1 against d1 and g1 against h1.

11.5 f3-c3 captures c2 against c1. The capturing pawn cannot be taken back.

11.6 f3-f1 captures e1 against d1 and g1 against h1. The capturing pawn cannot be taken back.

Chapter 12 · The main lines

12.1 i6-i4. After i6-h6, Black wins in three moves starting with h11-h7 (see the next exercise). The retreat i6-i5 still holds, but makes the hunt easier for Black. It is i6-i4 that resists best.

12.2 h11-h7! Threatens to capture the white king in two ways (h1-h5 or g5-h5) — for example h6-h3 — i2-i3: threatens to capture the white king with g4-g3 — f3-g3 — h1-h2: threatens to capture the white king in two ways (g4-h4 or h7-h4).

12.3 i5-k5: the king threatens k1. Against the best defence, i2-k2, White plays f3-k3 and captures k2 against the empty corner k1. The white king, hunted until now, becomes an attacking piece in turn.

Chapter 13 · Opening traps

13.1 d3-d1 captures e1 against f1, and the exchanges that follow leave Black a pawn up. After the right move f8-i8, the same capture would have been met by i8-i11, which takes h11: a mere exchange. After a4-a3, White is left with only an insufficient recapture.

13.2 d1-d4 captures d5 against d6, and the exchanges that follow leave Black a pawn up. White had to take d6 with e7-d7: that pawn is exactly the anvil of the capture d1-d4. By capturing on the other side, White leaves Black the last word.

13.3 h9-k9 captures k8 against k7, and the exchanges that follow leave Black a pawn up. With the king on b6, as in the main line, this capture would have lost: b6-b11 at once threatened the corner a11, with no defence. On d6 the king threatens nothing, and the capture becomes free.

13.4 No capture. White threatens b6-b11, which attacks the neighbouring corner a11: an unstoppable threat. Only three moves hold: **a7-b7**, **h9-b9** or **f10-b10**. Every capture therefore loses the game. In the main line, Black plays h9-b9.

Chapter 14 · Control and occupation

14.1 White pawns hanging: a8 (b9-a9); k8 (b9-k9). **Black pawns hanging:** g6 (f7-g7 or h7-g7). Black is to move: he can capture first, but must also think of his own threatened pawn.

14.2 White pawns hanging: e11 (f10-f11 or i11-f11). **Black pawns hanging:** a7 (f8-a8). Black is to move: he can capture first, but must also think of his own threatened pawn.

14.3 The possible exits from f6 are f7, f8, f5 and f4. Without being captured, the black king can come out on f7, f8, f5 and f4; the white king has none. Only Black can move the pair out safely: the same lines serve both kings, but the landing squares are controlled against one of them only.

14.4 The possible exits from f6 are e6, d6 and c6. Without being captured, the black king has none; the white king can come out on e6, d6 and c6. Only White can move the pair out safely: the same lines serve both kings, but the landing squares are controlled against one of them only.

14.5 g7-g6: the pawn occupies g6 and bars h6 and i6; the other exits are controlled against the black king: f7 (e7, g11, g6). The black king no longer has a safe exit, while the white king can still come out on f7.

14.6 e5-e6: the pawn occupies e6 and bars d6; the other exits are controlled against the black king: g6 (f7, g5, g8), f5 (e6, f3, g5, i4) and f4 (f3, f7, g5). The black king no longer has a safe exit, while the white king can still come out on g6, f5 and f4.

14.7 e9-e8: the other exits are controlled against the white king: g6 (g2, g7), f7 (e6, e8, g7) and f8 (e8, g7). The white king no longer has a safe exit, while the black king can still come out on g6 and f7.

Chapter 15 · The value of pawns

15.1 Black. Material is level. Black has nine more safe squares for his king (two and a quarter pawns), four fewer edge pawns (one pawn), two more corner guards (one pawn), two more barrages (one pawn), two more corner threats (one pawn), the move (half a pawn) and one fewer hanging pawn (a quarter of a pawn). Against that: two fewer locks (half a pawn) and four more glued pawns (one pawn). In total, Black is better by about five and a half pawns.

15.2 White. Black is one pawn up. White has eleven more safe squares for his king (two and three-quarter pawns), one more barrage (half a pawn), one more corner threat (half a pawn) and one fewer glued pawn (a quarter of a pawn). Against that: four fewer locks (one pawn), one fewer corner guard (half a pawn), a pawn down (one pawn) and the move to the opponent (half a pawn). In total, White is better by about one pawn.

15.3 Black. Material is level. Black has ten more safe squares for his king (two and a half pawns), two more corner guards (one pawn), two fewer edge pawns (half a pawn) and the move (half a pawn). Against that: three more glued pawns (three quarters of a pawn). In total, Black is better by about four pawns.

15.4 White. Material is level. White has nine more safe squares for his king (two and a quarter pawns), five fewer edge pawns (one and a quarter pawns), one more corner guard (half a pawn), one more barrage (half a pawn) and one fewer glued pawn (a quarter of a pawn). Against that: the move to the opponent (half a pawn). In total, White is better by about four pawns.

15.5 White. Material is level. White has five more safe squares for his king (one and a quarter pawns), three more locks (three quarters of a pawn), three fewer glued pawns (three quarters of a pawn), one more corner guard (half a pawn), one more barrage (half a pawn) and one fewer hanging pawn (a quarter of a pawn). Against that: two more edge pawns (half a pawn) and the move to the opponent (half a pawn). In total, White is better by about three pawns.

15.6 White. Black is one pawn up. White has seven fewer edge pawns (one and three-quarter pawns), seven more safe squares for his king (one and three-quarter pawns), two more corner guards (one pawn), two fewer glued pawns (half a pawn) and one more corner threat (half a pawn). Against that: one fewer lock (a quarter of a pawn), a pawn down (one pawn) and the move to the opponent (half a pawn). In total, White is better by about four pawns.

Chapter 16 · The functions of pawn structures

16.1 a2-c2: the black king can no longer stop on b1 or a2, where it would be captured against the empty corner a1: on b1 by c2-c1 and on a2 by a8-a3 or b3-a3. The pawn leaves the square next to the corner, where it locked nothing. Placed on c2, it can arrive on c1 and capture there against the corner: that is what locks.

16.2 k10-i10: the white king can no longer stop on k10 or j11, where it would be captured against the empty corner k11: on k10 by k5-k9 and on j11 by i10-i11. The pawn leaves the square next to the corner, where it locked nothing. Placed on i10, it can arrive on i11 and capture there against the corner: that is what locks.

16.3 a9-c9: the black king can no longer stop on a10 or b11, where it would be captured against the empty corner a11: on a10 by a4-a9 or c9-a9 and on b11 by c9-c11. The pawn leaves a9, the capture square itself: standing on it, it could not arrive there. Once the square is free, it is enough that a pawn can reach it.

16.4 g6-g5: the pawn occupies g5 and cuts the road to h5, i5 and k5. The black king, which could go to h5, i5 and k5, no longer has a safe square.

16.5 e6-e7: the pawn occupies e7 and cuts the road to d7, c7 and b7. The white king, which could go to e7, d7, c7 and b7, no longer has a safe square.

16.6 f10-g10: the pawn occupies g10 and cuts the road to g11; the remaining squares are controlled: f8 (by e8, g10 and g7) and g9 (by e8, g10 and g7). The black king, which could go to g9, g10 and g11, no longer has a safe square.

Chapter 17 · Sacrifices and initiative

17.1 j6-i6! Threatens to capture the black king with e6-g6 — for example d1-d6, which captures e6 against the empty throne — e7-e6: threatens to capture the black king with e6-g6. The pawn on e6 is lost, but the move spent taking it does not save Black: the final threat comes first.

17.2 b3-b2! A quiet move that prepares what follows — for example d1-c1 — b4-b3 — c1-d1 (captures e1 against f1) — b3-a3: the king threatens a1, and no black piece can reach a2. Taking e1 costs Black a tempo, and that is exactly the tempo he is missing.

17.3 d3-c3! A quiet move that prepares what follows — for example d1-c1 — b3-b5: threatens to capture the black king with b5-c5 — e5-c5 (captures b5 against a5) — c3-a3: the king threatens a1, and no black piece can reach a2. Taking b5 changes nothing: White keeps the initiative to the end.

17.4 k3-e3! A quiet move that prepares what follows — for example f4-e4, which captures d4 against c4 — c1-d1: threatens to capture the white king in two ways (h1-f1 or f2-f1) — e1-f1 — h1-g1: threatens to capture the white king in two ways (d1-e1 or e3-e1). The pawn on d4 is lost, but the move spent taking it does not save White: the final threat comes first.

17.5 j9-k9! A quiet move that prepares what follows — for example f11-f10, which captures e10 against d10 — b9-b11: threatens to capture the black king with b11-h11 — h9-h11 — i10-k10: the king threatens the neighbouring corner k11. Taking e10 costs Black a tempo, and that is exactly the tempo he is missing.

17.6 e1-e6! Threatens to capture the black king in three ways (c5-c6, a6-c6 or c7-c6) — for example d5-e5, which captures e6 against e7 — c5-d5: captures e5 against f5; threatens to capture the black king in two ways (c7-d7 or d8-d7) — e7-d7 — a6-c6: threatens to capture the black king with g6-e6. Taking e6 changes nothing: White keeps the initiative to the end. Other solution: c5-c6.

Chapter 18 · Moving the pair out

18.1 Yes: f6-f9! With a pawn up, leaving is the right plan. From f9, the black king can reach k11 in two moves: White must commit pieces to that road, and the extra pawn will make the difference elsewhere.

18.2 Yes: f6-c6! White is a pawn up. From c6, the white king can reach a11 in two moves and moves away from the centre, where Black could catch it in a pincer.

18.3 No. Every exit loses: after f6-f5 or f6-g6, Black creates an unstoppable threat at once (for example g5-h5), and after f6-f4 or f6-f3, he wins in two moves. The white king would be surrounded by black pawns. The right move is **f8-a8**, which captures a7.

18.4 No: j10-d10! which captures c10 and restores the balance. Leaving now would be a mistake: after f6-f9, g7-g9! puts the white king in serious trouble.

18.5 Not yet: f1-c1! captures b1 against the empty corner a1. A free pawn comes first; moving the pair out can wait until the next move.

Chapter 19 · Game plans

19.1 f8-h8! A quiet move that prepares what follows — for example f9-h9 — g7-g9: captures h9 against i9 — f2-f8 — h8-h11: the king threatens k11 and a11 at once: White can close only one line.

19.2 b8-b2! A quiet move that prepares what follows — for example c3-b3 — c10-c3: captures b3 against a3; threatens to capture the white king with j3-e3 — e2-e3 — a3-b3: the king threatens a1, and no white piece can reach a3 or a2.

19.3 f10-d10! A quiet move that prepares what follows — for example e6-d6 — c10-c9 — c2-d2 — d10-a10: the king threatens the neighbouring corner a11, and it cannot be captured, since no black pawn can reach a9.

19.4 i6-i8! Threatens to capture the black king in two ways (k6-k8 or k9-k8) — for example j8-j9 — k9-k10: threatens to capture the black king in two ways (k6-k9 or k10-k9) — j9-j10 — i9-j9: threatens to capture the black king with h11-j11. Other solution: i9-i8.

Chapter 20 · Long combinations

20.1 j6-j4! A quiet move that prepares what follows — for example i1-i4 — j4-k4 — h2-h3 — k3-j3: the king threatens k1, and no black piece can reach k3 or k2.

20.2 c6-c2! A quiet move that prepares what follows — for example g1-g3 — c2-b2 — i1-j1 — b2-b1: the king threatens the neighbouring corner a1.

20.3 f2-b2! A quiet move that prepares what follows — for example f9-f11 — j7-j11 — g2-h2 — b2-b1: the king threatens the neighbouring corner a1.

20.4 a3-j3! A quiet move that prepares what follows — for example e2-h2 — c2-g2: captures h2 against i2 — k3-k4 — h6-h1: the king threatens k1, and no black piece can reach i1 or j1.

20.5 j3-j2! A quiet move that prepares what follows — for example b2-b3 — j4-j3 — i2-h2 — j3-k3: the king threatens k1, and no white piece can reach k2.

20.6 h10-h1! Threatens to capture the white king with h1-f1 — for example e1-e2 — d3-e3: captures f3 against g3; threatens to capture the white king in two ways (d1-e1 or h1-e1) — e2-f2 — e3-f3: threatens to capture the white king in two ways (d1-f1 or h1-f1) — f2-e2 — b2-d2: threatens to capture the white king with f3-f2.

Chapter 21 · Calculating

21.1 d4-k4! A quiet move that prepares what follows — for example i1-i2 — j1-i1: captures i2 against i3; threatens to capture the black king with c1-g1 — h1-g1 — h2-k2: the king threatens the neighbouring corner k1, and it cannot be captured, since no black pawn can reach k3. Other solution: g5-k5.

21.2 d6-d2! A quiet move that prepares what follows — for example a3-a4 — d2-g2 — c3-d3 — g2-g1: the king threatens k1, and no white piece can reach h1, i1 or j1.

21.3 f7-f2! A quiet move that prepares what follows — for example a3-f3 — f2-b2 — f3-f4 — b2-b1: the king threatens the neighbouring corner a1.

21.4 d6-d3! A quiet move that prepares what follows — for example f2-e2 — d3-c3 — f1-g1 — c3-c1: the king threatens a1, and no black piece can reach b1.

21.5 b8-a8! Threatens to capture the black king in two ways (a3-a6 or b6-a6) — for example a7-a5 — b6-b10 — h1-i1 — c9-a9: the king threatens a11, and no black piece can reach a10.

21.6 h5-h1! Threatens to capture the white king with h1-d1 — for example c1-c2 — b3-b2 — c2-c3 — b2-a2: the king threatens the neighbouring corner a1. Other solution: f11-f1.

Chapter 22 · The final phase

22.1 b8-c8 or **f9-c9**: these are the only two defences. Black was threatening c6-c11, an unstoppable threat: the king threatens a11, and no white piece can reach b11. Any other move loses.

22.2 a9-b9. Black was threatening b6-b11, an unstoppable threat: the king threatens the neighbouring corner a11. Any other move loses.

22.3 b3-b2. Black was threatening c2-a2, an unstoppable threat: the king threatens the neighbouring corner a1. Any other move loses.

22.4 i5-i2! A quiet move that prepares what follows — for example h11-h3 — k3-j3: the king threatens k1, and no black piece can reach k3 or k2.

22.5 f8-c8! A quiet move that prepares what follows — for example e1-c1 — c8-c11: the king threatens a11, and no black piece can reach b11.

22.6 c1-c3! A quiet move that prepares what follows — for example b1-b3 — c3-k3: the king threatens k1, and no white piece can reach k2.

Chapter 23 · Counting tempi

23.1 White: two (for example d6-a6 then d4-d8). **Black: three** (for example c1-c6, a3-a4 then c6-a6). White, to move, is a tempo ahead: he is favoured in the race.

23.2 Black: three (for example a6-a8, d8-d9 then a8-k8). **White: two** (for example e1-d1 then d1-d2). Black is a tempo behind: he must first slow down the enemy king before starting his own race.

23.3 Black: two (for example a3-b3 then h9-a9). **White: two** (for example h6-h8 then b10-h10). The counts are equal: having the move gives Black the edge.

23.4 White: two (for example j9-j10 then i9-k9). **Black: three** (for example i11-g11, g11-g2 then g2-a2). White, to move, is a tempo ahead: he is favoured in the race.

23.5 Black: three (for example c1-c2, h5-h1 then h1-a1). **White: two** (for example i2-i5 then g3-g5). Black is a tempo behind: he must first slow down the enemy king before starting his own race.

23.6 Black: two (for example i1-e1 then c3-e3). **White: two** (for example e2-g2 then g2-g11). The counts are equal: having the move gives Black the edge.

Chapter 24 · Races, locks and defence

24.1 e11-e1. Black was threatening b6-b1, an unstoppable threat: the king threatens the neighbouring corner a1, and it cannot be captured, since no white pawn can reach c1. Any other move loses.

24.2 a3-b3. Black was threatening b5-b1, an unstoppable threat: the king threatens the neighbouring corner a1, and it cannot be captured, since no white pawn can reach c1. Any other move loses.

24.3 e10-b10. Black was threatening b11-b10, an unstoppable threat: the king threatens a11, and no white piece can reach b11. Any other move loses.

24.4 i7-i8. White was threatening h8-k8, an unstoppable threat: the king threatens k11, and no black piece can reach k9 or k10. Any other move loses. The defence is a counter-threat: threatens to capture the white king in two ways (g3-g8 or c8-g8).

24.5 j4-k4. Black was threatening b10-b11, an unstoppable threat: the king threatens the neighbouring corner a11. Any other move loses. The defence is a counter-threat: the king threatens k1.

24.6 d3-a3. White was threatening g4-k4, an unstoppable threat: the king threatens k1, and no black piece can reach k3 or k2. Any other move loses. The defence is a counter-threat: the king threatens a1.